

Chapter 2

HEAT CONDUCTION EQUATION

Introduction

2-1C Heat transfer is a *vector* quantity since it has direction as well as magnitude. Therefore, we must specify both direction and magnitude in order to describe heat transfer completely at a point. Temperature, on the other hand, is a scalar quantity.

2-2C The term *steady* implies *no change with time* at any point within the medium while *transient* implies *variation with time* or *time dependence*. Therefore, the temperature or heat flux remains unchanged with time during steady heat transfer through a medium at any location although both quantities may vary from one location to another. During transient heat transfer, the temperature and heat flux may vary with time as well as location. Heat transfer is one-dimensional if it occurs primarily in one direction. It is two-dimensional if heat transfer in the third dimension is negligible.

2-3C Heat transfer to a canned drink can be modeled as two-dimensional since temperature differences (and thus heat transfer) will exist in the radial and axial directions (but there will be symmetry about the center line and no heat transfer in the azimuthal direction). This would be a transient heat transfer process since the temperature at any point within the drink will change with time during heating. Also, we would use the cylindrical coordinate system to solve this problem since a cylinder is best described in cylindrical coordinates. Also, we would place the origin somewhere on the center line, possibly at the center of the bottom surface.

2-4C Heat transfer to a potato in an oven can be modeled as one-dimensional since temperature differences (and thus heat transfer) will exist in the radial direction only because of symmetry about the center point. This would be a transient heat transfer process since the temperature at any point within the potato will change with time during cooking. Also, we would use the spherical coordinate system to solve this problem since the entire outer surface of a spherical body can be described by a constant value of the radius in spherical coordinates. We would place the origin at the center of the potato.

2-5C Assuming the egg to be round, heat transfer to an egg in boiling water can be modeled as one-dimensional since temperature differences (and thus heat transfer) will primarily exist in the radial direction only because of symmetry about the center point. This would be a transient heat transfer process since the temperature at any point within the egg will change with time during cooking. Also, we would use the spherical coordinate system to solve this problem since the entire outer surface of a spherical body can be described by a constant value of the radius in spherical coordinates. We would place the origin at the center of the egg.

2-6C Heat transfer to a hot dog can be modeled as two-dimensional since temperature differences (and thus heat transfer) will exist in the radial and axial directions (but there will be symmetry about the center line and no heat transfer in the azimuthal direction. This would be a transient heat transfer process since the temperature at any point within the hot dog will change with time during cooking. Also, we would use the cylindrical coordinate system to solve this problem since a cylinder is best described in cylindrical coordinates. Also, we would place the origin somewhere on the center line, possibly at the center of the hot dog. Heat transfer in a very long hot dog could be considered to be one-dimensional in preliminary calculations.

2-7C Heat transfer to a roast beef in an oven would be transient since the temperature at any point within the roast will change with time during cooking. Also, by approximating the roast as a spherical object, this heat transfer process can be modeled as one-dimensional since temperature differences (and thus heat transfer) will primarily exist in the radial direction because of symmetry about the center point.

2-8C Heat loss from a hot water tank in a house to the surrounding medium can be considered to be a steady heat transfer problem. Also, it can be considered to be two-dimensional since temperature differences (and thus heat transfer) will exist in the radial and axial directions (but there will be symmetry about the center line and no heat transfer in the azimuthal direction.)

2-9C Yes, the heat flux vector at a point P on an isothermal surface of a medium has to be perpendicular to the surface at that point.

2-10C Isotropic materials have the same properties in all directions, and we do not need to be concerned about the variation of properties with direction for such materials. The properties of anisotropic materials such as the fibrous or composite materials, however, may change with direction.

2-11C In heat conduction analysis, the conversion of electrical, chemical, or nuclear energy into heat (or thermal) energy in solids is called heat generation.

2-12C The phrase “thermal energy generation” is equivalent to “heat generation,” and they are used interchangeably. They imply the conversion of some other form of energy into thermal energy. The phrase “energy generation,” however, is vague since the form of energy generated is not clear.

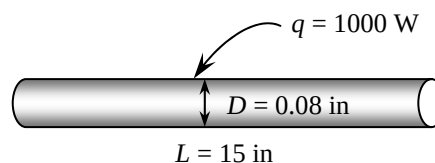
2-13 Heat transfer through the walls, door, and the top and bottom sections of an oven is transient in nature since the thermal conditions in the kitchen and the oven, in general, change with time. However, we would analyze this problem as a steady heat transfer problem under the worst anticipated conditions such as the highest temperature setting for the oven, and the anticipated lowest temperature in the kitchen (the so called “design” conditions). If the heating element of the oven is large enough to keep the oven at the desired temperature setting under the presumed worst conditions, then it is large enough to do so under all conditions by cycling on and off.

Heat transfer from the oven is three-dimensional in nature since heat will be entering through all six sides of the oven. However, heat transfer through any wall or floor takes place in the direction normal to the surface, and thus it can be analyzed as being one-dimensional. Therefore, this problem can be simplified greatly by considering the heat transfer as being one-dimensional at each of the four sides as well as the top and bottom sections, and then by adding the calculated values of heat transfers at each surface.

2-14E The power consumed by the resistance wire of an iron is given. The heat generation and the heat flux are to be determined.

Assumptions Heat is generated uniformly in the resistance wire.

Analysis A 1000 W iron will convert electrical energy into heat in the wire at a rate of 1000 W. Therefore, the rate of heat generation in a resistance wire is simply equal to the power rating of a resistance heater. Then the rate of heat generation in the wire per unit volume is determined by dividing the total rate of heat generation by the volume of the wire to be



$$\dot{g} = \frac{\dot{G}}{V_{\text{wire}}} = \frac{\dot{G}}{(\pi D^2 / 4)L} = \frac{1000 \text{ W}}{[\pi(0.08/12 \text{ ft})^2 / 4](15/12 \text{ ft})} \left(\frac{3.412 \text{ Btu/h}}{1 \text{ W}} \right) = 7.820 \times 10^7 \text{ Btu/h} \cdot \text{ft}^3$$

Similarly, heat flux on the outer surface of the wire as a result of this heat generation is determined by dividing the total rate of heat generation by the surface area of the wire to be

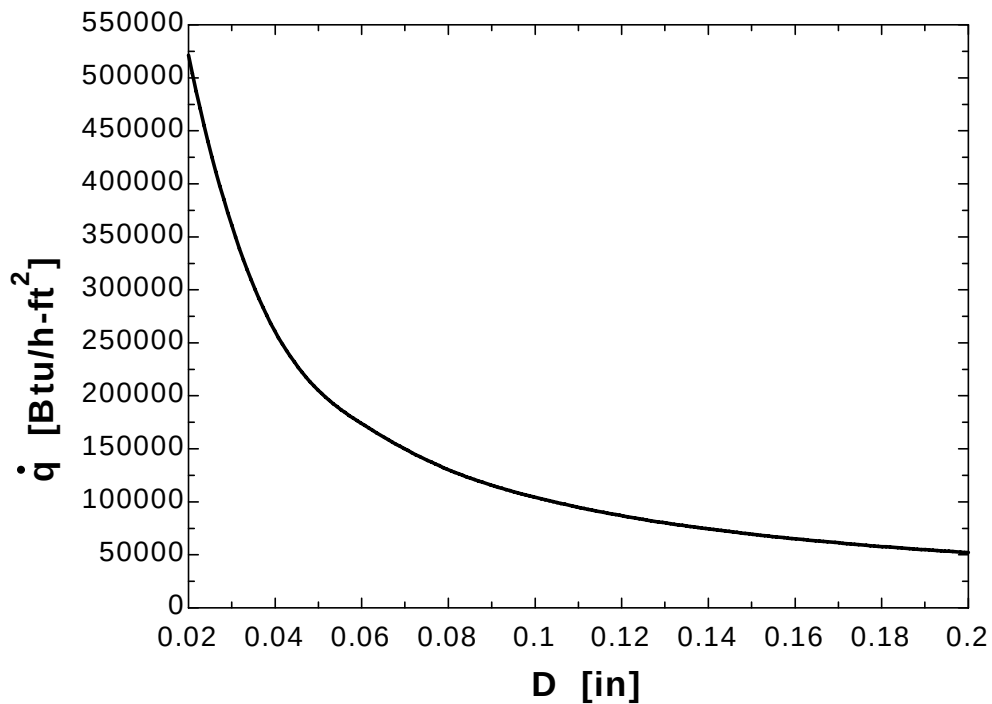
$$\dot{q} = \frac{\dot{G}}{A_{\text{wire}}} = \frac{\dot{G}}{\pi DL} = \frac{1000 \text{ W}}{\pi(0.08/12 \text{ ft})(15/12 \text{ ft})} \left(\frac{3.412 \text{ Btu/h}}{1 \text{ W}} \right) = 1.303 \times 10^5 \text{ Btu/h} \cdot \text{ft}^2$$

Discussion Note that heat generation is expressed per unit volume in Btu/h·ft³ whereas heat flux is expressed per unit surface area in Btu/h·ft².

2-15E

"GIVEN" $\dot{E} = 1000$ "[W]" $L = 15$ "[in]" $D = 0.08$ [in], parameter to be varied**"ANALYSIS"** $\dot{q} = \dot{E} / V_{\text{wire}} \cdot \text{Convert}(W, \text{Btu/h})$ $V_{\text{wire}} = \pi \cdot D^2 / 4 \cdot L \cdot \text{Convert}(\text{in}^3, \text{ft}^3)$ $\dot{q} = \dot{E} / A_{\text{wire}} \cdot \text{Convert}(W, \text{Btu/h})$ $A_{\text{wire}} = \pi \cdot D \cdot L \cdot \text{Convert}(\text{in}^2, \text{ft}^2)$

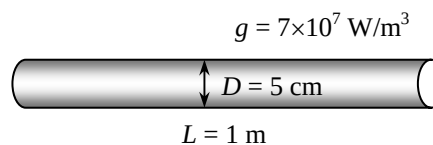
D [in]	$\dot{q}$ [Btu/h.ft ²]
0.02	521370
0.04	260685
0.06	173790
0.08	130342
0.1	104274
0.12	86895
0.14	74481
0.16	65171
0.18	57930
0.2	52137



2-16 The rate of heat generation per unit volume in the uranium rods is given. The total rate of heat generation in each rod is to be determined.

Assumptions Heat is generated uniformly in the uranium rods.

Analysis The total rate of heat generation in the rod is determined by multiplying the rate of heat generation per unit volume by the volume of the rod



$$\dot{G} = \dot{g}V_{\text{rod}} = \dot{g}(\pi D^2 / 4)L = (7 \times 10^7 \text{ W/m}^3)[\pi(0.05 \text{ m})^2 / 4](1 \text{ m}) = 1.374 \times 10^5 \text{ W} = \mathbf{137.4 \text{ kW}}$$

2-17 The variation of the absorption of solar energy in a solar pond with depth is given. A relation for the total rate of heat generation in a water layer at the top of the pond is to be determined.

Assumptions Absorption of solar radiation by water is modeled as heat generation.

Analysis The total rate of heat generation in a water layer of surface area A and thickness L at the top of the pond is determined by integration to be

$$\dot{G} = \int_V \dot{g}dV = \int_{x=0}^L \dot{g}_0 e^{-bx} (Adx) = A\dot{g}_0 \left. \frac{e^{-bx}}{-b} \right|_0^L = \frac{A\dot{g}_0(1 - e^{-bL})}{b}$$

2-18 The rate of heat generation per unit volume in a stainless steel plate is given. The heat flux on the surface of the plate is to be determined.

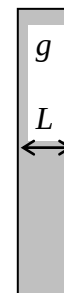
Assumptions Heat is generated uniformly in steel plate.

Analysis We consider a unit surface area of 1 m^2 . The total rate of heat generation in this section of the plate is

$$\dot{G} = \dot{g}V_{\text{plate}} = \dot{g}(A \times L) = (5 \times 10^6 \text{ W/m}^3)(1 \text{ m}^2)(0.03 \text{ m}) = 1.5 \times 10^5 \text{ W}$$

Noting that this heat will be dissipated from both sides of the plate, the heat flux on either surface of the plate becomes

$$\dot{q} = \frac{\dot{G}}{A_{\text{plate}}} = \frac{1.5 \times 10^5 \text{ W}}{2 \times 1 \text{ m}^2} = \mathbf{75,000 \text{ W/m}^2}$$



Heat Conduction Equation

2-19 The one-dimensional transient heat conduction equation for a plane wall with constant thermal conductivity and heat generation is $\frac{\partial^2 T}{\partial x^2} + \frac{\dot{g}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$. Here T is the temperature, x is the space variable, g is the heat generation per unit volume, k is the thermal conductivity, α is the thermal diffusivity, and t is the time.

2-20 The one-dimensional transient heat conduction equation for a plane wall with constant thermal conductivity and heat generation is $\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{\dot{g}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$. Here T is the temperature, r is the space

variable, g is the heat generation per unit volume, k is the thermal conductivity, α is the thermal diffusivity, and t is the time.

2-21 We consider a thin element of thickness Δx in a large plane wall (see Fig. 2-13 in the text). The density of the wall is ρ , the specific heat is C , and the area of the wall normal to the direction of heat transfer is A . In the absence of any heat generation, an *energy balance* on this thin element of thickness Δx during a small time interval Δt can be expressed as

$$\dot{Q}_x - \dot{Q}_{x+\Delta x} = \frac{\Delta E_{\text{element}}}{\Delta t}$$

where

$$\Delta E_{\text{element}} = E_{t+\Delta t} - E_t = mC(T_{t+\Delta t} - T_t) = \rho CA\Delta x(T_{t+\Delta t} - T_t)$$

Substituting,

$$\dot{Q}_x - \dot{Q}_{x+\Delta x} = \rho CA\Delta x \frac{T_{t+\Delta t} - T_t}{\Delta t}$$

Dividing by $A\Delta x$ gives

$$-\frac{1}{A} \frac{\dot{Q}_{x+\Delta x} - \dot{Q}_x}{\Delta x} = \rho C \frac{T_{t+\Delta t} - T_t}{\Delta t}$$

Taking the limit as $\Delta x \rightarrow 0$ and $\Delta t \rightarrow 0$ yields

$$\frac{1}{A} \frac{\partial}{\partial x} \left(kA \frac{\partial T}{\partial x} \right) = \rho C \frac{\partial T}{\partial t}$$

since, from the definition of the derivative and Fourier's law of heat conduction,

$$\lim_{\Delta x \rightarrow 0} \frac{\dot{Q}_{x+\Delta x} - \dot{Q}_x}{\Delta x} = \frac{\partial \dot{Q}}{\partial x} = \frac{\partial}{\partial x} \left(-kA \frac{\partial T}{\partial x} \right)$$

Noting that the area A of a plane wall is constant, the one-dimensional transient heat conduction equation in a plane wall with constant thermal conductivity k becomes

$$\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

where the property $\alpha = k / \rho C$ is the thermal diffusivity of the material.

2-22 We consider a thin cylindrical shell element of thickness Δr in a long cylinder (see Fig. 2-15 in the text). The density of the cylinder is ρ , the specific heat is C , and the length is L . The area of the cylinder normal to the direction of heat transfer at any location is $A = 2\pi rL$ where r is the value of the radius at that location. Note that the heat transfer area A depends on r in this case, and thus it varies with location. An *energy balance* on this thin cylindrical shell element of thickness Δr during a small time interval Δt can be expressed as

$$\dot{Q}_r - \dot{Q}_{r+\Delta r} + \dot{G}_{\text{element}} = \frac{\Delta E_{\text{element}}}{\Delta t}$$

where

$$\Delta E_{\text{element}} = E_{t+\Delta t} - E_t = mC(T_{t+\Delta t} - T_t) = \rho CA\Delta r(T_{t+\Delta t} - T_t)$$

$$\dot{G}_{\text{element}} = \dot{g}V_{\text{element}} = \dot{g}A\Delta r$$

Substituting,

$$\dot{Q}_r - \dot{Q}_{r+\Delta r} + \dot{g}A\Delta r = \rho CA\Delta r \frac{T_{t+\Delta t} - T_t}{\Delta t}$$

where $A = 2\pi rL$. Dividing the equation above by $A\Delta r$ gives

$$-\frac{1}{A} \frac{\dot{Q}_{r+\Delta r} - \dot{Q}_r}{\Delta r} + \dot{g} = \rho C \frac{T_{t+\Delta t} - T_t}{\Delta t}$$

Taking the limit as $\Delta r \rightarrow 0$ and $\Delta t \rightarrow 0$ yields

$$\frac{1}{A} \frac{\partial}{\partial r} \left(kA \frac{\partial T}{\partial r} \right) + \dot{g} = \rho C \frac{\partial T}{\partial t}$$

since, from the definition of the derivative and Fourier's law of heat conduction,

$$\lim_{\Delta r \rightarrow 0} \frac{\dot{Q}_{r+\Delta r} - \dot{Q}_r}{\Delta r} = \frac{\partial \dot{Q}}{\partial r} = \frac{\partial}{\partial r} \left(-kA \frac{\partial T}{\partial r} \right)$$

Noting that the heat transfer area in this case is $A = 2\pi rL$ and the thermal conductivity is constant, the one-dimensional transient heat conduction equation in a cylinder becomes

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \dot{g} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

where $\alpha = k / \rho C$ is the thermal diffusivity of the material.

2-23 We consider a thin spherical shell element of thickness Δr in a sphere (see Fig. 2-17 in the text).. The density of the sphere is ρ , the specific heat is C , and the length is L . The area of the sphere normal to the direction of heat transfer at any location is $A = 4\pi r^2$ where r is the value of the radius at that location. Note that the heat transfer area A depends on r in this case, and thus it varies with location. When there is no heat generation, an *energy balance* on this thin spherical shell element of thickness Δr during a small time interval Δt can be expressed as

$$\dot{Q}_r - \dot{Q}_{r+\Delta r} = \frac{\Delta E_{\text{element}}}{\Delta t}$$

where

$$\Delta E_{\text{element}} = E_{t+\Delta t} - E_t = mC(T_{t+\Delta t} - T_t) = \rho CA\Delta r(T_{t+\Delta t} - T_t)$$

Substituting,

$$\dot{Q}_r - \dot{Q}_{r+\Delta r} + \dot{q}A\Delta r = \rho CA\Delta r \frac{T_{t+\Delta t} - T_t}{\Delta t}$$

where $A = 4\pi r^2$. Dividing the equation above by $A\Delta r$ gives

$$-\frac{1}{A} \frac{\dot{Q}_{r+\Delta r} - \dot{Q}_r}{\Delta r} = \rho C \frac{T_{t+\Delta t} - T_t}{\Delta t}$$

Taking the limit as $\Delta r \rightarrow 0$ and $\Delta t \rightarrow 0$ yields

$$\frac{1}{A} \frac{\partial}{\partial r} \left(kA \frac{\partial T}{\partial r} \right) = \rho C \frac{\partial T}{\partial t}$$

since, from the definition of the derivative and Fourier's law of heat conduction,

$$\lim_{\Delta r \rightarrow 0} \frac{\dot{Q}_{r+\Delta r} - \dot{Q}_r}{\Delta r} = \frac{\partial \dot{Q}}{\partial r} = \frac{\partial}{\partial r} \left(-kA \frac{\partial T}{\partial r} \right)$$

Noting that the heat transfer area in this case is $A = 4\pi r^2$ and the thermal conductivity k is constant, the one-dimensional transient heat conduction equation in a sphere becomes

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

where $\alpha = k / \rho C$ is the thermal diffusivity of the material.

2-24 For a medium in which the heat conduction equation is given in its simplest by $\frac{\partial^2 T}{\partial x^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$:

(a) Heat transfer is transient, (b) it is one-dimensional, (c) there is no heat generation, and (d) the thermal conductivity is constant.

2-25 For a medium in which the heat conduction equation is given in its simplest by

$$\frac{1}{r} \frac{d}{dr} \left(rk \frac{dT}{dr} \right) + \dot{q} = 0 :$$

(a) Heat transfer is steady, (b) it is one-dimensional, (c) there is heat generation, and (d) the thermal conductivity is variable.

2-26 For a medium in which the heat conduction equation is given by $\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) = \frac{1}{\alpha} \frac{\partial T}{\partial t}$

(a) Heat transfer is transient, (b) it is one-dimensional, (c) there is no heat generation, and (d) the thermal conductivity is constant.

2-27 For a medium in which the heat conduction equation is given in its simplest by $r \frac{d^2 T}{dr^2} + \frac{dT}{dr} = 0$:

(a) Heat transfer is steady, (b) it is one-dimensional, (c) there is heat generation, and (d) the thermal conductivity is constant.

2-28 We consider a small rectangular element of length Δx , width Δy , and height $\Delta z = 1$ (similar to the one in Fig. 2-21). The density of the body is ρ and the specific heat is C . Noting that heat conduction is two-dimensional and assuming no heat generation, an *energy balance* on this element during a small time interval Δt can be expressed as

$$\left(\begin{array}{c} \text{Rate of heat} \\ \text{conduction at the} \\ \text{surfaces at } x \text{ and } y \end{array} \right) - \left(\begin{array}{c} \text{Rate of heat conduction} \\ \text{at the surfaces at} \\ x + \Delta x \text{ and } y + \Delta y \end{array} \right) = \left(\begin{array}{c} \text{Rate of change of} \\ \text{the energy content} \\ \text{of the element} \end{array} \right)$$

or
$$\dot{Q}_x + \dot{Q}_y - \dot{Q}_{x+\Delta x} - \dot{Q}_{y+\Delta y} = \frac{\Delta E_{\text{element}}}{\Delta t}$$

Noting that the volume of the element is $V_{\text{element}} = \Delta x \Delta y \Delta z = \Delta x \Delta y \times 1$, the change in the energy content of the element can be expressed as

$$\Delta E_{\text{element}} = E_{t+\Delta t} - E_t = mC(T_{t+\Delta t} - T_t) = \rho C \Delta x \Delta y (T_{t+\Delta t} - T_t)$$

Substituting,
$$\dot{Q}_x + \dot{Q}_y - \dot{Q}_{x+\Delta x} - \dot{Q}_{y+\Delta y} = \rho C \Delta x \Delta y \frac{T_{t+\Delta t} - T_t}{\Delta t}$$

Dividing by $\Delta x \Delta y$ gives

$$-\frac{1}{\Delta y} \frac{\dot{Q}_{x+\Delta x} - \dot{Q}_x}{\Delta x} - \frac{1}{\Delta x} \frac{\dot{Q}_{y+\Delta y} - \dot{Q}_y}{\Delta y} = \rho C \frac{T_{t+\Delta t} - T_t}{\Delta t}$$

Taking the thermal conductivity k to be constant and noting that the heat transfer surface areas of the element for heat conduction in the x and y directions are $A_x = \Delta y \times 1$ and $A_y = \Delta x \times 1$, respectively, and taking the limit as Δx , Δy , and $\Delta t \rightarrow 0$ yields

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

since, from the definition of the derivative and Fourier's law of heat conduction,

$$\begin{aligned} \lim_{\Delta x \rightarrow 0} \frac{1}{\Delta y \Delta z} \frac{\dot{Q}_{x+\Delta x} - \dot{Q}_x}{\Delta x} &= \frac{1}{\Delta y \Delta z} \frac{\partial \dot{Q}_x}{\partial x} = \frac{1}{\Delta y \Delta z} \frac{\partial}{\partial x} \left(-k \Delta y \Delta z \frac{\partial T}{\partial x} \right) = -\frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) = -k \frac{\partial^2 T}{\partial x^2} \\ \lim_{\Delta y \rightarrow 0} \frac{1}{\Delta x \Delta z} \frac{\dot{Q}_{y+\Delta y} - \dot{Q}_y}{\Delta y} &= \frac{1}{\Delta x \Delta z} \frac{\partial \dot{Q}_y}{\partial y} = \frac{1}{\Delta x \Delta z} \frac{\partial}{\partial y} \left(-k \Delta x \Delta z \frac{\partial T}{\partial y} \right) = -\frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) = -k \frac{\partial^2 T}{\partial y^2} \end{aligned}$$

Here the property $\alpha = k / \rho C$ is the thermal diffusivity of the material.

2-29 We consider a thin ring shaped volume element of width Δz and thickness Δr in a cylinder. The density of the cylinder is ρ and the specific heat is C . In general, an *energy balance* on this ring element during a small time interval Δt can be expressed as

$$(\dot{Q}_r - \dot{Q}_{r+\Delta r}) + (\dot{Q}_z - \dot{Q}_{z+\Delta z}) = \frac{\Delta E_{\text{element}}}{\Delta t}$$

But the change in the energy content of the element can be expressed as

$$\Delta E_{\text{element}} = E_{t+\Delta t} - E_t = mC(T_{t+\Delta t} - T_t) = \rho C(2\pi r \Delta r) \Delta z (T_{t+\Delta t} - T_t)$$

Substituting,

$$(\dot{Q}_r - \dot{Q}_{r+\Delta r}) + (\dot{Q}_z - \dot{Q}_{z+\Delta z}) = \rho C(2\pi r \Delta r) \Delta z \frac{T_{t+\Delta t} - T_t}{\Delta t}$$

Dividing the equation above by $(2\pi r \Delta r) \Delta z$ gives

$$-\frac{1}{2\pi r \Delta z} \frac{\dot{Q}_{r+\Delta r} - \dot{Q}_r}{\Delta r} - \frac{1}{2\pi r \Delta r} \frac{\dot{Q}_{z+\Delta z} - \dot{Q}_z}{\Delta z} = \rho C \frac{T_{t+\Delta t} - T_t}{\Delta t}$$

Noting that the heat transfer surface areas of the element for heat conduction in the r and z directions are $A_r = 2\pi r \Delta z$ and $A_z = 2\pi r \Delta r$, respectively, and taking the limit as Δr , Δz and $\Delta t \rightarrow 0$ yields

$$\frac{1}{r} \frac{\partial}{\partial r} \left(kr \frac{\partial T}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \phi} \left(k \frac{\partial T}{\partial \phi} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) = \rho C \frac{\partial T}{\partial t}$$

since, from the definition of the derivative and Fourier's law of heat conduction,

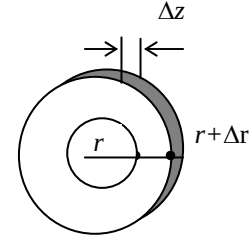
$$\begin{aligned} \lim_{\Delta r \rightarrow 0} \frac{1}{2\pi r \Delta z} \frac{\dot{Q}_{r+\Delta r} - \dot{Q}_r}{\Delta r} &= \frac{1}{2\pi r \Delta z} \frac{\partial Q}{\partial r} = \frac{1}{2\pi r \Delta z} \frac{\partial}{\partial r} \left(-k(2\pi r \Delta z) \frac{\partial T}{\partial r} \right) = -\frac{1}{r} \frac{\partial}{\partial r} \left(kr \frac{\partial T}{\partial r} \right) \\ \lim_{\Delta z \rightarrow 0} \frac{1}{2\pi r \Delta r} \frac{\dot{Q}_{z+\Delta z} - \dot{Q}_z}{\Delta z} &= \frac{1}{2\pi r \Delta r} \frac{\partial Q_z}{\partial z} = \frac{1}{2\pi r \Delta r} \frac{\partial}{\partial z} \left(-k(2\pi r \Delta r) \frac{\partial T}{\partial z} \right) = -\frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) \end{aligned}$$

For the case of constant thermal conductivity the equation above reduces to

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{\partial^2 T}{\partial z^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

where $\alpha = k / \rho C$ is the thermal diffusivity of the material. For the case of steady heat conduction with no heat generation it reduces to

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{\partial^2 T}{\partial z^2} = 0$$



2-30 Consider a thin disk element of thickness Δz and diameter D in a long cylinder (Fig. P2-30). The density of the cylinder is ρ , the specific heat is C , and the area of the cylinder normal to the direction of heat transfer is $A = \pi D^2 / 4$, which is constant. An *energy balance* on this thin element of thickness Δz during a small time interval Δt can be expressed as

$$\left(\begin{array}{c} \text{Rate of heat} \\ \text{conduction at} \\ \text{the surface at } z \end{array} \right) - \left(\begin{array}{c} \text{Rate of heat} \\ \text{conduction at the} \\ \text{surface at } z + \Delta z \end{array} \right) + \left(\begin{array}{c} \text{Rate of heat} \\ \text{generation inside} \\ \text{the element} \end{array} \right) = \left(\begin{array}{c} \text{Rate of change of} \\ \text{the energy content} \\ \text{of the element} \end{array} \right)$$

or,

$$\dot{Q}_z - \dot{Q}_{z+\Delta z} + \dot{G}_{\text{element}} = \frac{\Delta E_{\text{element}}}{\Delta t}$$

But the change in the energy content of the element and the rate of heat generation within the element can be expressed as

$$\Delta E_{\text{element}} = E_{t+\Delta t} - E_t = mC(T_{t+\Delta t} - T_t) = \rho CA\Delta z(T_{t+\Delta t} - T_t)$$

and

$$\dot{G}_{\text{element}} = \dot{g}V_{\text{element}} = \dot{g}A\Delta z$$

Substituting,

$$\dot{Q}_z - \dot{Q}_{z+\Delta z} + \dot{g}A\Delta z = \rho CA\Delta z \frac{T_{t+\Delta t} - T_t}{\Delta t}$$

Dividing by $A\Delta z$ gives

$$-\frac{1}{A} \frac{\dot{Q}_{z+\Delta z} - \dot{Q}_z}{\Delta z} + \dot{g} = \rho C \frac{T_{t+\Delta t} - T_t}{\Delta t}$$

Taking the limit as $\Delta z \rightarrow 0$ and $\Delta t \rightarrow 0$ yields

$$\frac{1}{A} \frac{\partial}{\partial z} \left(kA \frac{\partial T}{\partial z} \right) + \dot{g} = \rho C \frac{\partial T}{\partial t}$$

since, from the definition of the derivative and Fourier's law of heat conduction,

$$\lim_{\Delta z \rightarrow 0} \frac{\dot{Q}_{z+\Delta z} - \dot{Q}_z}{\Delta z} = \frac{\partial \dot{Q}}{\partial z} = \frac{\partial}{\partial z} \left(-kA \frac{\partial T}{\partial z} \right)$$

Noting that the area A and the thermal conductivity k are constant, the one-dimensional transient heat conduction equation in the axial direction in a long cylinder becomes

$$\frac{\partial^2 T}{\partial z^2} + \frac{\dot{g}}{k} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

where the property $\alpha = k / \rho C$ is the thermal diffusivity of the material.

2-31 For a medium in which the heat conduction equation is given by $\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$:

(a) Heat transfer is transient, (b) it is two-dimensional, (c) there is no heat generation, and (d) the thermal conductivity is constant.

2-32 For a medium in which the heat conduction equation is given by $\frac{1}{r} \frac{\partial}{\partial r} \left(kr \frac{\partial T}{\partial r} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) + \dot{g} = 0$:

(a) Heat transfer is steady, (b) it is two-dimensional, (c) there is heat generation, and (d) the thermal conductivity is variable.

2-33 For a medium in which the heat conduction equation is given by

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 T}{\partial \phi^2} = \frac{1}{\alpha} \frac{\partial T}{\partial t} :$$

(a) Heat transfer is transient, (b) it is two-dimensional, (c) there is no heat generation, and (d) the thermal conductivity is constant.

Boundary and Initial Conditions; Formulation of Heat Conduction Problems

2-34C The mathematical expressions of the thermal conditions at the boundaries are called the **boundary conditions**. To describe a heat transfer problem completely, *two boundary conditions* must be given for *each direction* of the coordinate system along which heat transfer is significant. Therefore, we need to specify four boundary conditions for two-dimensional problems.

2-35C The mathematical expression for the temperature distribution of the medium initially is called the **initial condition**. We need only one initial condition for a heat conduction problem regardless of the dimension since the conduction equation is first order in time (it involves the first derivative of temperature with respect to time). Therefore, we need only 1 initial condition for a two-dimensional problem.

2-36C A heat transfer problem that is symmetric about a plane, line, or point is said to have thermal symmetry about that plane, line, or point. The thermal symmetry boundary condition is a mathematical expression of this thermal symmetry. It is equivalent to *insulation* or *zero heat flux* boundary condition, and is expressed at a point x_0 as $\partial T(x_0, t) / \partial x = 0$.

2-37C The boundary condition at a perfectly insulated surface (at $x = 0$, for example) can be expressed as

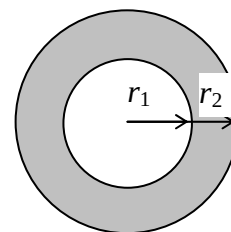
$$-k \frac{\partial T(0, t)}{\partial x} = 0 \quad \text{or} \quad \frac{\partial T(0, t)}{\partial x} = 0$$

which indicates zero heat flux.

2-38C Yes, the temperature profile in a medium must be perpendicular to an insulated surface since the slope $\partial T / \partial x = 0$ at that surface.

2-39C We try to avoid the radiation boundary condition in heat transfer analysis because it is a non-linear expression that causes mathematical difficulties while solving the problem; often making it impossible to obtain analytical solutions.

2-40 A spherical container of inner radius r_1 , outer radius r_2 , and thermal conductivity k is given. The boundary condition on the inner surface of the container for steady one-dimensional conduction is to be expressed for the following cases:



(a) Specified temperature of 50°C: $T(r_1) = 50^\circ\text{C}$

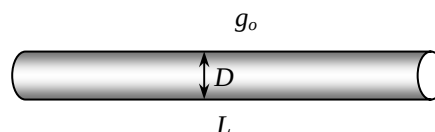
(b) Specified heat flux of 30 W/m² towards the center: $k \frac{dT(r_1)}{dr} = 30 \text{ W/m}^2$

(c) Convection to a medium at T_∞ with a heat transfer coefficient of h : $k \frac{dT(r_1)}{dr} = h[T(r_1) - T_\infty]$

2-41 Heat is generated in a long wire of radius r_0 covered with a plastic insulation layer at a constant rate of $\dot{g}_0$. The heat flux boundary condition at the interface (radius r_0) in terms of the heat generated is to be expressed. The total heat generated in the wire and the heat flux at the interface are

$$\dot{G} = \dot{g}_0 V_{\text{wire}} = \dot{g}_0 (\pi r_0^2 L)$$

$$\dot{q}_s = \frac{\dot{Q}_s}{A} = \frac{\dot{G}}{A} = \frac{\dot{g}_0 (\pi r_0^2 L)}{(2\pi r_0) L} = \frac{\dot{g}_0 r_0}{2}$$

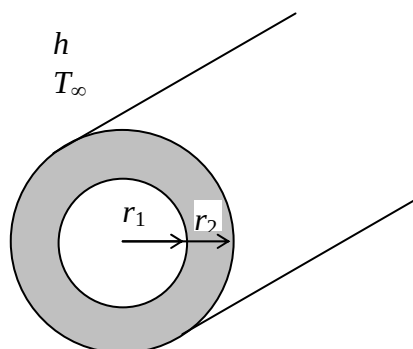


Assuming steady one-dimensional conduction in the radial direction, the heat flux boundary condition can be expressed as

$$-k \frac{dT(r_0)}{dr} = \frac{\dot{g}_0 r_0}{2}$$

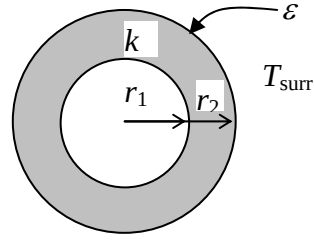
2-42 A long pipe of inner radius r_1 , outer radius r_2 , and thermal conductivity k is considered. The outer surface of the pipe is subjected to convection to a medium at T_∞ with a heat transfer coefficient of h . Assuming steady one-dimensional conduction in the radial direction, the convection boundary condition on the outer surface of the pipe can be expressed as

$$-k \frac{dT(r_2)}{dr} = h[T(r_2) - T_\infty]$$



2-43 A spherical shell of inner radius r_1 , outer radius r_2 , and thermal conductivity k is considered. The outer surface of the shell is subjected to radiation to surrounding surfaces at T_{surr} . Assuming no convection and steady one-dimensional conduction in the radial direction, the radiation boundary condition on the outer surface of the shell can be expressed as

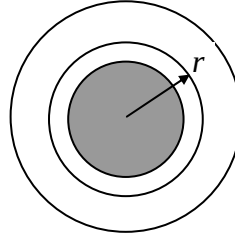
$$-k \frac{dT(r_2)}{dr} = \varepsilon \sigma [T(r_2)^4 - T_{\text{surr}}^4]$$



2-44 A spherical container consists of two spherical layers A and B that are at perfect contact. The radius of the interface is r_0 . Assuming transient one-dimensional conduction in the radial direction, the boundary conditions at the interface can be expressed as

$$T_A(r_0, t) = T_B(r_0, t)$$

and
$$-k_A \frac{\partial T_A(r_0, t)}{\partial x} = -k_B \frac{\partial T_B(r_0, t)}{\partial x}$$



2-45 Heat conduction through the bottom section of a steel pan that is used to boil water on top of an electric range is considered (Fig. P2-45). Assuming constant thermal conductivity and one-dimensional heat transfer, the mathematical formulation (the differential equation and the boundary conditions) of this heat conduction problem is to be obtained for steady operation.

Assumptions 1 Heat transfer is given to be steady and one-dimensional. 2 Thermal conductivity is given to be constant. 3 There is no heat generation in the medium. 4 The top surface at $x = L$ is subjected to convection and the bottom surface at $x = 0$ is subjected to uniform heat flux.

Analysis The heat flux at the bottom of the pan is

$$\dot{q}_s = \frac{\dot{Q}_s}{A_s} = \frac{\dot{G}}{\pi D^2 / 4} = \frac{0.85 \times (1000 \text{ W})}{\pi (0.20 \text{ m})^2 / 4} = 27,056 \text{ W / m}^2$$

Then the differential equation and the boundary conditions for this heat conduction problem can be expressed as

$$\begin{aligned} \frac{d^2 T}{dx^2} &= 0 \\ -k \frac{dT(0)}{dx} &= \dot{q}_s = 27,056 \text{ W / m}^2 \\ -k \frac{dT(L)}{dx} &= h[T(L) - T_\infty] \end{aligned}$$

2-46E A 1.5-kW resistance heater wire is used for space heating. Assuming constant thermal conductivity and one-dimensional heat transfer, the mathematical formulation (the differential equation and the boundary conditions) of this heat conduction problem is to be obtained for steady operation.

Assumptions 1 Heat transfer is given to be steady and one-dimensional. 2 Thermal conductivity is given to be constant. 3 Heat is generated uniformly in the wire.

Analysis The heat flux at the surface of the wire is

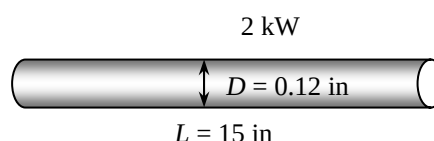
$$\dot{q}_s = \frac{\dot{Q}_s}{A_s} = \frac{\dot{G}}{2\pi r_0 L} = \frac{1200 \text{ W}}{2\pi(0.06 \text{ in})(15 \text{ in})} = 212.2 \text{ W/in}^2$$

Noting that there is thermal symmetry about the center line and there is uniform heat flux at the outer surface, the differential equation and the boundary conditions for this heat conduction problem can be expressed as

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dT}{dr} \right) + \frac{\dot{g}}{k} = 0$$

$$\frac{dT(0)}{dr} = 0$$

$$-k \frac{dT(r_0)}{dr} = \dot{q}_s = 212.2 \text{ W/in}^2$$



2-47 Heat conduction through the bottom section of an aluminum pan that is used to cook stew on top of an electric range is considered (Fig. P2-47). Assuming variable thermal conductivity and one-dimensional heat transfer, the mathematical formulation (the differential equation and the boundary conditions) of this heat conduction problem is to be obtained for steady operation.

Assumptions 1 Heat transfer is given to be steady and one-dimensional. 2 Thermal conductivity is given to be variable. 3 There is no heat generation in the medium. 4 The top surface at $x = L$ is subjected to specified temperature and the bottom surface at $x = 0$ is subjected to uniform heat flux.

Analysis The heat flux at the bottom of the pan is

$$\dot{q}_s = \frac{\dot{Q}_s}{A_s} = \frac{\dot{G}}{\pi D^2 / 4} = \frac{0.90 \times (900 \text{ W})}{\pi(0.18 \text{ m})^2 / 4} = 31,831 \text{ W/m}^2$$

Then the differential equation and the boundary conditions

for this heat conduction problem can be expressed as

$$\frac{d}{dx} \left(k \frac{dT}{dx} \right) = 0$$

$$-k \frac{dT(0)}{dx} = \dot{q}_s = 31,831 \text{ W/m}^2$$

$$T(L) = T_L = 108^\circ \text{C}$$

2-48 Water flows through a pipe whose outer surface is wrapped with a thin electric heater that consumes 300 W per m length of the pipe. The exposed surface of the heater is heavily insulated so that the entire heat generated in the heater is transferred to the pipe. Heat is transferred from the inner surface of the pipe to the water by convection. Assuming constant thermal conductivity and one-dimensional heat transfer, the mathematical formulation (the differential equation and the boundary conditions) of the heat conduction in the pipe is to be obtained for steady operation.

Assumptions 1 Heat transfer is given to be steady and one-dimensional. 2 Thermal conductivity is given to be constant. 3 There is no heat generation in the medium. 4 The outer surface at $r = r_2$ is subjected to uniform heat flux and the inner surface at $r = r_1$ is subjected to convection.

Analysis The heat flux at the outer surface of the pipe is

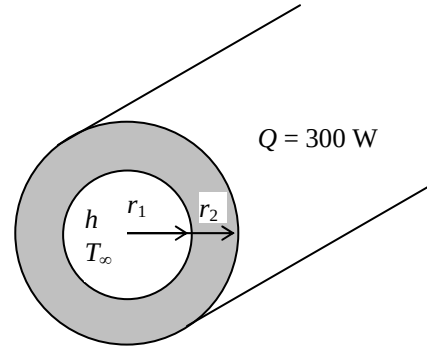
$$\dot{q}_s = \frac{\dot{Q}_s}{A_s} = \frac{\dot{Q}_s}{2\pi r_2 L} = \frac{300 \text{ W}}{2\pi(0.065 \text{ cm})(1 \text{ m})} = 734.6 \text{ W/m}^2$$

Noting that there is thermal symmetry about the center line and there is uniform heat flux at the outer surface, the differential equation and the boundary conditions for this heat conduction problem can be expressed as

$$\frac{d}{dr} \left(r \frac{dT}{dr} \right) = 0$$

$$k \frac{dT(r_1)}{dr} = h[T(r_1) - T_\infty]$$

$$k \frac{dT(r_2)}{dr} = \dot{q}_s = 734.6 \text{ W/m}^2$$



2-49 A spherical metal ball that is heated in an oven to a temperature of T_i throughout is dropped into a large body of water at T_∞ where it is cooled by convection. Assuming constant thermal conductivity and transient one-dimensional heat transfer, the mathematical formulation (the differential equation and the boundary and initial conditions) of this heat conduction problem is to be obtained.

Assumptions 1 Heat transfer is given to be transient and one-dimensional. 2 Thermal conductivity is given to be constant. 3 There is no heat generation in the medium. 4 The outer surface at $r = r_0$ is subjected to convection.

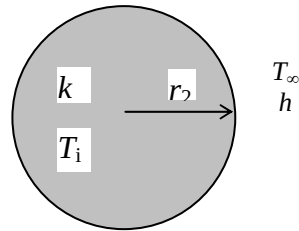
Analysis Noting that there is thermal symmetry about the midpoint and convection at the outer surface, the differential equation and the boundary conditions for this heat conduction problem can be expressed as

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

$$\frac{\partial T(0, t)}{\partial r} = 0$$

$$-k \frac{\partial T(r_0, t)}{\partial r} = h[T(r_0) - T_\infty]$$

$$T(r, 0) = T_i$$



2-50 A spherical metal ball that is heated in an oven to a temperature of T_i throughout is allowed to cool in ambient air at T_∞ by convection and radiation. Assuming constant thermal conductivity and transient one-dimensional heat transfer, the mathematical formulation (the differential equation and the boundary and initial conditions) of this heat conduction problem is to be obtained.

Assumptions 1 Heat transfer is given to be transient and one-dimensional. 2 Thermal conductivity is given to be variable. 3 There is no heat generation in the medium. 4 The outer surface at $r = r_0$ is subjected to convection and radiation.

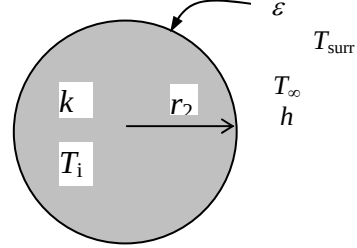
Analysis Noting that there is thermal symmetry about the midpoint and convection and radiation at the outer surface and expressing all temperatures in Rankine, the differential equation and the boundary conditions for this heat conduction problem can be expressed as

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(kr^2 \frac{\partial T}{\partial r} \right) = \rho C \frac{\partial T}{\partial t}$$

$$\frac{\partial T(0, t)}{\partial r} = 0$$

$$-k \frac{\partial T(r_0, t)}{\partial r} = h[T(r_0) - T_\infty] + \varepsilon \sigma [T(r_0)^4 - T_{\text{surr}}^4]$$

$$T(r, 0) = T_i$$



2-51 The outer surface of the North wall of a house exchanges heat with both convection and radiation., while the interior surface is subjected to convection only. Assuming the heat transfer through the wall to be steady and one-dimensional, the mathematical formulation (the differential equation and the boundary and initial conditions) of this heat conduction problem is to be obtained.

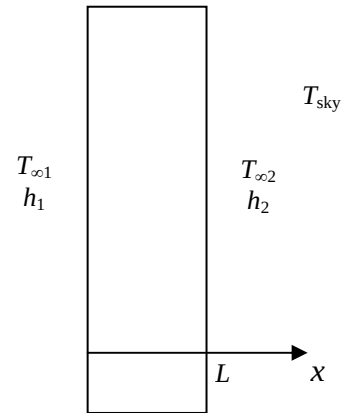
Assumptions 1 Heat transfer is given to be steady and one-dimensional. 2 Thermal conductivity is given to be constant. 3 There is no heat generation in the medium. 4 The outer surface at $x = L$ is subjected to convection and radiation while the inner surface at $x = 0$ is subjected to convection only.

Analysis Expressing all the temperatures in Kelvin, the differential equation and the boundary conditions for this heat conduction problem can be expressed as

$$\frac{d^2 T}{dx^2} = 0$$

$$-k \frac{dT(0)}{dx} = h_1 [T_{\infty 1} - T(0)]$$

$$-k \frac{dT(L)}{dx} = h_1 [T(L) - T_{\infty 2}] + \varepsilon_2 \sigma [T(L)^4 - T_{\text{sky}}^4]$$



Solution of Steady One-Dimensional Heat Conduction Problems

2-52C Yes, this claim is reasonable since in the absence of any heat generation the rate of heat transfer through a plain wall in steady operation must be constant. But the value of this constant must be zero since one side of the wall is perfectly insulated. Therefore, there can be no temperature difference between different parts of the wall; that is, the temperature in a plane wall must be uniform in steady operation.

2-53C Yes, the temperature in a plane wall with constant thermal conductivity and no heat generation will vary linearly during steady one-dimensional heat conduction even when the wall loses heat by radiation from its surfaces. This is because the steady heat conduction equation in a plane wall is $d^2T/dx^2 = 0$ whose solution is $T(x) = C_1x + C_2$ regardless of the boundary conditions. The solution function represents a straight line whose slope is C_1 .

2-54C Yes, in the case of constant thermal conductivity and no heat generation, the temperature in a solid cylindrical rod whose ends are maintained at constant but different temperatures while the side surface is perfectly insulated will vary linearly during steady one-dimensional heat conduction. This is because the steady heat conduction equation in this case is $d^2T/dx^2 = 0$ whose solution is $T(x) = C_1x + C_2$ which represents a straight line whose slope is C_1 .

2-55C Yes, this claim is reasonable since no heat is entering the cylinder and thus there can be no heat transfer from the cylinder in steady operation. This condition will be satisfied only when there are no temperature differences within the cylinder and the outer surface temperature of the cylinder is the equal to the temperature of the surrounding medium.

2-56 A large plane wall is subjected to specified temperature on the left surface and convection on the right surface. The mathematical formulation, the variation of temperature, and the rate of heat transfer are to be determined for steady one-dimensional heat transfer.

Assumptions 1 Heat conduction is steady and one-dimensional. 2 Thermal conductivity is constant. 3 There is no heat generation.

Properties The thermal conductivity is given to be $k = 2.3 \text{ W/m}\cdot^\circ\text{C}$.

Analysis (a) Taking the direction normal to the surface of the wall to be the x direction with $x = 0$ at the left surface, the mathematical formulation of this problem can be expressed as

$$\frac{d^2T}{dx^2} = 0$$

and

$$T(0) = T_1 = 80^\circ\text{C}$$

$$-k \frac{dT(L)}{dx} = h[T(L) - T_\infty]$$

(b) Integrating the differential equation twice with respect to x yields

$$\frac{dT}{dx} = C_1$$

$$T(x) = C_1x + C_2$$

where C_1 and C_2 are arbitrary constants. Applying the boundary conditions give

$$x = 0: \quad T(0) = C_1 \times 0 + C_2 \rightarrow C_2 = T_1$$

$$x = L: \quad -kC_1 = h[(C_1L + C_2) - T_\infty] \rightarrow C_1 = -\frac{h(C_2 - T_\infty)}{k + hL} \rightarrow C_1 = -\frac{h(T_1 - T_\infty)}{k + hL}$$

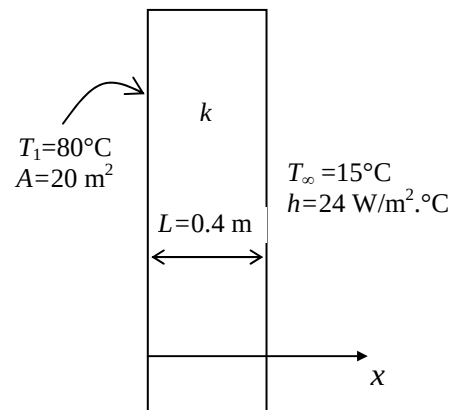
Substituting C_1 and C_2 into the general solution, the variation of temperature is determined to be

$$\begin{aligned} T(x) &= -\frac{h(T_1 - T_\infty)}{k + hL}x + T_1 \\ &= -\frac{(24 \text{ W/m}^2\cdot^\circ\text{C})(80 - 15)^\circ\text{C}}{(2.3 \text{ W/m}\cdot^\circ\text{C}) + (24 \text{ W/m}^2\cdot^\circ\text{C})(0.4 \text{ m})}x + 80^\circ\text{C} \\ &= 80 - 131.1x \end{aligned}$$

(c) The rate of heat conduction through the wall is

$$\begin{aligned} \dot{Q}_{\text{wall}} &= -kA \frac{dT}{dx} = -kAC_1 = kA \frac{h(T_1 - T_\infty)}{k + hL} \\ &= (2.3 \text{ W/m}\cdot^\circ\text{C})(20 \text{ m}^2) \frac{(24 \text{ W/m}^2\cdot^\circ\text{C})(80 - 15)^\circ\text{C}}{(2.3 \text{ W/m}\cdot^\circ\text{C}) + (24 \text{ W/m}^2\cdot^\circ\text{C})(0.4 \text{ m})} \\ &= \mathbf{6030 \text{ W}} \end{aligned}$$

Note that under steady conditions the rate of heat conduction through a plain wall is constant.



2-57 The top and bottom surfaces of a solid cylindrical rod are maintained at constant temperatures of 20°C and 95°C while the side surface is perfectly insulated. The rate of heat transfer through the rod is to be determined for the cases of copper, steel, and granite rod.

Assumptions 1 Heat conduction is steady and one-dimensional. 2 Thermal conductivity is constant. 3 There is no heat generation.

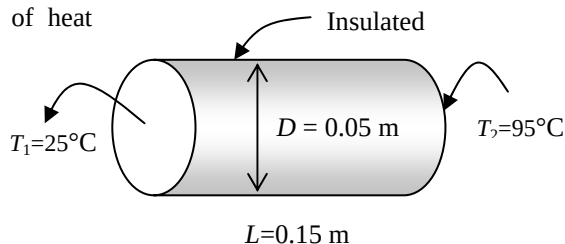
Properties The thermal conductivities are given to be $k = 380 \text{ W/m}\cdot^\circ\text{C}$ for copper, $k = 18 \text{ W/m}\cdot^\circ\text{C}$ for steel, and $k = 1.2 \text{ W/m}\cdot^\circ\text{C}$ for granite.

Analysis Noting that the heat transfer area (the area normal to the direction of heat transfer) is constant, the rate of heat transfer along the rod is determined from

$$\dot{Q} = kA \frac{T_1 - T_2}{L}$$

where $L = 0.15 \text{ m}$ and the heat transfer area A is

$$A = \pi D^2 / 4 = \pi (0.05 \text{ m})^2 / 4 = 1.964 \times 10^{-3} \text{ m}^2$$



Then the heat transfer rate for each case is determined as follows:

$$(a) \text{ Copper: } \dot{Q} = kA \frac{T_1 - T_2}{L} = (380 \text{ W/m}\cdot^\circ\text{C})(1.964 \times 10^{-3} \text{ m}^2) \frac{(95 - 20)^\circ\text{C}}{0.15 \text{ m}} = \mathbf{373.1 \text{ W}}$$

$$(b) \text{ Steel: } \dot{Q} = kA \frac{T_1 - T_2}{L} = (18 \text{ W/m}\cdot^\circ\text{C})(1.964 \times 10^{-3} \text{ m}^2) \frac{(95 - 20)^\circ\text{C}}{0.15 \text{ m}} = \mathbf{17.7 \text{ W}}$$

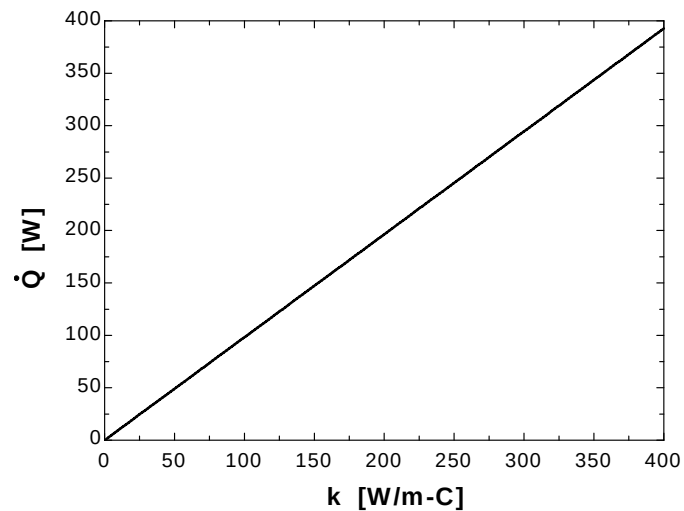
$$(c) \text{ Granite: } \dot{Q} = kA \frac{T_1 - T_2}{L} = (1.2 \text{ W/m}\cdot^\circ\text{C})(1.964 \times 10^{-3} \text{ m}^2) \frac{(95 - 20)^\circ\text{C}}{0.15 \text{ m}} = \mathbf{1.2 \text{ W}}$$

Discussion: The steady rate of heat conduction can differ by orders of magnitude, depending on the thermal conductivity of the material.

2-58

"GIVEN" $L=0.15$ "[m]" $D=0.05$ "[m]" $T_1=20$ "[C]" $T_2=95$ "[C]" $k=1.2$ [W/m-C], parameter to be varied**"ANALYSIS"** $A=\pi D^2/4$ $\dot{Q}=kA(T_2-T_1)/L$

k [W/m.C]	Q [W]
1	0.9817
22	21.6
43	42.22
64	62.83
85	83.45
106	104.1
127	124.7
148	145.3
169	165.9
190	186.5
211	207.1
232	227.8
253	248.4
274	269
295	289.6
316	310.2
337	330.8
358	351.5
379	372.1
400	392.7



2-59 The base plate of a household iron is subjected to specified heat flux on the left surface and to specified temperature on the right surface. The mathematical formulation, the variation of temperature in the plate, and the inner surface temperature are to be determined for steady one-dimensional heat transfer.

Assumptions 1 Heat conduction is steady and one-dimensional since the surface area of the base plate is large relative to its thickness, and the thermal conditions on both sides of the plate are uniform. 2 Thermal conductivity is constant. 3 There is no heat generation in the plate. 4 Heat loss through the upper part of the iron is negligible.

Properties The thermal conductivity is given to be $k = 20 \text{ W/m}\cdot^\circ\text{C}$.

Analysis (a) Noting that the upper part of the iron is well insulated and thus the entire heat generated in the resistance wires is transferred to the base plate, the heat flux through the inner surface is determined to be

$$\dot{q}_0 = \frac{\dot{Q}_0}{A_{\text{base}}} = \frac{800 \text{ W}}{160 \times 10^{-4} \text{ m}^2} = 50,000 \text{ W/m}^2$$

Taking the direction normal to the surface of the wall to be the x direction with $x = 0$ at the left surface, the mathematical formulation of this problem can be expressed as

$$\frac{d^2T}{dx^2} = 0$$

$$\text{and} \quad -k \frac{dT(0)}{dx} = \dot{q}_0 = 50,000 \text{ W/m}^2$$

$$T(L) = T_2 = 85^\circ\text{C}$$

(b) Integrating the differential equation twice with respect to x yields

$$\frac{dT}{dx} = C_1$$

$$T(x) = C_1x + C_2$$

where C_1 and C_2 are arbitrary constants. Applying the boundary conditions give

$$x = 0: \quad -kC_1 = \dot{q}_0 \rightarrow C_1 = -\frac{\dot{q}_0}{k}$$

$$x = L: \quad T(L) = C_1L + C_2 = T_2 \rightarrow C_2 = T_2 - C_1L \rightarrow C_2 = T_2 + \frac{\dot{q}_0L}{k}$$

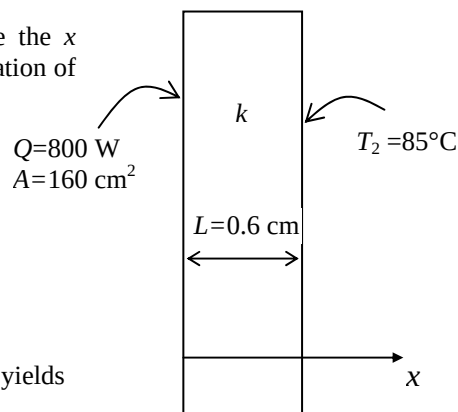
Substituting C_1 and C_2 into the general solution, the variation of temperature is determined to be

$$\begin{aligned} T(x) &= -\frac{\dot{q}_0}{k}x + T_2 + \frac{\dot{q}_0L}{k} = \frac{\dot{q}_0(L-x)}{k} + T_2 \\ &= \frac{(50,000 \text{ W/m}^2)(0.006 - x)\text{m}}{20 \text{ W/m}\cdot^\circ\text{C}} + 85^\circ\text{C} \\ &= 2500(0.006 - x) + 85 \end{aligned}$$

(c) The temperature at $x = 0$ (the inner surface of the plate) is

$$T(0) = 2500(0.006 - 0) + 85 = \mathbf{100^\circ\text{C}}$$

Note that the inner surface temperature is higher than the exposed surface temperature, as expected.



2-60 The base plate of a household iron is subjected to specified heat flux on the left surface and to specified temperature on the right surface. The mathematical formulation, the variation of temperature in the plate, and the inner surface temperature are to be determined for steady one-dimensional heat transfer.

Assumptions 1 Heat conduction is steady and one-dimensional since the surface area of the base plate is large relative to its thickness, and the thermal conditions on both sides of the plate are uniform. 2 Thermal conductivity is constant. 3 There is no heat generation in the plate. 4 Heat loss through the upper part of the iron is negligible.

Properties The thermal conductivity is given to be $k = 20 \text{ W/m}\cdot^\circ\text{C}$.

Analysis (a) Noting that the upper part of the iron is well insulated and thus the entire heat generated in the resistance wires is transferred to the base plate, the heat flux through the inner surface is determined to be

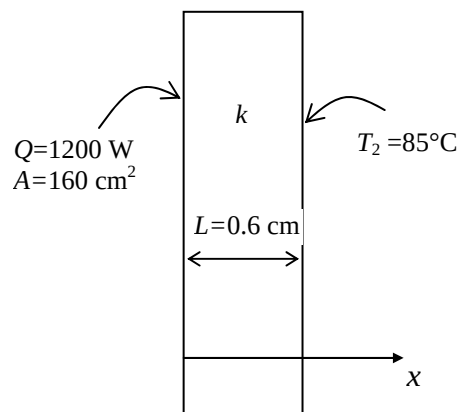
$$\dot{q}_0 = \frac{\dot{Q}_0}{A_{\text{base}}} = \frac{1200 \text{ W}}{160 \times 10^{-4} \text{ m}^2} = 75,000 \text{ W/m}^2$$

Taking the direction normal to the surface of the wall to be the x direction with $x = 0$ at the left surface, the mathematical formulation of this problem can be expressed as

$$\frac{d^2 T}{dx^2} = 0$$

$$\text{and} \quad -k \frac{dT(0)}{dx} = \dot{q}_0 = 75,000 \text{ W/m}^2$$

$$T(L) = T_2 = 85^\circ\text{C}$$



(b) Integrating the differential equation twice with respect to x yields

$$\frac{dT}{dx} = C_1$$

$$T(x) = C_1 x + C_2$$

where C_1 and C_2 are arbitrary constants. Applying the boundary conditions give

$$x = 0: \quad -kC_1 = \dot{q}_0 \rightarrow C_1 = -\frac{\dot{q}_0}{k}$$

$$x = L: \quad T(L) = C_1 L + C_2 = T_2 \rightarrow C_2 = T_2 - C_1 L \rightarrow C_2 = T_2 + \frac{\dot{q}_0 L}{k}$$

Substituting C_1 and C_2 into the general solution, the variation of temperature is determined to be

$$\begin{aligned} T(x) &= -\frac{\dot{q}_0}{k} x + T_2 + \frac{\dot{q}_0 L}{k} = \frac{\dot{q}_0 (L - x)}{k} + T_2 \\ &= \frac{(75,000 \text{ W/m}^2)(0.006 - x) \text{ m}}{20 \text{ W/m}\cdot^\circ\text{C}} + 85^\circ\text{C} \\ &= 3750(0.006 - x) + 85 \end{aligned}$$

(c) The temperature at $x = 0$ (the inner surface of the plate) is

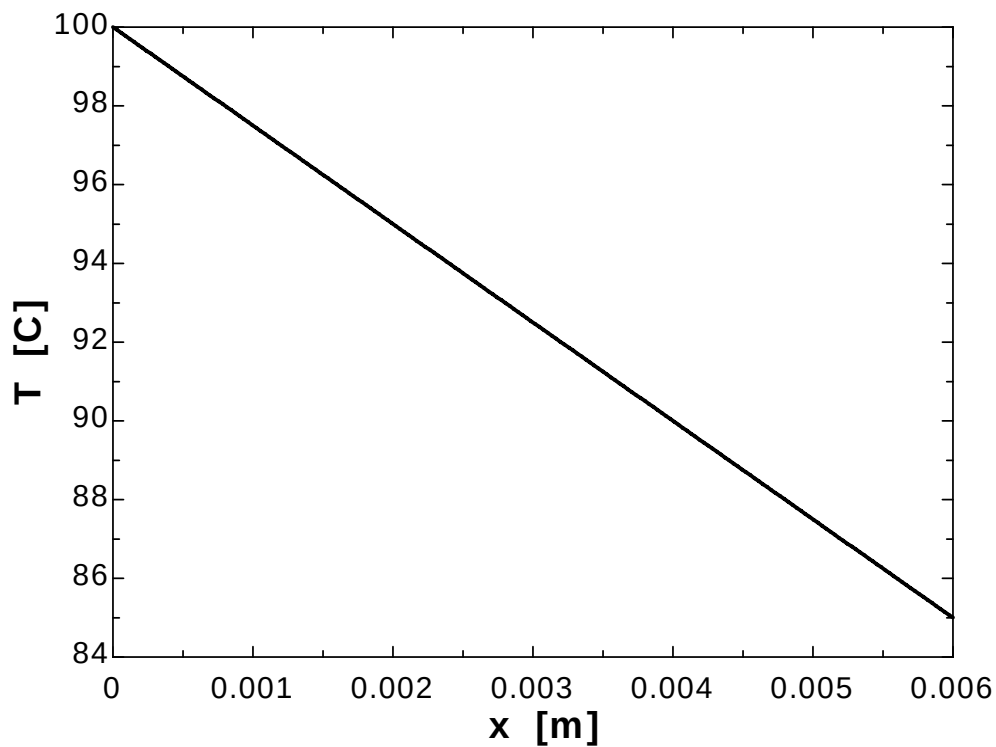
$$T(0) = 3750(0.006 - 0) + 85 = 107.5^\circ\text{C}$$

Note that the inner surface temperature is higher than the exposed surface temperature, as expected.

2-61

"GIVEN" $\dot{Q}=800 \text{ [W]}$ $L=0.006 \text{ [m]}$ $A_{\text{base}}=160\text{E-}4 \text{ [m}^2\text{]}$ $k=20 \text{ [W/m-C]}$ $T_2=85 \text{ [C]}$ **"ANALYSIS"** $q_{\text{dot}_0}=\dot{Q}/A_{\text{base}}$ $T=q_{\text{dot}_0}(L-x)/k+T_2$ "Variation of temperature"**"x is the parameter to be varied"**

0	100
0.0006667	98.33
0.001333	96.67
0.002	95
0.002667	93.33
0.003333	91.67
0.004	90
0.004667	88.33
0.005333	86.67
0.006	85



2-62E A steam pipe is subjected to convection on the inner surface and to specified temperature on the outer surface. The mathematical formulation, the variation of temperature in the pipe, and the rate of heat loss are to be determined for steady one-dimensional heat transfer.

Assumptions 1 Heat conduction is steady and one-dimensional since the pipe is long relative to its thickness, and there is thermal symmetry about the center line. **2** Thermal conductivity is constant. **3** There is no heat generation in the pipe.

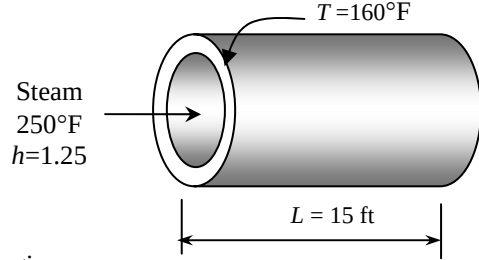
Properties The thermal conductivity is given to be $k = 7.2 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}$.

Analysis (a) Noting that heat transfer is one-dimensional in the radial r direction, the mathematical formulation of this problem can be expressed as

$$\frac{d}{dr} \left(r \frac{dT}{dr} \right) = 0$$

$$\text{and} \quad -k \frac{dT(r_1)}{dr} = h[T_\infty - T(r_1)]$$

$$T(r_2) = T_2 = 160^\circ\text{F}$$



(b) Integrating the differential equation once with respect to r gives

$$r \frac{dT}{dr} = C_1$$

Dividing both sides of the equation above by r to bring it to a readily integrable form and then integrating,

$$\frac{dT}{dr} = \frac{C_1}{r}$$

$$T(r) = C_1 \ln r + C_2$$

where C_1 and C_2 are arbitrary constants. Applying the boundary conditions give

$$r = r_1: \quad -k \frac{C_1}{r_1} = h[T_\infty - (C_1 \ln r_1 + C_2)]$$

$$r = r_2: \quad T(r_2) = C_1 \ln r_2 + C_2 = T_2$$

Solving for C_1 and C_2 simultaneously gives

$$C_1 = \frac{T_2 - T_\infty}{\ln \frac{r_2}{r_1} + \frac{k}{hr_1}} \quad \text{and} \quad C_2 = T_2 - C_1 \ln r_2 = T_2 - \frac{T_2 - T_\infty}{\ln \frac{r_2}{r_1} + \frac{k}{hr_1}} \ln r_2$$

Substituting C_1 and C_2 into the general solution, the variation of temperature is determined to be

$$\begin{aligned} T(r) &= C_1 \ln r + T_2 - C_1 \ln r_2 = C_1 (\ln r - \ln r_2) + T_2 = \frac{T_2 - T_\infty}{\ln \frac{r_2}{r_1} + \frac{k}{hr_1}} \ln \frac{r}{r_2} + T_2 \\ &= \frac{(160 - 250)^\circ\text{F}}{\ln \frac{2.4}{2} + \frac{7.2 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}}{(12.5 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F})(2/12 \text{ ft})}} \ln \frac{r}{2.4 \text{ in}} + 160^\circ\text{F} = -24.74 \ln \frac{r}{2.4 \text{ in}} + 160^\circ\text{F} \end{aligned}$$

(c) The rate of heat conduction through the pipe is

$$\begin{aligned} \dot{Q} &= -kA \frac{dT}{dr} = -k(2\pi rL) \frac{C_1}{r} = -2\pi Lk \frac{T_2 - T_\infty}{\ln \frac{r_2}{r_1} + \frac{k}{hr_1}} \\ &= -2\pi(15 \text{ ft})(7.2 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}) \frac{(160 - 250)^\circ\text{F}}{\ln \frac{2.4}{2} + \frac{7.2 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}}{(12.5 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F})(2/12 \text{ ft})}} = \mathbf{16,800 \text{ Btu/h}} \end{aligned}$$

2-63 A spherical container is subjected to specified temperature on the inner surface and convection on the outer surface. The mathematical formulation, the variation of temperature, and the rate of heat transfer are to be determined for steady one-dimensional heat transfer.

Assumptions 1 Heat conduction is steady and one-dimensional since there is no change with time and there is thermal symmetry about the midpoint. **2** Thermal conductivity is constant. **3** There is no heat generation.

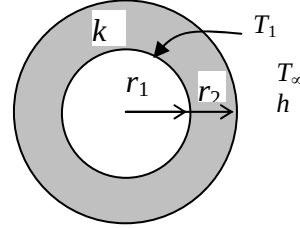
Properties The thermal conductivity is given to be $k = 30 \text{ W/m} \cdot ^\circ\text{C}$.

Analysis (a) Noting that heat transfer is one-dimensional in the radial r direction, the mathematical formulation of this problem can be expressed as

$$\frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = 0$$

and $T(r_1) = T_1 = 0^\circ\text{C}$

$$-k \frac{dT(r_2)}{dr} = h[T(r_2) - T_\infty]$$



(b) Integrating the differential equation once with respect to r gives

$$r^2 \frac{dT}{dr} = C_1$$

Dividing both sides of the equation above by r to bring it to a readily integrable form and then integrating,

$$\frac{dT}{dr} = \frac{C_1}{r^2}$$

$$T(r) = -\frac{C_1}{r} + C_2$$

where C_1 and C_2 are arbitrary constants. Applying the boundary conditions give

$$r = r_1: \quad T(r_1) = -\frac{C_1}{r_1} + C_2 = T_1$$

$$r = r_2: \quad -k \frac{C_1}{r_2^2} = h \left(-\frac{C_1}{r_2} + C_2 - T_\infty \right)$$

Solving for C_1 and C_2 simultaneously gives

$$C_1 = \frac{r_2(T_1 - T_\infty)}{1 - \frac{r_2}{r_1} - \frac{k}{hr_2}} \quad \text{and} \quad C_2 = T_1 + \frac{C_1}{r_1} = T_1 + \frac{T_1 - T_\infty}{1 - \frac{r_2}{r_1} - \frac{k}{hr_2}} \frac{r_2}{r_1}$$

Substituting C_1 and C_2 into the general solution, the variation of temperature is determined to be

$$\begin{aligned} T(r) &= -\frac{C_1}{r} + T_1 + \frac{C_1}{r_1} = C_1 \left(\frac{1}{r_1} - \frac{1}{r} \right) + T_1 = \frac{T_1 - T_\infty}{1 - \frac{r_2}{r_1} - \frac{k}{hr_2}} \left(\frac{r_2}{r_1} - \frac{r_2}{r} \right) + T_1 \\ &= \frac{(0 - 25)^\circ\text{C}}{1 - \frac{2.1}{2} - \frac{30 \text{ W/m} \cdot ^\circ\text{C}}{(18 \text{ W/m}^2 \cdot ^\circ\text{C})(2.1 \text{ m})}} \left(\frac{2.1}{2} - \frac{2.1}{r} \right) + 0^\circ\text{C} = 29.63(1.05 - 2.1/r) \end{aligned}$$

(c) The rate of heat conduction through the wall is

$$\begin{aligned}\dot{Q} &= -kA \frac{dT}{dx} = -k(4\pi r^2) \frac{C_1}{r^2} = -4\pi k C_1 = -4\pi k \frac{r_2(T_1 - T_\infty)}{1 - \frac{r_2}{r_1} - \frac{k}{hr_2}} \\ &= -4\pi(30 \text{ W/m} \cdot ^\circ\text{C}) \frac{(2.1 \text{ m})(0 - 25)^\circ\text{C}}{1 - \frac{2.1}{2} - \frac{30 \text{ W/m} \cdot ^\circ\text{C}}{(18 \text{ W/m}^2 \cdot ^\circ\text{C})(2.1 \text{ m})}} = \mathbf{23,460 \text{ W}}\end{aligned}$$

2-64 A large plane wall is subjected to specified heat flux and temperature on the left surface and no conditions on the right surface. The mathematical formulation, the variation of temperature in the plate, and the right surface temperature are to be determined for steady one-dimensional heat transfer.

Assumptions 1 Heat conduction is steady and one-dimensional since the wall is large relative to its thickness, and the thermal conditions on both sides of the wall are uniform. 2 Thermal conductivity is constant. 3 There is no heat generation in the wall.

Properties The thermal conductivity is given to be $k = 2.5 \text{ W/m}\cdot^\circ\text{C}$.

Analysis (a) Taking the direction normal to the surface of the wall to be the x direction with $x = 0$ at the left surface, the mathematical formulation of this problem can be expressed as

$$\frac{d^2T}{dx^2} = 0$$

$$\text{and} \quad -k \frac{dT(0)}{dx} = \dot{q}_0 = 700 \text{ W/m}^2$$

$$T(0) = T_1 = 80^\circ\text{C}$$

(b) Integrating the differential equation twice with respect to x yields

$$\frac{dT}{dx} = C_1$$

$$T(x) = C_1x + C_2$$

where C_1 and C_2 are arbitrary constants. Applying the boundary conditions give

$$\text{Heat flux at } x = 0: \quad -kC_1 = \dot{q}_0 \rightarrow C_1 = -\frac{\dot{q}_0}{k}$$

$$\text{Temperature at } x = 0: \quad T(0) = C_1 \times 0 + C_2 = T_1 \rightarrow C_2 = T_1$$

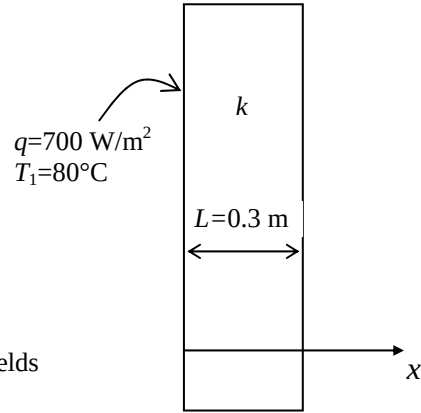
Substituting C_1 and C_2 into the general solution, the variation of temperature is determined to be

$$T(x) = -\frac{\dot{q}_0}{k}x + T_1 = -\frac{700 \text{ W/m}^2}{2.5 \text{ W/m}\cdot^\circ\text{C}}x + 80^\circ\text{C} = -280x + 80$$

(c) The temperature at $x = L$ (the right surface of the wall) is

$$T(L) = -280 \times (0.3 \text{ m}) + 80 = -4^\circ\text{C}$$

Note that the right surface temperature is lower as expected.



2-65 A large plane wall is subjected to specified heat flux and temperature on the left surface and no conditions on the right surface. The mathematical formulation, the variation of temperature in the plate, and the right surface temperature are to be determined for steady one-dimensional heat transfer.

Assumptions 1 Heat conduction is steady and one-dimensional since the wall is large relative to its thickness, and the thermal conditions on both sides of the wall are uniform. 2 Thermal conductivity is constant. 3 There is no heat generation in the wall.

Properties The thermal conductivity is given to be $k = 2.5 \text{ W/m}\cdot^\circ\text{C}$.

Analysis (a) Taking the direction normal to the surface of the wall to be the x direction with $x = 0$ at the left surface, the mathematical formulation of this problem can be expressed as

$$\frac{d^2T}{dx^2} = 0$$

$$\text{and} \quad -k \frac{dT(0)}{dx} = \dot{q}_0 = 950 \text{ W/m}^2$$

$$T(0) = T_1 = 85^\circ\text{C}$$

(b) Integrating the differential equation twice with respect to x yields

$$\frac{dT}{dx} = C_1$$

$$T(x) = C_1x + C_2$$

where C_1 and C_2 are arbitrary constants. Applying the boundary conditions give

$$\text{Heat flux at } x = 0: \quad -kC_1 = \dot{q}_0 \rightarrow C_1 = -\frac{\dot{q}_0}{k}$$

$$\text{Temperature at } x = 0: \quad T(0) = C_1 \times 0 + C_2 = T_1 \rightarrow C_2 = T_1$$

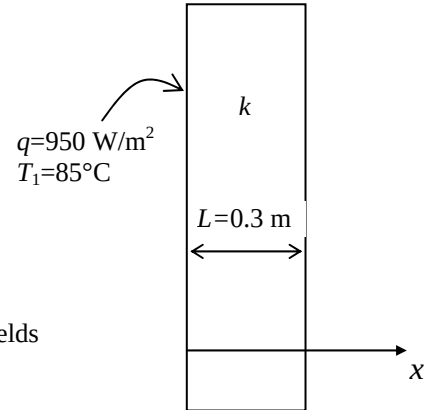
Substituting C_1 and C_2 into the general solution, the variation of temperature is determined to be

$$T(x) = -\frac{\dot{q}_0}{k}x + T_1 = -\frac{950 \text{ W/m}^2}{2.5 \text{ W/m}\cdot^\circ\text{C}}x + 85^\circ\text{C} = -380x + 85$$

(c) The temperature at $x = L$ (the right surface of the wall) is

$$T(L) = -380 \times (0.3 \text{ m}) + 85 = -29^\circ\text{C}$$

Note that the right surface temperature is lower as expected.



2-66E A large plate is subjected to convection, radiation, and specified temperature on the top surface and no conditions on the bottom surface. The mathematical formulation, the variation of temperature in the plate, and the bottom surface temperature are to be determined for steady one-dimensional heat transfer.

Assumptions 1 Heat conduction is steady and one-dimensional since the plate is large relative to its thickness, and the thermal conditions on both sides of the plate are uniform. 2 Thermal conductivity is constant. 3 There is no heat generation in the plate.

Properties The thermal conductivity and emissivity are given to be $k = 7.2 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}$ and $\varepsilon = 0.6$.

Analysis (a) Taking the direction normal to the surface of the plate to be the x direction with $x = 0$ at the bottom surface, and the mathematical formulation of this problem can be expressed as

$$\frac{d^2T}{dx^2} = 0$$

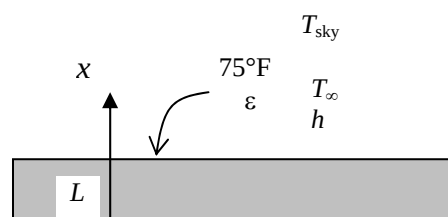
$$\text{and} \quad -k \frac{dT(L)}{dx} = h[T(L) - T_\infty] + \varepsilon\sigma[T(L)^4 - T_{\text{sky}}^4] = h[T_2 - T_\infty] + \varepsilon\sigma[(T_2 + 460)^4 - T_{\text{sky}}^4]$$

$$T(L) = T_2 = 75^\circ\text{F}$$

(b) Integrating the differential equation twice with respect to x yields

$$\frac{dT}{dx} = C_1$$

$$T(x) = C_1x + C_2$$



where C_1 and C_2 are arbitrary constants. Applying the boundary conditions give

$$\begin{aligned} \text{Convection at } x = L: \quad & -kC_1 = h[T_2 - T_\infty] + \varepsilon\sigma[(T_2 + 460)^4 - T_{\text{sky}}^4] \\ \rightarrow C_1 = & -\{h[T_2 - T_\infty] + \varepsilon\sigma[(T_2 + 460)^4 - T_{\text{sky}}^4]\} / k \end{aligned}$$

$$\text{Temperature at } x = L: \quad T(L) = C_1 \times L + C_2 = T_2 \rightarrow C_2 = T_2 - C_1L$$

Substituting C_1 and C_2 into the general solution, the variation of temperature is determined to be

$$\begin{aligned} T(x) &= C_1x + (T_2 - C_1L) = T_2 - (L - x)C_1 = T_2 + \frac{h[T_2 - T_\infty] + \varepsilon\sigma[(T_2 + 460)^4 - T_{\text{sky}}^4]}{k}(L - x) \\ &= 75^\circ\text{F} + \frac{(12 \text{ Btu/h} \cdot \text{ft}^2 \cdot ^\circ\text{F})(75 - 90)^\circ\text{F} + 0.6(0.1714 \times 10^{-8} \text{ Btu/h} \cdot \text{ft}^2 \cdot \text{R}^4)[(535 \text{ R})^4 - (510 \text{ R})^4]}{7.2 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}}(4/12 - x) \text{ ft} \\ &= 75 - 23.0(1/3 - x) \end{aligned}$$

(c) The temperature at $x = 0$ (the bottom surface of the plate) is

$$T(0) = 75 - 23.0 \times (1/3 - 0) = \mathbf{67.3^\circ\text{F}}$$

2-67E A large plate is subjected to convection and specified temperature on the top surface and no conditions on the bottom surface. The mathematical formulation, the variation of temperature in the plate, and the bottom surface temperature are to be determined for steady one-dimensional heat transfer.

Assumptions 1 Heat conduction is steady and one-dimensional since the plate is large relative to its thickness, and the thermal conditions on both sides of the plate are uniform. 2 Thermal conductivity is constant. 3 There is no heat generation in the plate.

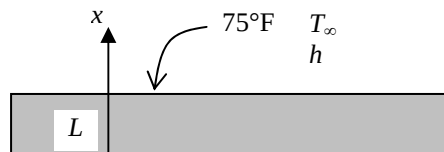
Properties The thermal conductivity is given to be $k = 7.2 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$.

Analysis (a) Taking the direction normal to the surface of the plate to be the x direction with $x = 0$ at the bottom surface, the mathematical formulation of this problem can be expressed as

$$\frac{d^2T}{dx^2} = 0$$

and $-k \frac{dT(L)}{dx} = h[T(L) - T_\infty] = h(T_2 - T_\infty)$

$$T(L) = T_2 = 75^\circ\text{F}$$



(b) Integrating the differential equation twice with respect to x yields

$$\frac{dT}{dx} = C_1$$

$$T(x) = C_1x + C_2$$

where C_1 and C_2 are arbitrary constants. Applying the boundary conditions give

$$\text{Convection at } x = L: -kC_1 = h(T_2 - T_\infty) \rightarrow C_1 = -h(T_2 - T_\infty) / k$$

$$\text{Temperature at } x = L: T(L) = C_1 \times L + C_2 = T_2 \rightarrow C_2 = T_2 - C_1L$$

Substituting C_1 and C_2 into the general solution, the variation of temperature is determined to be

$$T(x) = C_1x + (T_2 - C_1L) = T_2 - (L - x)C_1 = T_2 + \frac{h(T_2 - T_\infty)}{k}(L - x)$$

$$= 75^\circ\text{F} + \frac{(12 \text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F})(75 - 90)^\circ\text{F}}{7.2 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}}(4/12 - x) \text{ ft}$$

$$= 75 - 25(1/3 - x)$$

(c) The temperature at $x = 0$ (the bottom surface of the plate) is

$$T(0) = 75 - 25 \times (1/3 - 0) = \mathbf{66.7^\circ\text{F}}$$

2-68 A compressed air pipe is subjected to uniform heat flux on the outer surface and convection on the inner surface. The mathematical formulation, the variation of temperature in the pipe, and the surface temperatures are to be determined for steady one-dimensional heat transfer.

Assumptions 1 Heat conduction is steady and one-dimensional since the pipe is long relative to its thickness, and there is thermal symmetry about the center line. **2** Thermal conductivity is constant. **3** There is no heat generation in the pipe.

Properties The thermal conductivity is given to be $k = 14 \text{ W/m}\cdot^\circ\text{C}$.

Analysis (a) Noting that the 85% of the 300 W generated by the strip heater is transferred to the pipe, the heat flux through the outer surface is determined to be

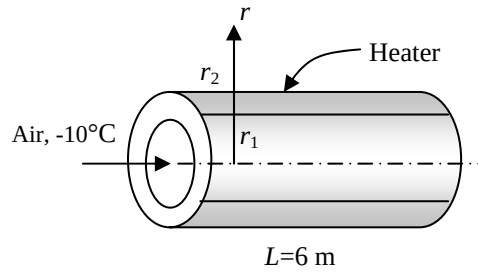
$$\dot{q}_s = \frac{\dot{Q}_s}{A_2} = \frac{\dot{Q}_s}{2\pi r_2 L} = \frac{0.85 \times 300 \text{ W}}{2\pi(0.04 \text{ m})(6 \text{ m})} = 169.1 \text{ W/m}^2$$

Noting that heat transfer is one-dimensional in the radial r direction and heat flux is in the negative r direction, the mathematical formulation of this problem can be expressed as

$$\frac{d}{dr} \left(r \frac{dT}{dr} \right) = 0$$

$$\text{and} \quad -k \frac{dT(r_1)}{dr} = h[T_\infty - T(r_1)]$$

$$k \frac{dT(r_2)}{dr} = \dot{q}_s$$



(b) Integrating the differential equation once with respect to r gives

$$r \frac{dT}{dr} = C_1$$

Dividing both sides of the equation above by r to bring it to a readily integrable form and then integrating,

$$\frac{dT}{dr} = \frac{C_1}{r}$$

$$T(r) = C_1 \ln r + C_2$$

where C_1 and C_2 are arbitrary constants. Applying the boundary conditions give

$$r = r_2: \quad k \frac{C_1}{r_2} = \dot{q}_s \rightarrow C_1 = \frac{\dot{q}_s r_2}{k}$$

$$r = r_1: \quad -k \frac{C_1}{r_1} = h[T_\infty - (C_1 \ln r_1 + C_2)] \rightarrow C_2 = T_\infty - \left(\ln r_1 - \frac{k}{hr_1} \right) C_1 = T_\infty - \left(\ln r_1 - \frac{k}{hr_1} \right) \frac{\dot{q}_s r_2}{k}$$

Substituting C_1 and C_2 into the general solution, the variation of temperature is determined to be

$$\begin{aligned} T(r) &= C_1 \ln r + T_\infty - \left(\ln r_1 - \frac{k}{hr_1} \right) C_1 = T_\infty + \left(\ln r - \ln r_1 + \frac{k}{hr_1} \right) C_1 = T_\infty + \left(\ln \frac{r}{r_1} + \frac{k}{hr_1} \right) \frac{\dot{q}_s r_2}{k} \\ &= -10^\circ\text{C} + \left(\ln \frac{r}{r_1} + \frac{14 \text{ W/m}\cdot^\circ\text{C}}{(30 \text{ W/m}^2 \cdot ^\circ\text{C})(0.037 \text{ m})} \right) \frac{(169.1 \text{ W/m}^2)(0.04 \text{ m})}{14 \text{ W/m}\cdot^\circ\text{C}} = -10 + 0.483 \left(\ln \frac{r}{r_1} + 12.61 \right) \end{aligned}$$

(c) The inner and outer surface temperatures are determined by direct substitution to be

$$\text{Inner surface } (r = r_1): \quad T(r_1) = -10 + 0.483 \left(\ln \frac{r_1}{r_1} + 12.61 \right) = -10 + 0.483(0 + 12.61) = \mathbf{-3.91^\circ\text{C}}$$

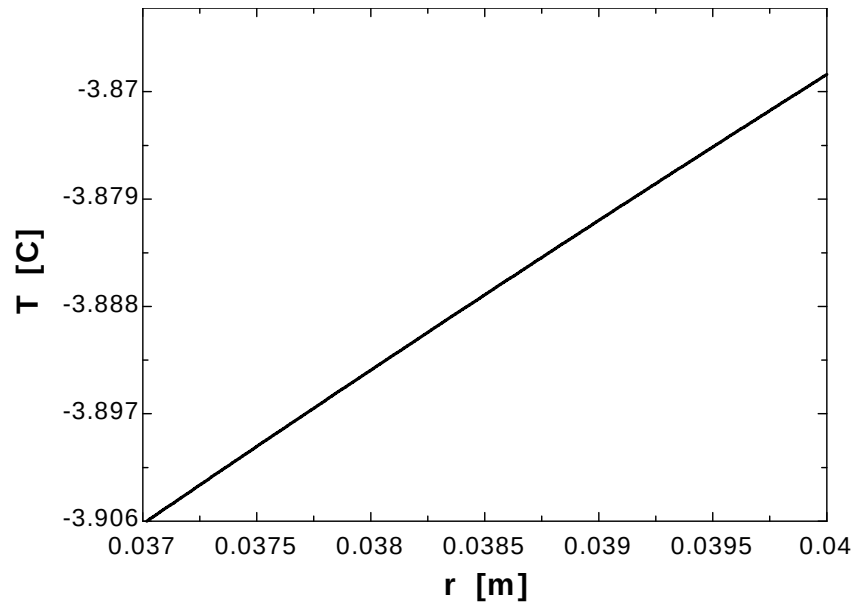
$$\text{Outer surface } (r = r_2): \quad T(r_1) = -10 + 0.483 \left(\ln \frac{r_2}{r_1} + 12.61 \right) = -10 + 0.483 \left(\ln \frac{0.04}{0.037} + 12.61 \right) = \mathbf{-3.87^\circ\text{C}}$$

Note that the pipe is essentially isothermal at a temperature of about -3.9°C .

2-69

"GIVEN" $L=6$ "[m]" $r_1=0.037$ "[m]" $r_2=0.04$ "[m]" $k=14$ "[W/m-C]" $\dot{Q}=300$ "[W]" $T_{\infty}=-10$ "[C]" $h=30$ "[W/m^2-C]" $f_{\text{loss}}=0.15$ **"ANALYSIS"** $\dot{q}_s=((1-f_{\text{loss}})\dot{Q})/A$ $A=2\pi r_2 L$ $T=T_{\infty}+(\ln(r/r_1)+k/(h r_1))(\dot{q}_s r_2)/k$ **"Variation of temperature"****"r is the parameter to be varied"**

r [m]	T [C]
0.037	3.906
0.03733	3.902
0.03767	3.898
0.038	3.893
0.03833	3.889
0.03867	3.885
0.039	3.881
0.03933	3.877
0.03967	3.873
0.04	3.869



2-70 A spherical container is subjected to uniform heat flux on the outer surface and specified temperature on the inner surface. The mathematical formulation, the variation of temperature in the pipe, and the outer surface temperature, and the maximum rate of hot water supply are to be determined for steady one-dimensional heat transfer.

Assumptions 1 Heat conduction is steady and one-dimensional since there is no change with time and there is thermal symmetry about the mid point. 2 Thermal conductivity is constant. 3 There is no heat generation in the container.

Properties The thermal conductivity is given to be $k = 1.5 \text{ W/m}\cdot^\circ\text{C}$. The specific heat of water at the average temperature of $(100+20)/2 = 60^\circ\text{C}$ is $4.185 \text{ kJ/kg}\cdot^\circ\text{C}$ (Table A-9).

Analysis (a) Noting that the 90% of the 500 W generated by the strip heater is transferred to the container, the heat flux through the outer surface is determined to be

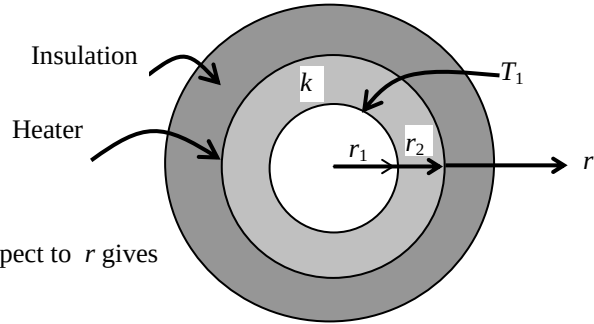
$$\dot{q}_s = \frac{\dot{Q}_s}{A_2} = \frac{\dot{Q}_s}{4\pi r_2^2} = \frac{0.90 \times 500 \text{ W}}{4\pi (0.41 \text{ m})^2} = 213.0 \text{ W/m}^2$$

Noting that heat transfer is one-dimensional in the radial r direction and heat flux is in the negative r direction, the mathematical formulation of this problem can be expressed as

$$\frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = 0$$

and $T(r_1) = T_1 = 100^\circ\text{C}$

$$k \frac{dT(r_2)}{dr} = \dot{q}_s$$



(b) Integrating the differential equation once with respect to r gives

$$r^2 \frac{dT}{dr} = C_1$$

Dividing both sides of the equation above by r^2 and then integrating,

$$\frac{dT}{dr} = \frac{C_1}{r^2}$$

$$T(r) = -\frac{C_1}{r} + C_2$$

where C_1 and C_2 are arbitrary constants. Applying the boundary conditions give

$$r = r_2: \quad k \frac{C_1}{r_2^2} = \dot{q}_s \rightarrow C_1 = \frac{\dot{q}_s r_2^2}{k}$$

$$r = r_1: \quad T(r_1) = T_1 = -\frac{C_1}{r_1} + C_2 \rightarrow C_2 = T_1 + \frac{C_1}{r_1} = T_1 + \frac{\dot{q}_s r_2^2}{k r_1}$$

Substituting C_1 and C_2 into the general solution, the variation of temperature is determined to be

$$\begin{aligned} T(r) &= -\frac{C_1}{r} + C_2 = -\frac{C_1}{r} + T_1 + \frac{C_1}{r_1} = T_1 + \left(\frac{1}{r_1} - \frac{1}{r} \right) C_1 = T_1 + \left(\frac{1}{r_1} - \frac{1}{r} \right) \frac{\dot{q}_s r_2^2}{k} \\ &= 100^\circ\text{C} + \left(\frac{1}{0.40 \text{ m}} - \frac{1}{r} \right) \frac{(213 \text{ W/m}^2)(0.41 \text{ m})^2}{1.5 \text{ W/m}\cdot^\circ\text{C}} = 100 + 23.87 \left(2.5 - \frac{1}{r} \right) \end{aligned}$$

(c) The outer surface temperature is determined by direct substitution to be

$$\text{Outer surface } (r = r_2): \quad T(r_2) = 100 + 23.87 \left(2.5 - \frac{1}{r_2} \right) = 100 + 23.87 \left(2.5 - \frac{1}{0.41} \right) = \mathbf{101.5^\circ\text{C}}$$

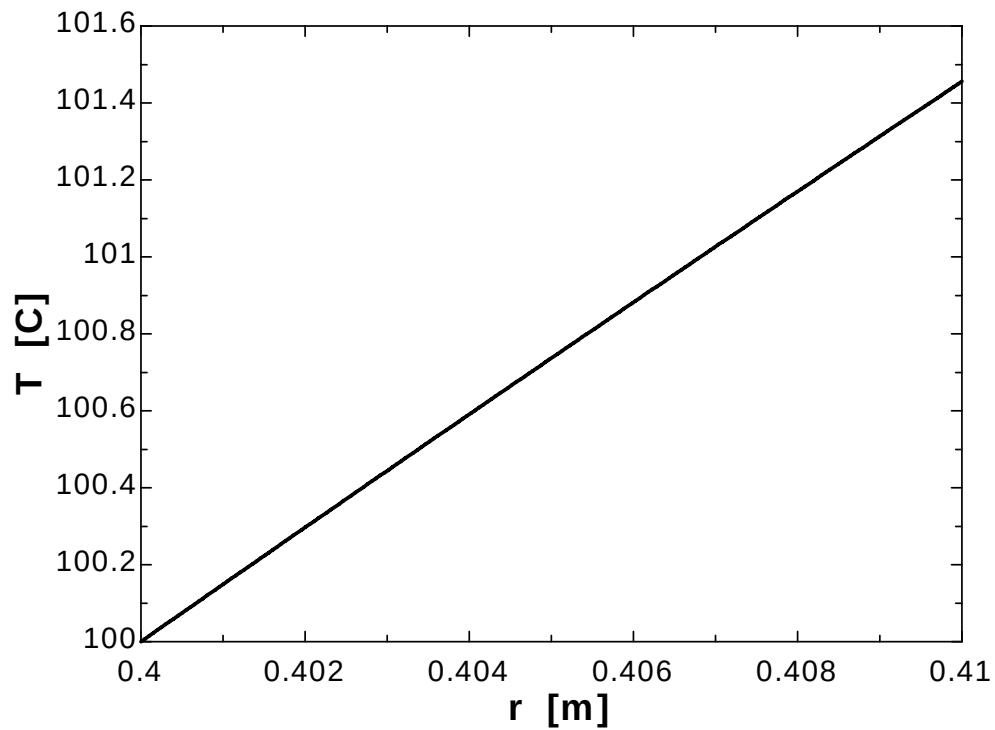
Noting that the maximum rate of heat supply to the water is $0.9 \times 500 \text{ W} = 450 \text{ W}$, water can be heated from 20 to 100°C at a rate of

$$\dot{Q} = \dot{m}C_p\Delta T \rightarrow \dot{m} = \frac{\dot{Q}}{C_p\Delta T} = \frac{0.450 \text{ kJ / s}}{(4.185 \text{ kJ / kg}\cdot^\circ\text{C})(100 - 20)^\circ\text{C}} = 0.00134 \text{ kg / s} = \mathbf{4.84 \text{ kg / h}}$$

2-71

"GIVEN" $r_1=0.40$ "[m]" $r_2=0.41$ "[m]" $k=1.5$ "[W/m-C]" $T_1=100$ "[C]" $\dot{Q}=500$ "[W]" $f_{\text{loss}}=0.10$ **"ANALYSIS"** $q_{\text{dot}_s} = ((1-f_{\text{loss}})*\dot{Q})/A$ $A=4*\pi*r_2^2$ $T=T_1+(1/r_1-1/r)*(q_{\text{dot}_s}*r_2^2)/k$ "Variation of temperature"**"r is the parameter to be varied"**

r [m]	T [C]
0.4	100
0.4011	100.2
0.4022	100.3
0.4033	100.5
0.4044	100.7
0.4056	100.8
0.4067	101
0.4078	101.1
0.4089	101.3
0.41	101.5



Heat Generation in Solids

2-72C No. Heat generation in a solid is simply the conversion of some form of energy into sensible heat energy. For example resistance heating in wires is conversion of electrical energy to heat.

2-73C Heat generation in a solid is simply conversion of some form of energy into sensible heat energy. Some examples of heat generations are resistance heating in wires, exothermic chemical reactions in a solid, and nuclear reactions in nuclear fuel rods.

2-74C The rate of heat generation inside an iron becomes equal to the rate of heat loss from the iron when steady operating conditions are reached and the temperature of the iron stabilizes.

2-75C No, it is not possible since the highest temperature in the plate will occur at its center, and heat cannot flow “uphill.”

2-76C The cylinder will have a higher center temperature since the cylinder has less surface area to lose heat from per unit volume than the sphere.

2-77 A 2-kW resistance heater wire with a specified surface temperature is used to boil water. The center temperature of the wire is to be determined.

Assumptions **1** Heat transfer is steady since there is no change with time. **2** Heat transfer is one-dimensional since there is thermal symmetry about the center line and no change in the axial direction. **3** Thermal conductivity is constant. **4** Heat generation in the heater is uniform.

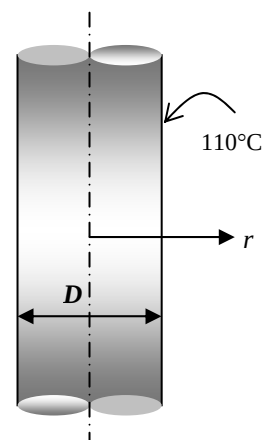
Properties The thermal conductivity is given to be $k = 20 \text{ W/m}\cdot^\circ\text{C}$.

Analysis The resistance heater converts electric energy into heat at a rate of 2 kW. The rate of heat generation per unit volume of the wire is

$$\dot{g} = \frac{\dot{Q}_{gen}}{V_{wire}} = \frac{\dot{Q}_{gen}}{\pi r_o^2 L} = \frac{2000 \text{ W}}{\pi (0.0025 \text{ m})^2 (0.7 \text{ m})} = 1.455 \times 10^8 \text{ W/m}^3$$

The center temperature of the wire is then determined from Eq. 2-71 to be

$$T_o = T_s + \frac{\dot{g} r_o^2}{4k} = 110^\circ\text{C} + \frac{(1.455 \times 10^8 \text{ W/m}^3)(0.0025 \text{ m})^2}{4(20 \text{ W/m}\cdot^\circ\text{C})} = 121.4^\circ\text{C}$$



2-78 Heat is generated in a long solid cylinder with a specified surface temperature. The variation of temperature in the cylinder is given by

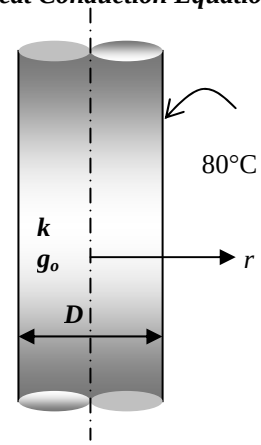
$$T(r) = \frac{\dot{g}r_0^2}{k} \left[1 - \left(\frac{r}{r_0} \right)^2 \right] + T_s$$

(a) Heat conduction is steady since there is no time t variable involved.

(b) Heat conduction is a one-dimensional.

(c) Using Eq. (1), the heat flux on the surface of the cylinder at $r = r_0$ is determined from its definition to be

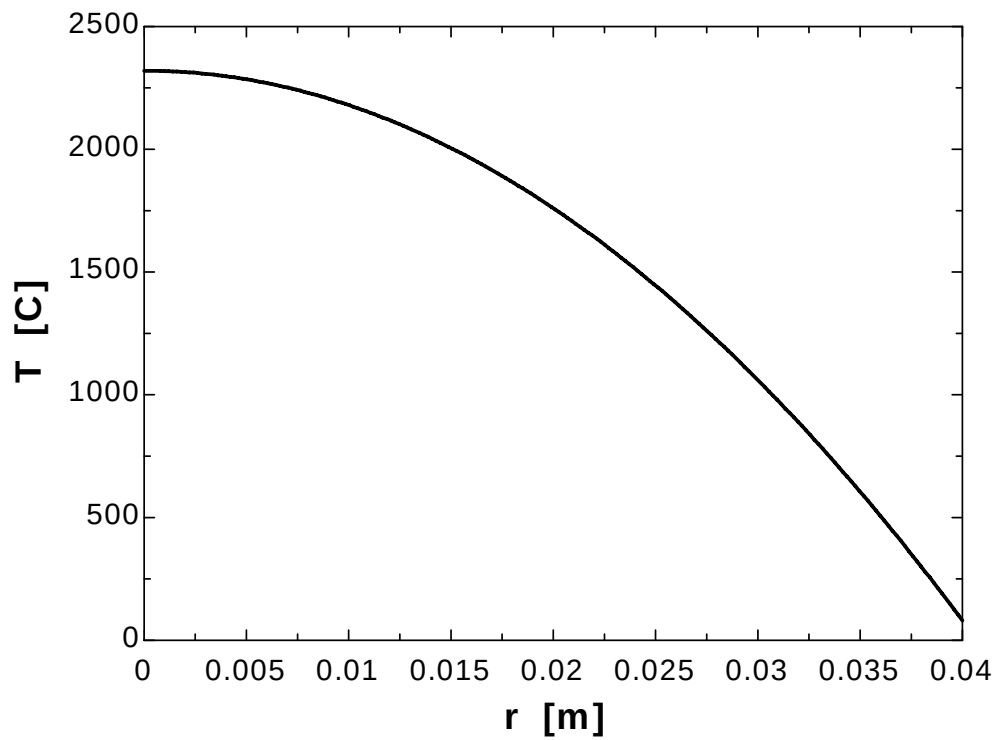
$$\dot{q}_s = -k \frac{dT(r_0)}{dr} = -k \left[\frac{\dot{g}r_0^2}{k} \left(-\frac{2r}{r_0^2} \right) \right]_{r=r_0} = -k \left[\frac{\dot{g}r_0^2}{k} \left(-\frac{2r_0}{r_0^2} \right) \right] = 2\dot{g}r_0 = 2(35 \text{ W/cm}^3)(4 \text{ cm}) = \mathbf{280 \text{ W/cm}^2}$$



2-79

"GIVEN" $r_0 = 0.04$ "[m]" $k = 25$ "[W/m-C]" $\dot{g}_0 = 35 \times 10^6$ "[W/m³]" $T_s = 80$ "[C]"**"ANALYSIS"** $T = (\dot{g}_0 r_0^2 / k) (1 - (r/r_0)^2) + T_s$ "Variation of temperature"**"r is the parameter to be varied"**

r [m]	T [C]
0	2320
0.004444	2292
0.008889	2209
0.01333	2071
0.01778	1878
0.02222	1629
0.02667	1324
0.03111	964.9
0.03556	550.1
0.04	80



2-80E A long homogeneous resistance heater wire with specified convection conditions at the surface is used to boil water. The mathematical formulation, the variation of temperature in the wire, and the temperature at the centerline of the wire are to be determined.

Assumptions 1 Heat transfer is steady since there is no indication of any change with time. 2 Heat transfer is one-dimensional since there is thermal symmetry about the center line and no change in the axial direction. 3 Thermal conductivity is constant. 4 Heat generation in the wire is uniform.

Properties The thermal conductivity is given to be $k = 8.6 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$.

Analysis Noting that heat transfer is steady and one-dimensional in the radial r direction, the mathematical formulation of this problem can be expressed as

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dT}{dr} \right) + \frac{\dot{g}}{k} = 0$$

$$\text{and } -k \frac{dT(r_0)}{dr} = h[T(r_0) - T_\infty] \quad (\text{convection at the outer surface})$$

$$\frac{dT(0)}{dr} = 0 \quad (\text{thermal symmetry about the centerline})$$

Multiplying both sides of the differential equation by r and rearranging gives

$$\frac{d}{dr} \left(r \frac{dT}{dr} \right) = -\frac{\dot{g}}{k} r$$

Integrating with respect to r gives

$$r \frac{dT}{dr} = -\frac{\dot{g}}{k} \frac{r^2}{2} + C_1 \quad (a)$$

It is convenient at this point to apply the second boundary condition since it is related to the first derivative of the temperature by replacing all occurrences of r and dT/dr in the equation above by zero. It yields

$$\text{B.C. at } r = 0: \quad 0 \times \frac{dT(0)}{dr} = -\frac{\dot{g}}{2k} \times 0 + C_1 \rightarrow C_1 = 0$$

Dividing both sides of Eq. (a) by r to bring it to a readily integrable form and integrating,

$$\frac{dT}{dr} = -\frac{\dot{g}}{2k} r$$

$$\text{and } T(r) = -\frac{\dot{g}}{4k} r^2 + C_2 \quad (b)$$

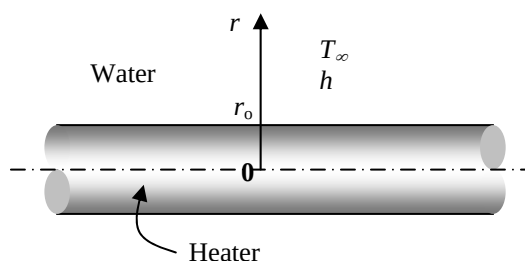
Applying the second boundary condition at $r = r_0$,

$$\text{B. C. at } r = r_0: \quad -k \frac{\dot{g} r_0}{2k} = h \left(-\frac{\dot{g}}{4k} r_0^2 + C_2 - T_\infty \right) \rightarrow C_2 = T_\infty + \frac{\dot{g} r_0}{2h} + \frac{\dot{g}}{4k} r_0^2$$

Substituting this C_2 relation into Eq. (b) and rearranging give

$$T(r) = T_\infty + \frac{\dot{g}}{4k} (r_0^2 - r^2) + \frac{\dot{g} r_0}{2h}$$

which is the desired solution for the temperature distribution in the wire as a function of r . Then the temperature at the center line ($r = 0$) is determined by substituting the known quantities to be



$$\begin{aligned}
 T(0) &= T_{\infty} + \frac{\dot{g}}{4k} r_0^2 + \frac{\dot{g}r_0}{2h} \\
 &= 212^{\circ}\text{F} + \frac{(1800 \text{ Btu/h}\cdot\text{in}^3)(0.25 \text{ in})^2}{4 \times (8.6 \text{ Btu/h}\cdot\text{ft}\cdot^{\circ}\text{F})} \left(\frac{12 \text{ in}}{1 \text{ ft}} \right) + \frac{(1800 \text{ Btu/h}\cdot\text{in}^3)(0.25 \text{ in})}{2 \times (820 \text{ Btu/h}\cdot\text{ft}^2\cdot^{\circ}\text{F})} \left(\frac{12 \text{ in}}{1 \text{ ft}} \right)^2 = \mathbf{290.8^{\circ}\text{F}}
 \end{aligned}$$

Thus the centerline temperature will be about 80°F above the temperature of the surface of the wire.

2-81E

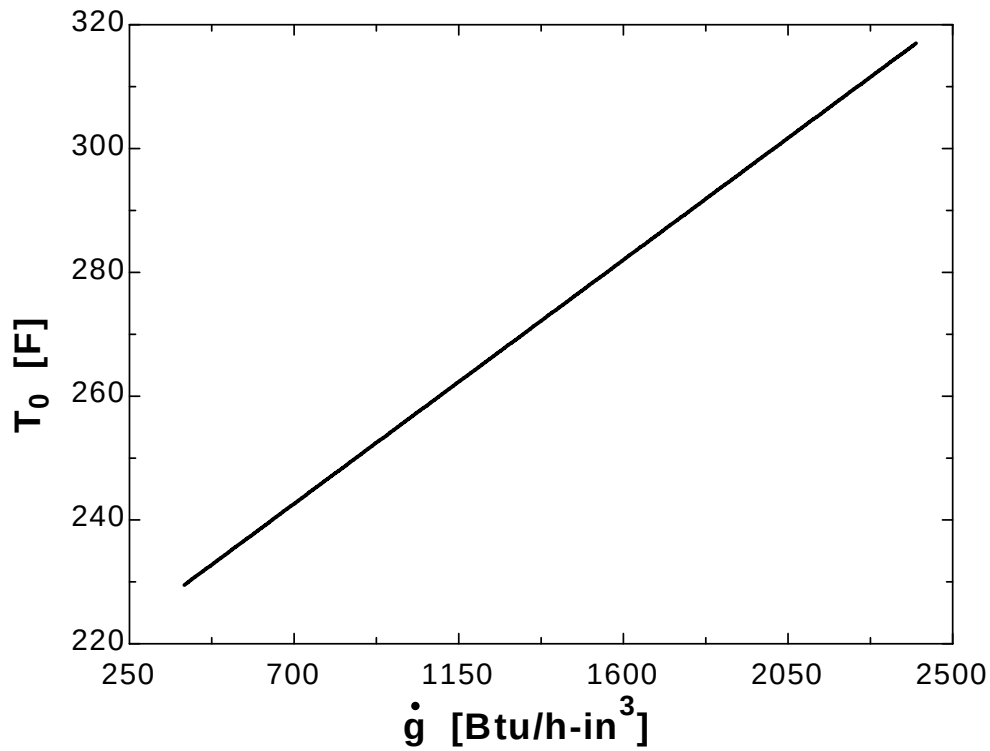
"GIVEN"

 $r_0 = 0.25/12$ "[ft]" $k = 8.6$ "[Btu/h-ft-F]" $\dot{g} = 1800$ [Btu/h-in³], parameter to be varied" $T_{\infty} = 212$ "[F]" $h = 820$ "[Btu/h-ft²-F]"

"ANALYSIS"

 $T_0 = T_{\infty} + (\dot{g}/\text{Convert}(\text{in}^3, \text{ft}^3))/(4*k)*(r_0^2 - r^2) + ((\dot{g}/\text{Convert}(\text{in}^3, \text{ft}^3))*r_0)/(2*h)$ "Variation of temperature" $r = 0$ "for centerline temperature"

$\dot{g}$ [Btu/h.in ³]	T_0 [F]
400	229.5
600	238.3
800	247
1000	255.8
1200	264.5
1400	273.3
1600	282
1800	290.8
2000	299.5
2200	308.3
2400	317



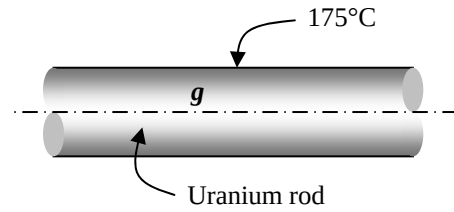
2-82 A nuclear fuel rod with a specified surface temperature is used as the fuel in a nuclear reactor. The center temperature of the rod is to be determined.

Assumptions 1 Heat transfer is steady since there is no indication of any change with time. 2 Heat transfer is one-dimensional since there is thermal symmetry about the center line and no change in the axial direction. 3 Thermal conductivity is constant. 4 Heat generation in the rod is uniform.

Properties The thermal conductivity is given to be $k = 29.5 \text{ W/m}\cdot^\circ\text{C}$.

Analysis The center temperature of the rod is determined from

$$T_o = T_s + \frac{\dot{g}r_o^2}{4k} = 175^\circ\text{C} + \frac{(7 \times 10^7 \text{ W/m}^3)(0.025 \text{ m})^2}{4(29.5 \text{ W/m}\cdot^\circ\text{C})} = \mathbf{545.8^\circ\text{C}}$$



2-83 Both sides of a large stainless steel plate in which heat is generated uniformly are exposed to convection with the environment. The location and values of the highest and the lowest temperatures in the plate are to be determined.

Assumptions 1 Heat transfer is steady since there is no indication of any change with time. 2 Heat transfer is one-dimensional since the plate is large relative to its thickness, and there is thermal symmetry about the center plane 3 Thermal conductivity is constant. 4 Heat generation is uniform.

Properties The thermal conductivity is given to be $k = 15.1 \text{ W/m}\cdot^\circ\text{C}$.

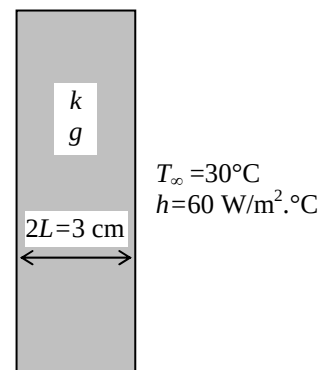
Analysis The lowest temperature will occur at surfaces of plate while the highest temperature will occur at the midplane. Their values are determined directly from

$$T_s = T_\infty + \frac{\dot{q}L}{h} = 30^\circ\text{C} + \frac{(5 \times 10^5 \text{ W/m}^3)(0.015 \text{ m})}{60 \text{ W/m}^2\cdot^\circ\text{C}} = \mathbf{155^\circ\text{C}}$$

$$T_o = T_s + \frac{\dot{q}L^2}{2k} = 155^\circ\text{C} + \frac{(5 \times 10^5 \text{ W/m}^3)(0.015 \text{ m})^2}{2(15.1 \text{ W/m}\cdot^\circ\text{C})} = \mathbf{158.7^\circ\text{C}}$$

$$T_\infty = 30^\circ\text{C}$$

$$h = 60 \text{ W/m}^2\cdot^\circ\text{C}$$



2-84 Heat is generated uniformly in a large brass plate. One side of the plate is insulated while the other side is subjected to convection. The location and values of the highest and the lowest temperatures in the plate are to be determined.

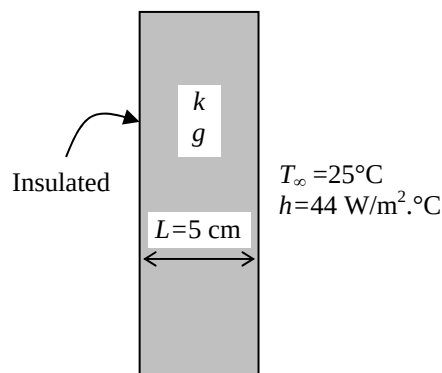
Assumptions 1 Heat transfer is steady since there is no indication of any change with time. 2 Heat transfer is one-dimensional since the plate is large relative to its thickness, and there is thermal symmetry about the center plane 3 Thermal conductivity is constant. 4 Heat generation is uniform.

Properties The thermal conductivity is given to be $k = 111 \text{ W/m}\cdot^\circ\text{C}$.

Analysis This insulated plate whose thickness is L is equivalent to one-half of an uninsulated plate whose thickness is $2L$ since the midplane of the uninsulated plate can be treated as insulated surface. The highest temperature will occur at the insulated surface while the lowest temperature will occur at the surface which is exposed to the environment. Note that L in the following relations is the full thickness of the given plate since the insulated side represents the center surface of a plate whose thickness is doubled. The desired values are determined directly from

$$T_s = T_\infty + \frac{\dot{q}L}{h} = 25^\circ\text{C} + \frac{(2 \times 10^5 \text{ W/m}^3)(0.05 \text{ m})}{44 \text{ W/m}^2\cdot^\circ\text{C}} = \mathbf{252.3^\circ\text{C}}$$

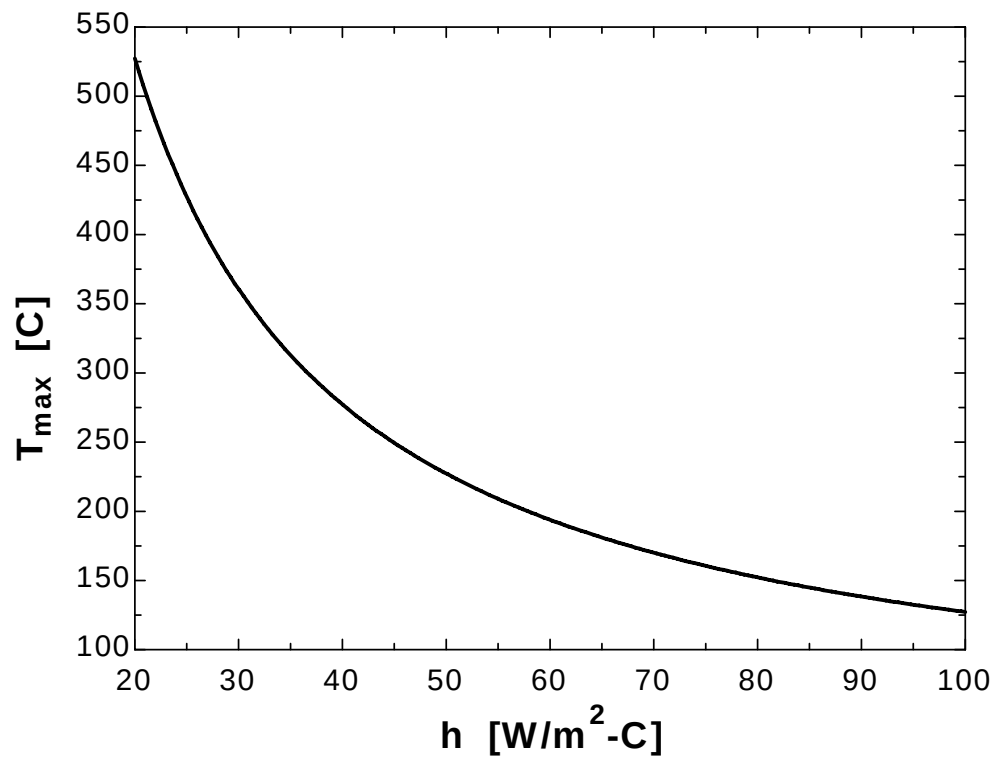
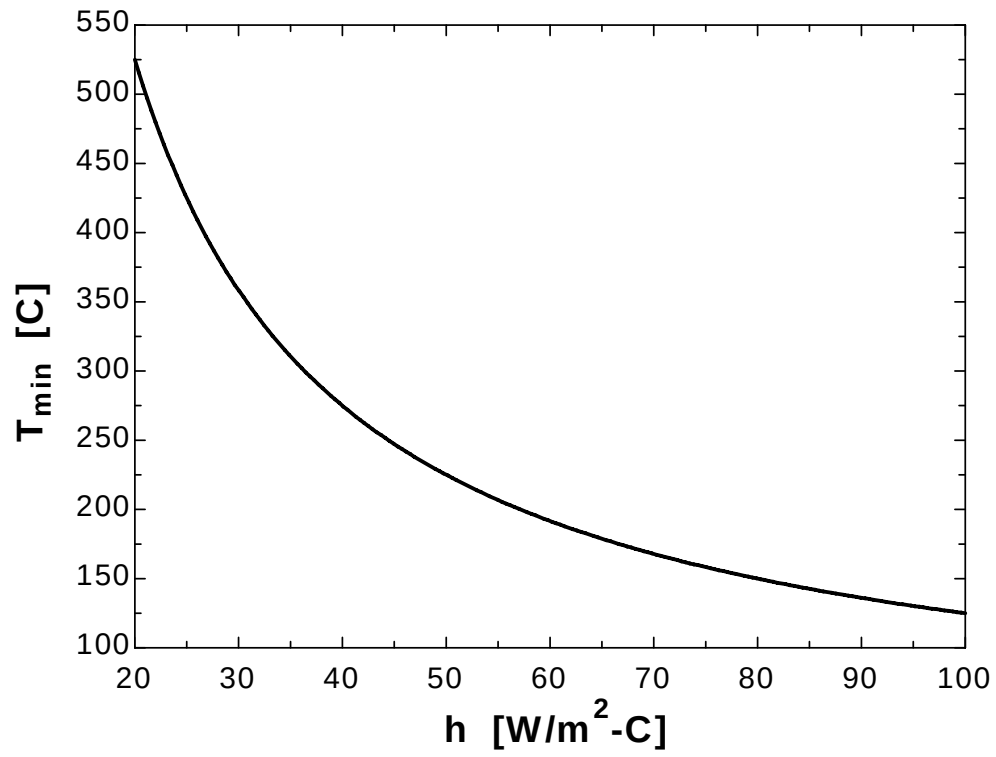
$$T_o = T_s + \frac{\dot{q}L^2}{2k} = 252.3^\circ\text{C} + \frac{(2 \times 10^5 \text{ W/m}^3)(0.05 \text{ m})^2}{2(111 \text{ W/m}\cdot^\circ\text{C})} = \mathbf{254.5^\circ\text{C}}$$



2-85

"GIVEN" $L=0.05$ "[m]" $k=111$ "[W/m-C]" $g_{\text{dot}}=2E5$ "[W/m^3]" $T_{\text{infinity}}=25$ "[C]" $h=44$ [W/m^2-C], parameter to be varied"**"ANALYSIS"** $T_{\text{min}}=T_{\text{infinity}}+(g_{\text{dot}}*L)/h$ $T_{\text{max}}=T_{\text{min}}+(g_{\text{dot}}*L^2)/(2*k)$

h [W/m ² .C]	T_{min} [C]	T_{max} [C]
20	525	527.3
25	425	427.3
30	358.3	360.6
35	310.7	313
40	275	277.3
45	247.2	249.5
50	225	227.3
55	206.8	209.1
60	191.7	193.9
65	178.8	181.1
70	167.9	170.1
75	158.3	160.6
80	150	152.3
85	142.6	144.9
90	136.1	138.4
95	130.3	132.5
100	125	127.3



2-86 A long resistance heater wire is subjected to convection at its outer surface. The surface temperature of the wire is to be determined using the applicable relations directly and by solving the applicable differential equation.

Assumptions 1 Heat transfer is steady since there is no indication of any change with time. 2 Heat transfer is one-dimensional since there is thermal symmetry about the center line and no change in the axial direction. 3 Thermal conductivity is constant. 4 Heat generation in the wire is uniform.

Properties The thermal conductivity is given to be $k = 15.1 \text{ W/m}\cdot^\circ\text{C}$.

Analysis (a) The heat generation per unit volume of the wire is

$$\dot{q} = \frac{\dot{Q}_{gen}}{V_{wire}} = \frac{\dot{Q}_{gen}}{\pi r_o^2 L} = \frac{2000 \text{ W}}{\pi (0.001 \text{ m})^2 (6 \text{ m})} = 1.061 \times 10^8 \text{ W/m}^3$$

The surface temperature of the wire is then (Eq. 2-68)

$$T_s = T_\infty + \frac{\dot{q} r_o}{2h} = 30^\circ\text{C} + \frac{(1.061 \times 10^8 \text{ W/m}^3)(0.001 \text{ m})}{2(140 \text{ W/m}^2\cdot^\circ\text{C})} = 409^\circ\text{C}$$

(b) The mathematical formulation of this problem can be expressed as

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dT}{dr} \right) + \frac{\dot{q}}{k} = 0$$

and $-k \frac{dT(r_o)}{dr} = h[T(r_o) - T_\infty]$ (convection at the outer surface)

$$\frac{dT(0)}{dr} = 0 \quad (\text{thermal symmetry about the centerline})$$

Multiplying both sides of the differential equation by r and integrating gives

$$\frac{d}{dr} \left(r \frac{dT}{dr} \right) = -\frac{\dot{q}}{k} r \rightarrow r \frac{dT}{dr} = -\frac{\dot{q}}{k} \frac{r^2}{2} + C_1 \quad (a)$$

Applying the boundary condition at the center line,

$$\text{B.C. at } r = 0: \quad 0 \times \frac{dT(0)}{dr} = -\frac{\dot{q}}{2k} \times 0 + C_1 \rightarrow C_1 = 0$$

Dividing both sides of Eq. (a) by r to bring it to a readily integrable form and integrating,

$$\frac{dT}{dr} = -\frac{\dot{q}}{2k} r \rightarrow T(r) = -\frac{\dot{q}}{4k} r^2 + C_2 \quad (b)$$

Applying the boundary condition at $r = r_o$,

$$\text{B.C. at } r = r_o: \quad -k \frac{\dot{q} r_o}{2k} = h \left(-\frac{\dot{q}}{4k} r_o^2 + C_2 - T_\infty \right) \rightarrow C_2 = T_\infty + \frac{\dot{q} r_o}{2h} + \frac{\dot{q}}{4k} r_o^2$$

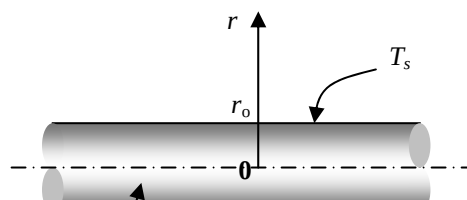
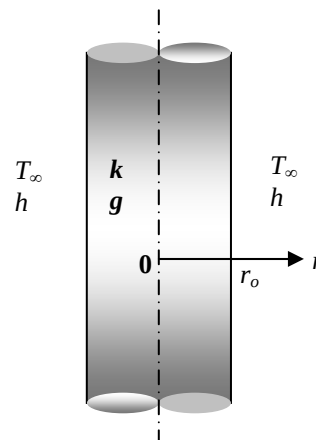
Substituting this C_2 relation into Eq. (b) and rearranging give

$$T(r) = T_\infty + \frac{\dot{q}}{4k} (r_o^2 - r^2) + \frac{\dot{q} r_o}{2h}$$

which is the temperature distribution in the wire as a function of r . Then the temperature of the wire at the surface ($r = r_o$) is determined by substituting the known quantities to be

$$T(r_o) = T_\infty + \frac{\dot{q}}{4k} (r_o^2 - r_o^2) + \frac{\dot{q} r_o}{2h} = T_\infty + \frac{\dot{q} r_o}{2h} = 30^\circ\text{C} + \frac{(1.061 \times 10^8 \text{ W/m}^3)(0.001 \text{ m})}{2(140 \text{ W/m}^2\cdot^\circ\text{C})} = 409^\circ\text{C}$$

Note that both approaches give the same result.



2-87E Heat is generated uniformly in a resistance heater wire. The temperature difference between the center and the surface of the wire is to be determined.

Assumptions 1 Heat transfer is steady since there is no change with time. 2 Heat transfer is one-dimensional since there is thermal symmetry about the center line and no change in the axial direction. 3 Thermal conductivity is constant. 4 Heat generation in the heater is uniform.

Properties The thermal conductivity is given to be $k = 5.8$ Btu/h·ft·°F.

Analysis The resistance heater converts electric energy into heat at a rate of 3 kW. The rate of heat generation per unit length of the wire is

$$\dot{q} = \frac{\dot{Q}_{gen}}{V_{wire}} = \frac{\dot{Q}_{gen}}{\pi r_o^2 L} = \frac{(3 \times 3412.14 \text{ Btu/h})}{\pi (0.04 / 12 \text{ ft})^2 (1 \text{ ft})} = 2.933 \times 10^8 \text{ Btu/h.ft}^3$$

Then the temperature difference between the centerline and the surface becomes

$$\Delta T_{max} = \frac{\dot{q} r_o^2}{4k} = \frac{(2.933 \times 10^8 \text{ Btu/h.ft}^3)(0.04 / 12 \text{ ft})^2}{4(5.8 \text{ Btu/h.ft.°F})} = \mathbf{140.4^\circ\text{F}}$$

2-88E Heat is generated uniformly in a resistance heater wire. The temperature difference between the center and the surface of the wire is to be determined.

Assumptions 1 Heat transfer is steady since there is no change with time. 2 Heat transfer is one-dimensional since there is thermal symmetry about the center line and no change in the axial direction. 3 Thermal conductivity is constant. 4 Heat generation in the heater is uniform.

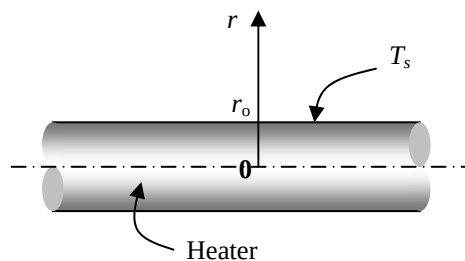
Properties The thermal conductivity is given to be $k = 4.5$ Btu/h·ft·°F.

Analysis The resistance heater converts electric energy into heat at a rate of 3 kW. The rate of heat generation per unit volume of the wire is

$$\dot{q} = \frac{\dot{Q}_{gen}}{V_{wire}} = \frac{\dot{Q}_{gen}}{\pi r_o^2 L} = \frac{(3 \times 3412.14 \text{ Btu/h})}{\pi (0.04 / 12 \text{ ft})^2 (1 \text{ ft})} = 2.933 \times 10^8 \text{ Btu/h.ft}^3$$

Then the temperature difference between the centerline and the surface becomes

$$\Delta T_{max} = \frac{\dot{q} r_o^2}{4k} = \frac{(2.933 \times 10^8 \text{ Btu/h.ft}^3)(0.04 / 12 \text{ ft})^2}{4(4.5 \text{ Btu/h.ft.°F})} = \mathbf{181.0^\circ\text{F}}$$



2-89 Heat is generated uniformly in a spherical radioactive material with specified surface temperature. The mathematical formulation, the variation of temperature in the sphere, and the center temperature are to be determined for steady one-dimensional heat transfer.

Assumptions 1 Heat transfer is steady since there is no indication of any changes with time. 2 Heat transfer is one-dimensional since there is thermal symmetry about the mid point. 3 Thermal conductivity is constant. 4 Heat generation is uniform.

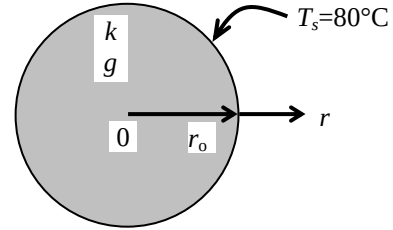
Properties The thermal conductivity is given to be $k = 15 \text{ W/m}\cdot^\circ\text{C}$.

Analysis (a) Noting that heat transfer is steady and one-dimensional in the radial r direction, the mathematical formulation of this problem can be expressed as

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) + \frac{\dot{g}}{k} = 0 \quad \text{with } \dot{g} = \text{constant}$$

and $T(r_0) = T_s = 80^\circ\text{C}$ (specified surface temperature)

$$\frac{dT(0)}{dr} = 0 \quad (\text{thermal symmetry about the mid point})$$



(b) Multiplying both sides of the differential equation by r^2 and rearranging gives

$$\frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = -\frac{\dot{g}}{k} r^2$$

Integrating with respect to r gives

$$r^2 \frac{dT}{dr} = -\frac{\dot{g}}{k} \frac{r^3}{3} + C_1 \quad (a)$$

Applying the boundary condition at the mid point,

$$\text{B.C. at } r = 0: \quad 0 \times \frac{dT(0)}{dr} = -\frac{\dot{g}}{3k} \times 0 + C_1 \rightarrow C_1 = 0$$

Dividing both sides of Eq. (a) by r^2 to bring it to a readily integrable form and integrating,

$$\frac{dT}{dr} = -\frac{\dot{g}}{3k} r$$

$$\text{and } T(r) = -\frac{\dot{g}}{6k} r^2 + C_2 \quad (b)$$

Applying the other boundary condition at $r = r_0$,

$$\text{B. C. at } r = r_0: \quad T_s = -\frac{\dot{g}}{6k} r_0^2 + C_2 \rightarrow C_2 = T_s + \frac{\dot{g}}{6k} r_0^2$$

Substituting this C_2 relation into Eq. (b) and rearranging give

$$T(r) = T_s + \frac{\dot{g}}{6k} (r_0^2 - r^2)$$

which is the desired solution for the temperature distribution in the wire as a function of r .

(c) The temperature at the center of the sphere ($r = 0$) is determined by substituting the known quantities to be

$$T(0) = T_s + \frac{\dot{g}}{6k} (r_0^2 - 0^2) = T_s + \frac{\dot{g} r_0^2}{6k} = 80^\circ\text{C} + \frac{(4 \times 10^7 \text{ W/m}^3)(0.04 \text{ m})^2}{6 \times (15 \text{ W/m}\cdot^\circ\text{C})} = 791^\circ\text{C}$$

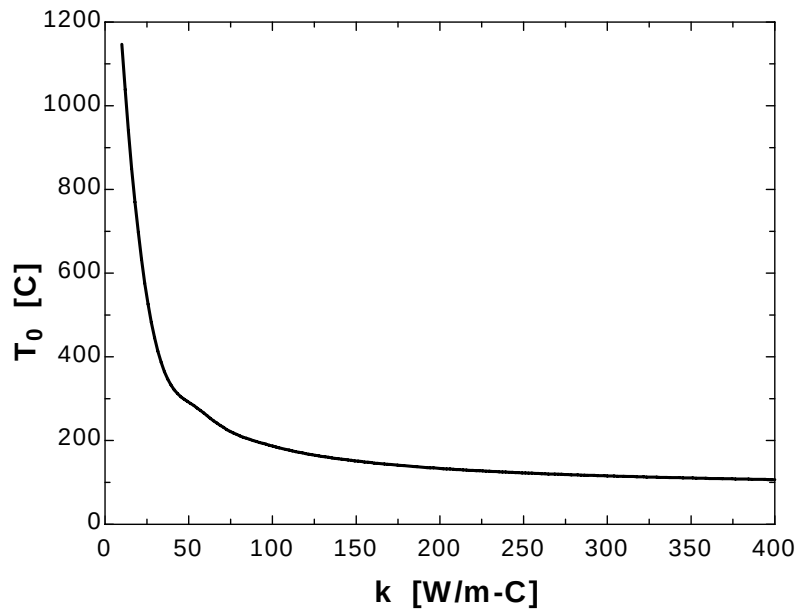
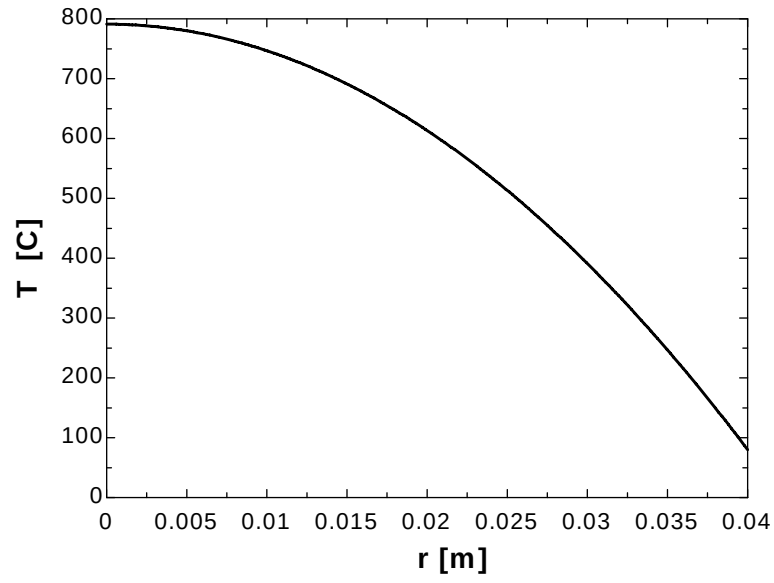
Thus the temperature at center will be about 711°C above the temperature of the outer surface of the sphere.

2-90

"GIVEN" $r_0 = 0.04$ "[m]" $g_{\text{dot}} = 4E7$ "[W/m³]" $T_s = 80$ "[C]" $k = 15$ "[W/m.C], Parameter to be varied"**"ANALYSIS"** $T = T_s + g_{\text{dot}} / (6 * k) * (r_0^2 - r^2)$ "Temperature distribution as a function of r"**"r is the parameter to be varied"** $T_0 = T_s + g_{\text{dot}} / (6 * k) * r_0^2$ "Temperature at the center (r=0)"

r [m]	T [C]
0	791.1
0.002105	789.1
0.004211	783.2
0.006316	773.4
0.008421	759.6
0.01053	741.9
0.01263	720.2
0.01474	694.6
0.01684	665
0.01895	631.6
0.02105	594.1
0.02316	552.8
0.02526	507.5
0.02737	458.2
0.02947	405
0.03158	347.9
0.03368	286.8
0.03579	221.8
0.03789	152.9
0.04	80

k [W/m.C]	T₀ [C]
10	1147
30.53	429.4
51.05	288.9
71.58	229
92.11	195.8
112.6	174.7
133.2	160.1
153.7	149.4
174.2	141.2
194.7	134.8
215.3	129.6
235.8	125.2
256.3	121.6
276.8	118.5
297.4	115.9
317.9	113.6
338.4	111.5
358.9	109.7
379.5	108.1
400	106.7



2-91 A long homogeneous resistance heater wire with specified surface temperature is used to boil water. The temperature of the wire 2 mm from the center is to be determined in steady operation.

Assumptions 1 Heat transfer is steady since there is no indication of any change with time. 2 Heat transfer is one-dimensional since there is thermal symmetry about the center line and no change in the axial direction. 3 Thermal conductivity is constant. 4 Heat generation in the wire is uniform.

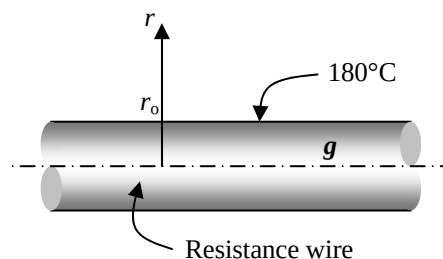
Properties The thermal conductivity is given to be $k = 8 \text{ W/m}\cdot^\circ\text{C}$.

Analysis Noting that heat transfer is steady and one-dimensional in the radial r direction, the mathematical formulation of this problem can be expressed as

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dT}{dr} \right) + \frac{\dot{g}}{k} = 0$$

and $T(r_0) = T_s = 180^\circ\text{C}$ (specified surface temperature)

$$\frac{dT(0)}{dr} = 0 \quad (\text{thermal symmetry about the centerline})$$



Multiplying both sides of the differential equation by r and rearranging gives

$$\frac{d}{dr} \left(r \frac{dT}{dr} \right) = -\frac{\dot{g}}{k} r$$

Integrating with respect to r gives

$$r \frac{dT}{dr} = -\frac{\dot{g}}{k} \frac{r^2}{2} + C_1 \quad (a)$$

It is convenient at this point to apply the boundary condition at the center since it is related to the first derivative of the temperature. It yields

B.C. at $r = 0$: $0 \times \frac{dT(0)}{dr} = -\frac{\dot{g}}{2k} \times 0 + C_1 \rightarrow C_1 = 0$

Dividing both sides of Eq. (a) by r to bring it to a readily integrable form and integrating,

$$\frac{dT}{dr} = -\frac{\dot{g}}{2k} r$$

and $T(r) = -\frac{\dot{g}}{4k} r^2 + C_2 \quad (b)$

Applying the other boundary condition at $r = r_0$,

B. C. at $r = r_0$: $T_s = -\frac{\dot{g}}{4k} r_0^2 + C_2 \rightarrow C_2 = T_s + \frac{\dot{g}}{4k} r_0^2$

Substituting this C_2 relation into Eq. (b) and rearranging give

$$T(r) = T_s + \frac{\dot{g}}{4k} (r_0^2 - r^2)$$

which is the desired solution for the temperature distribution in the wire as a function of r . The temperature 2 mm from the center line ($r = 0.002 \text{ m}$) is determined by substituting the known quantities to be

$$T(0.002 \text{ m}) = T_s + \frac{\dot{g}}{4k} (r_0^2 - r^2) = 180^\circ\text{C} + \frac{5 \times 10^7 \text{ W/m}^3}{4 \times (8 \text{ W/m}\cdot^\circ\text{C})} [(0.005 \text{ m})^2 - (0.002 \text{ m})^2] = 212.8^\circ\text{C}$$

Thus the temperature at that location will be about 33°C above the temperature of the outer surface of the wire.

2-92 Heat is generated in a large plane wall whose one side is insulated while the other side is maintained at a specified temperature. The mathematical formulation, the variation of temperature in the wall, and the temperature of the insulated surface are to be determined for steady one-dimensional heat transfer.

Assumptions 1 Heat transfer is steady since there is no indication of any change with time. 2 Heat transfer is one-dimensional since the wall is large relative to its thickness, and there is thermal symmetry about the center plane. 3 Thermal conductivity is constant. 4 Heat generation varies with location in the x direction.

Properties The thermal conductivity is given to be $k = 30 \text{ W/m}\cdot^\circ\text{C}$.

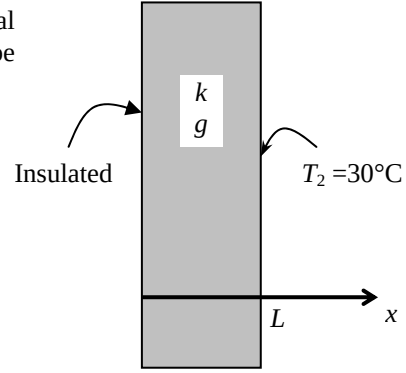
Analysis (a) Noting that heat transfer is steady and one-dimensional in x direction, the mathematical formulation of this problem can be expressed as

$$\frac{d^2 T}{dx^2} + \frac{\dot{g}(x)}{k} = 0$$

where $\dot{g} = \dot{g}_0 e^{-0.5x/L}$ and $\dot{g}_0 = 8 \times 10^6 \text{ W/m}^3$

and $\frac{dT(0)}{dx} = 0$ (insulated surface at $x = 0$)

$T(L) = T_2 = 30^\circ\text{C}$ (specified surface temperature)



(b) Rearranging the differential equation and integrating,

$$\frac{d^2 T}{dx^2} = -\frac{\dot{g}_0}{k} e^{-0.5x/L} \rightarrow \frac{dT}{dx} = -\frac{\dot{g}_0}{k} \frac{e^{-0.5x/L}}{-0.5/L} + C_1 \rightarrow \frac{dT}{dx} = \frac{2\dot{g}_0 L}{k} e^{-0.5x/L} + C_1$$

Integrating one more time,

$$T(x) = \frac{2\dot{g}_0 L}{k} \frac{e^{-0.5x/L}}{-0.5/L} + C_1 x + C_2 \rightarrow T(x) = -\frac{4\dot{g}_0 L^2}{k} e^{-0.5x/L} + C_1 x + C_2 \quad (1)$$

Applying the boundary conditions:

$$\text{B.C. at } x = 0: \quad \frac{dT(0)}{dx} = \frac{2\dot{g}_0 L}{k} e^{-0.5 \times 0/L} + C_1 \rightarrow 0 = \frac{2\dot{g}_0 L}{k} + C_1 \rightarrow C_1 = -\frac{2\dot{g}_0 L}{k}$$

$$\text{B.C. at } x = L: \quad T(L) = T_2 = -\frac{4\dot{g}_0 L^2}{k} e^{-0.5L/L} + C_1 L + C_2 \rightarrow C_2 = T_2 + \frac{4\dot{g}_0 L^2}{k} e^{-0.5} + \frac{2\dot{g}_0 L^2}{k}$$

Substituting the C_1 and C_2 relations into Eq. (1) and rearranging give

$$T(x) = T_2 + \frac{\dot{g}_0 L^2}{k} [4(e^{-0.5} - e^{-0.5x/L}) + (2 - x/L)]$$

which is the desired solution for the temperature distribution in the wall as a function of x .

(c) The temperature at the insulate surface ($x = 0$) is determined by substituting the known quantities to be

$$\begin{aligned} T(0) &= T_2 + \frac{\dot{g}_0 L^2}{k} [4(e^{-0.5} - e^0) + (2 - 0/L)] \\ &= 30^\circ\text{C} + \frac{(8 \times 10^6 \text{ W/m}^3)(0.05 \text{ m})^2}{(30 \text{ W/m}\cdot^\circ\text{C})} [4(e^{-0.5} - 1) + (2 - 0)] = \mathbf{314.1^\circ\text{C}} \end{aligned}$$

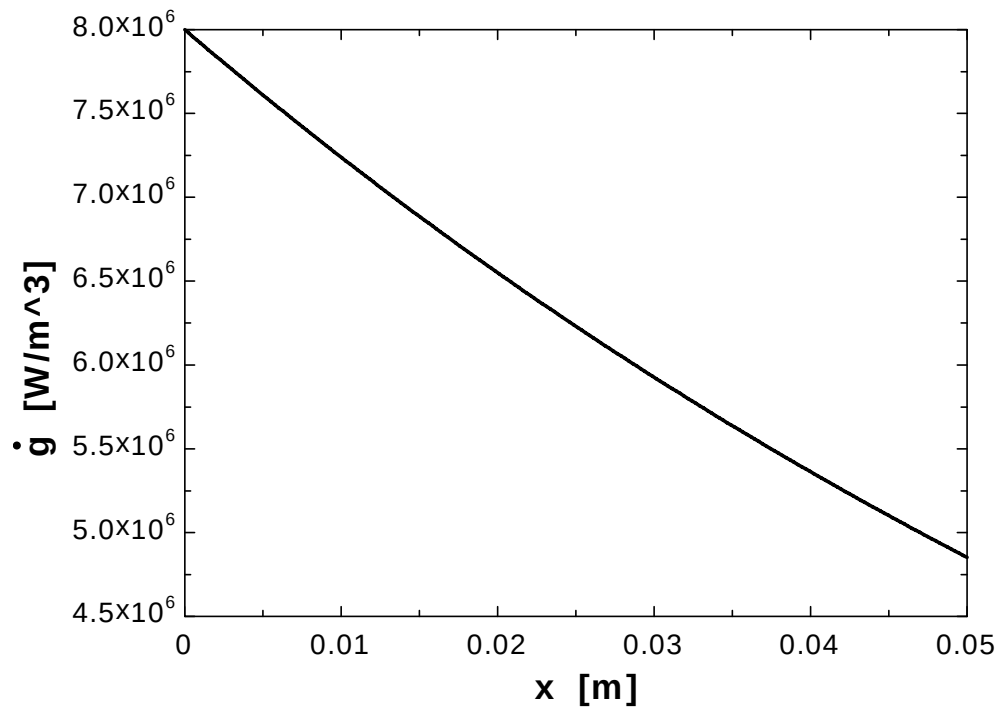
Therefore, there is a temperature difference of almost 300°C between the two sides of the plate.

2-93

"GIVEN" $L=0.05$ "[m]" $T_s=30$ "[C]" $k=30$ "[W/m-C]" $g_{\text{dot}_0}=8E6$ "[W/m³]"**"ANALYSIS"** $g_{\text{dot}}=g_{\text{dot}_0} \cdot \exp((-0.5 \cdot x)/L)$ "Heat generation as a function of x"

"x is the parameter to be varied"

x [m]	g [W/m ³]
0	8.000E+06
0.005	7.610E+06
0.01	7.239E+06
0.015	6.886E+06
0.02	6.550E+06
0.025	6.230E+06
0.03	5.927E+06
0.035	5.638E+06
0.04	5.363E+06
0.045	5.101E+06
0.05	4.852E+06



Variable Thermal Conductivity

2-94C During steady one-dimensional heat conduction in a plane wall, long cylinder, and sphere with constant thermal conductivity and no heat generation, the temperature in only the *plane wall* will vary linearly.

2-95C The thermal conductivity of a medium, in general, varies with temperature.

2-96C During steady one-dimensional heat conduction in a plane wall in which the thermal conductivity varies linearly, the error involved in heat transfer calculation by assuming constant thermal conductivity at the average temperature is (a) *none*.

2-97C No, the temperature variation in a plain wall will not be linear when the thermal conductivity varies with temperature.

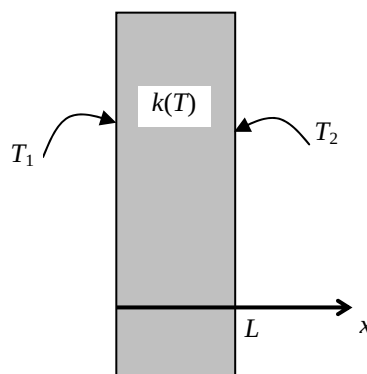
2-98C Yes, when the thermal conductivity of a medium varies linearly with temperature, the average thermal conductivity is always equivalent to the conductivity value at the average temperature.

2-99 A plate with variable conductivity is subjected to specified temperatures on both sides. The rate of heat transfer through the plate is to be determined.

Assumptions 1 Heat transfer is given to be steady and one-dimensional. 2 Thermal conductivity varies quadratically. 3 There is no heat generation.

Properties The thermal conductivity is given to be $k(T) = k_0(1 + \beta T^2)$.

Analysis When the variation of thermal conductivity with temperature $k(T)$ is known, the average value of the thermal conductivity in the temperature range between T_1 and T_2 can be determined from



$$k_{\text{ave}} = \frac{\int_{T_1}^{T_2} k(T) dT}{T_2 - T_1} = \frac{\int_{T_1}^{T_2} k_0(1 + \beta T^2) dT}{T_2 - T_1} = \frac{k_0 \left(T + \frac{\beta}{3} T^3 \right) \Big|_{T_1}^{T_2}}{T_2 - T_1} = \frac{k_0 \left[(T_2 - T_1) + \frac{\beta}{3} (T_2^3 - T_1^3) \right]}{T_2 - T_1}$$

$$= k_0 \left[1 + \frac{\beta}{3} (T_2^2 + T_1 T_2 + T_1^2) \right]$$

This relation is based on the requirement that the rate of heat transfer through a medium with constant average thermal conductivity k_{ave} equals the rate of heat transfer through the same medium with variable conductivity $k(T)$. Then the rate of heat conduction through the plate can be determined to be

$$\dot{Q} = k_{\text{ave}} A \frac{T_1 - T_2}{L} = k_0 \left[1 + \frac{\beta}{3} (T_2^2 + T_1 T_2 + T_1^2) \right] A \frac{T_1 - T_2}{L}$$

Discussion We would obtain the same result if we substituted the given $k(T)$ relation into the second part of Eq. 2-76, and performed the indicated integration.

2-100 A cylindrical shell with variable conductivity is subjected to specified temperatures on both sides. The variation of temperature and the rate of heat transfer through the shell are to be determined.

Assumptions 1 Heat transfer is given to be steady and one-dimensional. 2 Thermal conductivity varies linearly. 3 There is no heat generation.

Properties The thermal conductivity is given to be $k(T) = k_0(1 + \beta T)$.

Solution (a) The rate of heat transfer through the shell is expressed as

$$\dot{Q}_{\text{cylinder}} = 2\pi k_{\text{ave}} L \frac{T_1 - T_2}{\ln(r_2 / r_1)}$$

where L is the length of the cylinder, r_1 is the inner radius, and r_2 is the outer radius, and

$$k_{\text{ave}} = k(T_{\text{ave}}) = k_0 \left(1 + \beta \frac{T_2 + T_1}{2} \right)$$

is the average thermal conductivity.

(b) To determine the temperature distribution in the shell, we begin with the Fourier's law of heat conduction expressed as

$$\dot{Q} = -k(T)A \frac{dT}{dr}$$

where the rate of conduction heat transfer $\dot{Q}$ is constant and the heat conduction area $A = 2\pi rL$ is variable. Separating the variables in the above equation and integrating from $r = r_1$ where $T(r_1) = T_1$ to any r where $T(r) = T$, we get

$$\dot{Q} \int_{r_1}^r \frac{dr}{r} = -2\pi L \int_{T_1}^T k(T) dT$$

Substituting $k(T) = k_0(1 + \beta T)$ and performing the integrations gives

$$\dot{Q} \ln \frac{r}{r_1} = -2\pi L k_0 [(T - T_1) + \beta(T^2 - T_1^2) / 2]$$

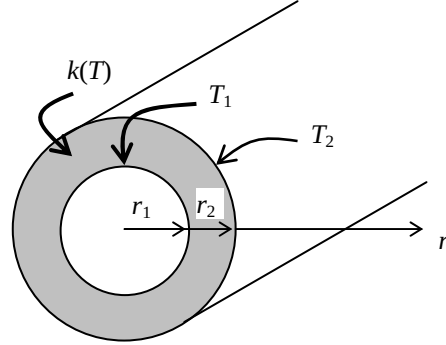
Substituting the $\dot{Q}$ expression from part (a) and rearranging give

$$T^2 + \frac{2}{\beta} T + \frac{2k_{\text{ave}}}{\beta k_0} \frac{\ln(r / r_1)}{\ln(r_2 / r_1)} (T_1 - T_2) - T_1^2 - \frac{2}{\beta} T_1 = 0$$

which is a *quadratic* equation in the unknown temperature T . Using the quadratic formula, the temperature distribution $T(r)$ in the cylindrical shell is determined to be

$$T(r) = -\frac{1}{\beta} \pm \sqrt{\frac{1}{\beta^2} - \frac{2k_{\text{ave}}}{\beta k_0} \frac{\ln(r / r_1)}{\ln(r_2 / r_1)} (T_1 - T_2) + T_1^2 + \frac{2}{\beta} T_1}$$

Discussion The proper sign of the square root term (+ or -) is determined from the requirement that the temperature at any point within the medium must remain between T_1 and T_2 .



2-101 A spherical shell with variable conductivity is subjected to specified temperatures on both sides. The variation of temperature and the rate of heat transfer through the shell are to be determined.

Assumptions 1 Heat transfer is given to be steady and one-dimensional. 2 Thermal conductivity varies linearly. 3 There is no heat generation.

Properties The thermal conductivity is given to be $k(T) = k_0(1 + \beta T)$.

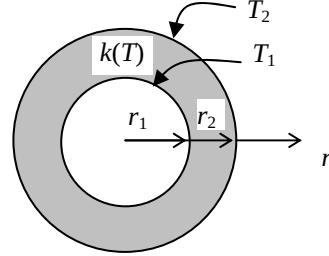
Solution (a) The rate of heat transfer through the shell is expressed as

$$\dot{Q}_{\text{sphere}} = 4\pi k_{\text{ave}} r_1 r_2 \frac{T_1 - T_2}{r_2 - r_1}$$

where r_1 is the inner radius, r_2 is the outer radius, and

$$k_{\text{ave}} = k(T_{\text{ave}}) = k_0 \left(1 + \beta \frac{T_2 + T_1}{2} \right)$$

is the average thermal conductivity.



(b) To determine the temperature distribution in the shell, we begin with the Fourier's law of heat conduction expressed as

$$\dot{Q} = -k(T) A \frac{dT}{dr}$$

where the rate of conduction heat transfer $\dot{Q}$ is constant and the heat conduction area $A = 4\pi r^2$ is variable. Separating the variables in the above equation and integrating from $r = r_1$ where $T(r_1) = T_1$ to any r where $T(r) = T$, we get

$$\dot{Q} \int_{r_1}^r \frac{dr}{r^2} = -4\pi \int_{T_1}^T k(T) dT$$

Substituting $k(T) = k_0(1 + \beta T)$ and performing the integrations gives

$$\dot{Q} \left(\frac{1}{r_1} - \frac{1}{r} \right) = -4\pi k_0 [(T - T_1) + \beta(T^2 - T_1^2) / 2]$$

Substituting the $\dot{Q}$ expression from part (a) and rearranging give

$$T^2 + \frac{2}{\beta} T + \frac{2k_{\text{ave}}}{\beta k_0} \frac{r_2(r - r_1)}{r(r_2 - r_1)} (T_1 - T_2) - T_1^2 - \frac{2}{\beta} T_1 = 0$$

which is a *quadratic* equation in the unknown temperature T . Using the quadratic formula, the temperature distribution $T(r)$ in the cylindrical shell is determined to be

$$T(r) = -\frac{1}{\beta} \pm \sqrt{\frac{1}{\beta^2} - \frac{2k_{\text{ave}}}{\beta k_0} \frac{r_2(r - r_1)}{r(r_2 - r_1)} (T_1 - T_2) + T_1^2 + \frac{2}{\beta} T_1}$$

Discussion The proper sign of the square root term (+ or -) is determined from the requirement that the temperature at any point within the medium must remain between T_1 and T_2 .

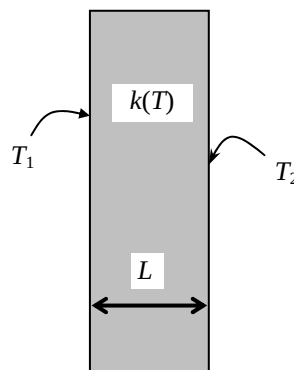
2-102 A plate with variable conductivity is subjected to specified temperatures on both sides. The rate of heat transfer through the plate is to be determined.

Assumptions 1 Heat transfer is given to be steady and one-dimensional. 2 Thermal conductivity varies linearly. 3 There is no heat generation.

Properties The thermal conductivity is given to be $k(T) = k_0(1 + \beta T)$.

Analysis The average thermal conductivity of the medium in this case is simply the conductivity value at the average temperature since the thermal conductivity varies linearly with temperature, and is determined to be

$$\begin{aligned} k_{\text{ave}} &= k(T_{\text{ave}}) = k_0 \left(1 + \beta \frac{T_2 + T_1}{2} \right) \\ &= (25 \text{ W/m} \cdot \text{K}) \left(1 + (8.7 \times 10^{-4} \text{ K}^{-1}) \frac{(500 + 350) \text{ K}}{2} \right) \\ &= 34.24 \text{ W/m} \cdot \text{K} \end{aligned}$$



Then the rate of heat conduction through the plate becomes

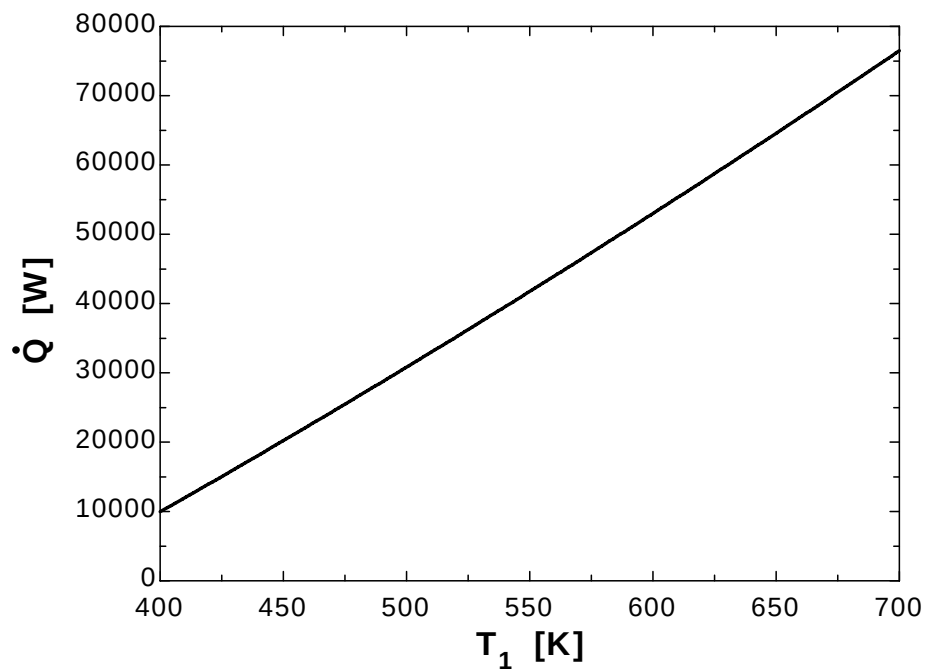
$$\dot{Q} = k_{\text{ave}} A \frac{T_1 - T_2}{L} = (34.24 \text{ W/m} \cdot \text{K})(1.5 \text{ m} \times 0.6 \text{ m}) \frac{(500 - 350) \text{ K}}{0.15 \text{ m}} = \mathbf{30,820 \text{ W}}$$

Discussion We would obtain the same result if we substituted the given $k(T)$ relation into the second part of Eq. 2-76, and performed the indicated integration.

2-103

"GIVEN" $A=1.5 \times 0.6 \text{ [m}^2\text{]}$ $L=0.15 \text{ [m]}$ $T_1=500 \text{ [K]}$, parameter to be varied $T_2=350 \text{ [K]}$ $k_0=25 \text{ [W/m-K]}$ $\beta=8.7 \times 10^{-4} \text{ [1/K]}$ **"ANALYSIS"** $k=k_0(1+\beta T)$ $T=1/2(T_1+T_2)$ $\dot{Q}=kA(T_1-T_2)/L$

$T_1 \text{ [K]}$	$\dot{Q} \text{ [W]}$
400	9947
425	15043
450	20220
475	25479
500	30819
525	36241
550	41745
575	47330
600	52997
625	58745
650	64575
675	70486
700	76479



Special Topic: Review of Differential equations

2-104C We utilize appropriate simplifying assumptions when deriving differential equations to obtain an equation that we can deal with and solve.

2-105C A **variable** is a quantity which may assume various values during a study. A variable whose value can be changed arbitrarily is called an **independent variable** (or argument). A variable whose value depends on the value of other variables and thus cannot be varied independently is called a **dependent variable** (or a function).

2-106C A differential equation may involve more than one dependent or independent variable. For example, the equation $\frac{\partial^2 T(x,t)}{\partial x^2} + \frac{\dot{q}}{k} = \frac{1}{\alpha} \frac{\partial T(x,t)}{\partial t}$ has one dependent (T) and 2 independent variables (x and t). the equation $\frac{\partial^2 T(x,t)}{\partial x^2} + \frac{\partial W(x,t)}{\partial x} = \frac{1}{\alpha} \frac{\partial T(x,t)}{\partial t} + \frac{1}{\alpha} \frac{\partial W(x,t)}{\partial t}$ has 2 dependent (T and W) and 2 independent variables (x and t).

2-107C Geometrically, the **derivative** of a function $y(x)$ at a point represents the *slope* of the tangent line to the graph of the function at that point. The derivative of a function that depends on two or more independent variables with respect to one variable while holding the other variables constant is called the partial derivative. Ordinary and partial derivatives are equivalent for functions that depend on a single independent variable.

2-108C The order of a derivative represents the number of times a function is differentiated, whereas the degree of a derivative represents how many times a derivative is multiplied by itself. For example, y''' is the third order derivative of y , whereas $(y')^3$ is the third degree of the first derivative of y .

2-109C For a function $f(x,y)$, the partial derivative $\partial f / \partial x$ will be equal to the ordinary derivative df / dx when f does not depend on y or this dependence is negligible.

2-110C For a function $f(x)$, the derivative df / dx does not have to be a function of x . The derivative will be a constant when the f is a linear function of x .

2-111C Integration is the inverse of derivation. Derivation increases the order of a derivative by one, integration reduces it by one.

2-112C A differential equation involves derivatives, an algebraic equation does not.

2-113C A differential equation that involves only ordinary derivatives is called an ordinary differential equation, and a differential equation that involves partial derivatives is called a partial differential equation.

2-114C The order of a differential equation is the order of the highest order derivative in the equation.

2-115C A differential equation is said to be **linear** if the dependent variable and all of its derivatives are of the first degree, and their coefficients depend on the independent variable only. In other words, a differential equation is linear if it can be written in a form which does not involve (1) any powers of the dependent variable or its derivatives such as y^3 or $(y')^2$, (2) any products of the dependent variable or its derivatives such as yy' or $y'y''$, and (3) any other nonlinear functions of the dependent variable such as $\sin y$ or e^y . Otherwise, it is **nonlinear**.

2-116C A linear homogeneous differential equation of order n is expressed in the most general form as

$$y^{(n)} + f_1(x)y^{(n-1)} + \cdots + f_{n-1}(x)y' + f_n(x)y = 0$$

Each term in a linear homogeneous equation contains the dependent variable or one of its derivatives after the equation is cleared of any common factors. The equation $y'' - 4x^2y = 0$ is linear and homogeneous since each term is linear in y , and contains the dependent variable or one of its derivatives.

2-117C A differential equation is said to have **constant coefficients** if the coefficients of all the terms which involve the dependent variable or its derivatives are constants. If, after cleared of any common factors, any of the terms with the dependent variable or its derivatives involve the independent variable as a coefficient, that equation is said to have **variable coefficients**. The equation $y'' - 4x^2y = 0$ has variable coefficients whereas the equation $y'' - 4y = 0$ has constant coefficients.

2-118C A linear differential equation that involves a single term with the derivatives can be solved by direct integration.

2-119C The general solution of a 3rd order linear and homogeneous differential equation will involve 3 arbitrary constants.

Review Problems

2-120 A small hot metal object is allowed to cool in an environment by convection. The differential equation that describes the variation of temperature of the ball with time is to be derived.

Assumptions 1 The temperature of the metal object changes uniformly with time during cooling so that $T = T(t)$. 2 The density, specific heat, and thermal conductivity of the body are constant. 3 There is no heat generation.

Analysis Consider a body of arbitrary shape of mass m , volume V , surface area A , density ρ , and specific heat C_p initially at a uniform temperature T_i . At time $t = 0$, the body is placed into a medium at temperature T_∞ , and heat transfer takes place between the body and its environment with a heat transfer coefficient h .

During a differential time interval dt , the temperature of the body rises by a differential amount dT . Noting that the temperature changes with time only, an energy balance of the solid for the time interval dt can be expressed as

$$\left(\begin{array}{c} \text{Heat transfer from the body} \\ \text{during } dt \end{array} \right) = \left(\begin{array}{c} \text{The decrease in the energy} \\ \text{of the body during } dt \end{array} \right)$$

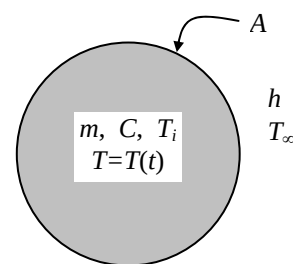
or
$$hA_s(T - T_\infty)dt = mC_p(-dT)$$

Noting that $m = \rho V$ and $dT = d(T - T_\infty)$ since $T_\infty = \text{constant}$, the equation

above can be rearranged as

$$\frac{d(T - T_\infty)}{T - T_\infty} = -\frac{hA_s}{\rho VC_p} dt$$

which is the desired differential equation.

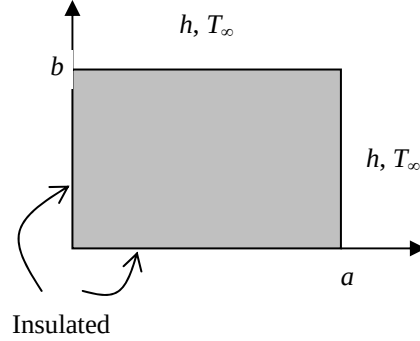


2-121 A long rectangular bar is initially at a uniform temperature of T_i . The surfaces of the bar at $x = 0$ and $y = 0$ are insulated while heat is lost from the other two surfaces by convection. The mathematical formulation of this heat conduction problem is to be expressed for transient two-dimensional heat transfer with no heat generation.

Assumptions 1 Heat transfer is transient and two-dimensional. 2 Thermal conductivity is constant. 3 There is no heat generation.

Analysis The differential equation and the boundary conditions for this heat conduction problem can be expressed as

$$\begin{aligned}\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} &= \frac{1}{\alpha} \frac{\partial T}{\partial t} \\ \frac{\partial T(x, 0, t)}{\partial x} &= 0 \\ \frac{\partial T(0, y, t)}{\partial x} &= 0 \\ -k \frac{\partial T(a, y, t)}{\partial x} &= h[T(a, y, t) - T_\infty] \\ -k \frac{\partial T(x, b, t)}{\partial y} &= h[T(x, b, t) - T_\infty] \\ T(x, y, 0) &= T_i\end{aligned}$$

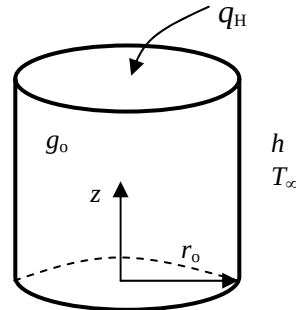


2-122 Heat is generated at a constant rate in a short cylinder. Heat is lost from the cylindrical surface at $r = r_0$ by convection to the surrounding medium at temperature T_∞ with a heat transfer coefficient of h . The bottom surface of the cylinder at $r = 0$ is insulated, the top surface at $z = H$ is subjected to uniform heat flux $\dot{q}_h$, and the cylindrical surface at $r = r_0$ is subjected to convection. The mathematical formulation of this problem is to be expressed for steady two-dimensional heat transfer.

Assumptions 1 Heat transfer is given to be steady and two-dimensional. 2 Thermal conductivity is constant. 3 Heat is generated uniformly.

Analysis The differential equation and the boundary conditions for this heat conduction problem can be expressed as

$$\begin{aligned}\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{k} &= 0 \\ \frac{\partial T(r, 0)}{\partial z} &= 0 \\ k \frac{\partial T(r, h)}{\partial z} &= \dot{q}_H \\ \frac{\partial T(0, z)}{\partial r} &= 0 \\ -k \frac{\partial T(r_0, z)}{\partial r} &= h[T(r_0, z) - T_\infty]\end{aligned}$$



2-123E A large plane wall is subjected to a specified temperature on the left (inner) surface and solar radiation and heat loss by radiation to space on the right (outer) surface. The temperature of the right surface of the wall and the rate of heat transfer are to be determined when steady operating conditions are reached.

Assumptions 1 Steady operating conditions are reached. 2 Heat transfer is one-dimensional since the wall is large relative to its thickness, and the thermal conditions on both sides of the wall are uniform. 3 Thermal properties are constant. 4 There is no heat generation in the wall.

Properties The properties of the plate are given to be $k = 1.2$ Btu/h·ft·°F and $\varepsilon = 0.80$, and $\alpha_s = 0.45$.

Analysis In steady operation, heat conduction through the wall must be equal to net heat transfer from the outer surface. Therefore, taking the outer surface temperature of the plate to be T_2 (absolute, in R),

$$kA_s \frac{T_1 - T_2}{L} = \varepsilon \sigma A_s T_2^4 - \alpha_s A_s \dot{q}_{\text{solar}}$$

Canceling the area A and substituting the known quantities,

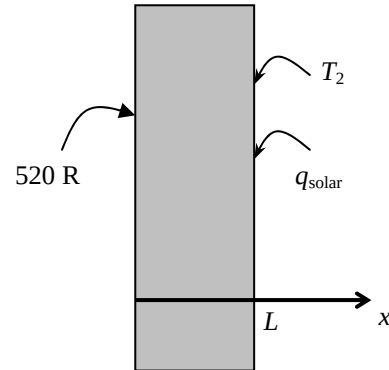
$$(1.2 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}) \frac{(520 \text{ R}) - T_2}{0.5 \text{ ft}} = 0.8(0.1714 \times 10^{-8} \text{ Btu/h} \cdot \text{ft}^2 \cdot \text{R}^4) T_2^4 - 0.45(300 \text{ Btu/h} \cdot \text{ft}^2)$$

Solving for T_2 gives the outer surface temperature to be $T_2 = 530.9 \text{ R}$

Then the rate of heat transfer through the wall becomes

$$\dot{q} = k \frac{T_1 - T_2}{L} = (1.2 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}) \frac{(520 - 530.9) \text{ R}}{0.5 \text{ ft}} = -26.2 \text{ Btu/h} \cdot \text{ft}^2 \quad (\text{per unit area})$$

Discussion The negative sign indicates that the direction of heat transfer is from the outside to the inside. Therefore, the structure is gaining heat.



2-124E A large plane wall is subjected to a specified temperature on the left (inner) surface and heat loss by radiation to space on the right (outer) surface. The temperature of the right surface of the wall and the rate of heat transfer are to be determined when steady operating conditions are reached.

Assumptions 1 Steady operating conditions are reached. 2 Heat transfer is one-dimensional since the wall is large relative to its thickness, and the thermal conditions on both sides of the wall are uniform. 3 Thermal properties are constant. 4 There is no heat generation in the wall.

Properties The properties of the plate are given to be $k = 1.2$ Btu/h·ft·°F and $\varepsilon = 0.80$.

Analysis In steady operation, heat conduction through the wall must be equal to net heat transfer from the outer surface. Therefore, taking the outer surface temperature of the plate to be T_2 (absolute, in R),

$$kA_s \frac{T_1 - T_2}{L} = \varepsilon \sigma A_s T_2^4$$

Canceling the area A and substituting the known quantities,

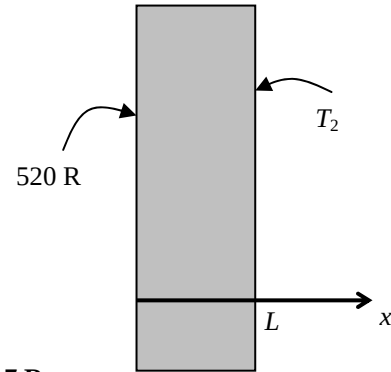
$$(1.2 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}) \frac{(520 \text{ R}) - T_2}{0.5 \text{ ft}} = 0.8(0.1714 \times 10^{-8} \text{ Btu/h} \cdot \text{ft}^2 \cdot \text{R}^4) T_2^4$$

Solving for T_2 gives the outer surface temperature to be $T_2 = 487.7 \text{ R}$

Then the rate of heat transfer through the wall becomes

$$\dot{q} = k \frac{T_1 - T_2}{L} = (1.2 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{F}) \frac{(520 - 487.7) \text{ R}}{0.5 \text{ ft}} = 77.5 \text{ Btu/h} \cdot \text{ft}^2 \quad (\text{per unit area})$$

Discussion The positive sign indicates that the direction of heat transfer is from the inside to the outside. Therefore, the structure is losing heat as expected.



2-125 A steam pipe is subjected to convection on both the inner and outer surfaces. The mathematical formulation of the problem and expressions for the variation of temperature in the pipe and on the outer surface temperature are to be obtained for steady one-dimensional heat transfer.

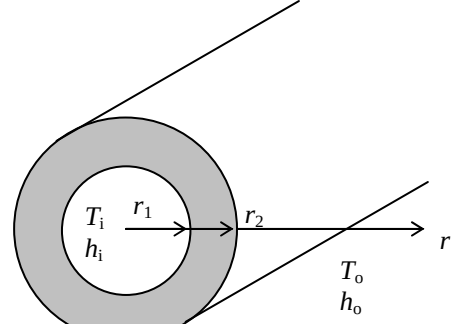
Assumptions **1** Heat conduction is steady and one-dimensional since the pipe is long relative to its thickness, and there is thermal symmetry about the center line. **2** Thermal conductivity is constant. **3** There is no heat generation in the pipe.

Analysis (a) Noting that heat transfer is steady and one-dimensional in the radial r direction, the mathematical formulation of this problem can be expressed as

$$\frac{d}{dr} \left(r \frac{dT}{dr} \right) = 0$$

$$\text{and} \quad -k \frac{dT(r_1)}{dr} = h_i [T_i - T(r_1)]$$

$$-k \frac{dT(r_2)}{dr} = h_o [T(r_2) - T_o]$$



(b) Integrating the differential equation once with respect to r gives

$$r \frac{dT}{dr} = C_1$$

Dividing both sides of the equation above by r to bring it to a readily integrable form and then integrating,

$$\frac{dT}{dr} = \frac{C_1}{r}$$

$$T(r) = C_1 \ln r + C_2$$

where C_1 and C_2 are arbitrary constants. Applying the boundary conditions give

$$r = r_1: \quad -k \frac{C_1}{r_1} = h_i [T_i - (C_1 \ln r_1 + C_2)]$$

$$r = r_2: \quad -k \frac{C_1}{r_2} = h_o [(C_1 \ln r_2 + C_2) - T_o]$$

Solving for C_1 and C_2 simultaneously gives

$$C_1 = \frac{T_o - T_i}{\ln \frac{r_2}{r_1} + \frac{k}{h_i r_1} + \frac{k}{h_o r_2}} \quad \text{and} \quad C_2 = T_i - C_1 \left(\ln r_1 - \frac{k}{h_i r_1} \right) = T_i - \frac{T_o - T_i}{\ln \frac{r_2}{r_1} + \frac{k}{h_i r_1} + \frac{k}{h_o r_2}} \left(\ln r_1 - \frac{k}{h_i r_1} \right)$$

Substituting C_1 and C_2 into the general solution and simplifying, we get the variation of temperature to be

$$T(r) = C_1 \ln r + T_i - C_1 \left(\ln r_1 - \frac{k}{h_i r_1} \right) = T_i + \frac{\ln \frac{r}{r_1} + \frac{k}{h_i r_1}}{\ln \frac{r_2}{r_1} + \frac{k}{h_i r_1} + \frac{k}{h_o r_2}}$$

(c) The outer surface temperature is determined by simply replacing r in the relation above by r_2 . We get

$$T(r_2) = T_i + \frac{\ln \frac{r_2}{r_1} + \frac{k}{h_i r_1}}{\ln \frac{r_2}{r_1} + \frac{k}{h_i r_1} + \frac{k}{h_o r_2}}$$

2-126 A spherical liquid nitrogen container is subjected to specified temperature on the inner surface and convection on the outer surface. The mathematical formulation, the variation of temperature, and the rate of evaporation of nitrogen are to be determined for steady one-dimensional heat transfer.

Assumptions 1 Heat conduction is steady and one-dimensional since there is no change with time and there is thermal symmetry about the midpoint. **2** Thermal conductivity is constant. **3** There is no heat generation.

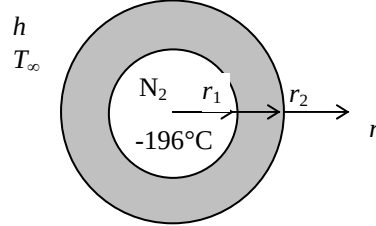
Properties The thermal conductivity of the tank is given to be $k = 18 \text{ W/m} \cdot ^\circ\text{C}$. Also, $h_{fg} = 198 \text{ kJ/kg}$ for nitrogen.

Analysis (a) Noting that heat transfer is one-dimensional in the radial r direction, the mathematical formulation of this problem can be expressed as

$$\frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = 0$$

and $T(r_1) = T_1 = -196^\circ\text{C}$

$$-k \frac{dT(r_2)}{dr} = h[T(r_2) - T_\infty]$$



(b) Integrating the differential equation once with respect to r gives

$$r^2 \frac{dT}{dr} = C_1$$

Dividing both sides of the equation above by r to bring it to a readily integrable form and then integrating,

$$\frac{dT}{dr} = \frac{C_1}{r^2} \rightarrow T(r) = -\frac{C_1}{r} + C_2$$

where C_1 and C_2 are arbitrary constants. Applying the boundary conditions give

$$r = r_1: \quad T(r_1) = -\frac{C_1}{r_1} + C_2 = T_1$$

$$r = r_2: \quad -k \frac{C_1}{r_2^2} = h \left(-\frac{C_1}{r_2} + C_2 - T_\infty \right)$$

Solving for C_1 and C_2 simultaneously gives

$$C_1 = \frac{r_2(T_1 - T_\infty)}{1 - \frac{r_2}{r_1} - \frac{k}{hr_2}} \quad \text{and} \quad C_2 = T_1 + \frac{C_1}{r_1} = T_1 + \frac{T_1 - T_\infty}{1 - \frac{r_2}{r_1} - \frac{k}{hr_2}} \frac{r_2}{r_1}$$

Substituting C_1 and C_2 into the general solution, the variation of temperature is determined to be

$$\begin{aligned} T(r) &= -\frac{C_1}{r} + T_1 + \frac{C_1}{r_1} = C_1 \left(\frac{1}{r_1} - \frac{1}{r} \right) + T_1 = \frac{T_1 - T_\infty}{1 - \frac{r_2}{r_1} - \frac{k}{hr_2}} \left(\frac{r_2}{r_1} - \frac{r_2}{r} \right) + T_1 \\ &= \frac{(-196 - 20)^\circ\text{C}}{1 - \frac{2.1}{2} - \frac{18 \text{ W/m} \cdot ^\circ\text{C}}{(25 \text{ W/m}^2 \cdot ^\circ\text{C})(2.1 \text{ m})}} \left(\frac{2.1}{2} - \frac{2.1}{r} \right) + (-196)^\circ\text{C} = 549.8(1.05 - 2.1/r) - 196 \end{aligned}$$

(c) The rate of heat transfer through the wall and the rate of evaporation of nitrogen are determined from

$$\begin{aligned} \dot{Q} &= -kA \frac{dT}{dr} = -k(4\pi r^2) \frac{C_1}{r^2} = -4\pi k C_1 = -4\pi k \frac{r_2(T_1 - T_\infty)}{1 - \frac{r_2}{r_1} - \frac{k}{hr_2}} \\ &= -4\pi(18 \text{ W/m} \cdot ^\circ\text{C}) \frac{(2.1 \text{ m})(-196 - 20)^\circ\text{C}}{1 - \frac{2.1}{2} - \frac{18 \text{ W/m} \cdot ^\circ\text{C}}{(25 \text{ W/m}^2 \cdot ^\circ\text{C})(2.1 \text{ m})}} = -261,200 \text{ W (to the tank since negative)} \\ \dot{m} &= \frac{\dot{Q}}{h_{fg}} = \frac{261,200 \text{ J/s}}{198,000 \text{ J/kg}} = \mathbf{1.32 \text{ kg/s}} \end{aligned}$$

2-127 A spherical liquid oxygen container is subjected to specified temperature on the inner surface and convection on the outer surface. The mathematical formulation, the variation of temperature, and the rate of evaporation of oxygen are to be determined for steady one-dimensional heat transfer.

Assumptions 1 Heat conduction is steady and one-dimensional since there is no change with time and there is thermal symmetry about the midpoint. **2** Thermal conductivity is constant. **3** There is no heat generation.

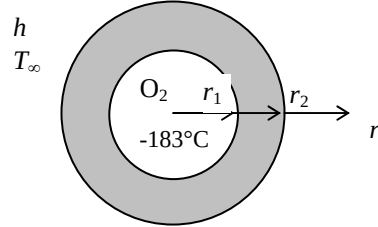
Properties The thermal conductivity of the tank is given to be $k = 18 \text{ W/m}\cdot^\circ\text{C}$. Also, $h_{fg} = 213 \text{ kJ/kg}$ for oxygen.

Analysis (a) Noting that heat transfer is one-dimensional in the radial r direction, the mathematical formulation of this problem can be expressed as

$$\frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = 0$$

and $T(r_1) = T_1 = -183^\circ\text{C}$

$$-k \frac{dT(r_2)}{dr} = h[T(r_2) - T_\infty]$$



(b) Integrating the differential equation once with respect to r gives

$$r^2 \frac{dT}{dr} = C_1$$

Dividing both sides of the equation above by r to bring it to a readily integrable form and then integrating,

$$\frac{dT}{dr} = \frac{C_1}{r^2} \rightarrow T(r) = -\frac{C_1}{r} + C_2$$

where C_1 and C_2 are arbitrary constants. Applying the boundary conditions give

$$r = r_1: \quad T(r_1) = -\frac{C_1}{r_1} + C_2 = T_1$$

$$r = r_2: \quad -k \frac{C_1}{r_2^2} = h \left(-\frac{C_1}{r_2} + C_2 - T_\infty \right)$$

Solving for C_1 and C_2 simultaneously gives

$$C_1 = \frac{r_2(T_1 - T_\infty)}{1 - \frac{r_2}{r_1} - \frac{k}{hr_2}} \quad \text{and} \quad C_2 = T_1 + \frac{C_1}{r_1} = T_1 + \frac{T_1 - T_\infty}{1 - \frac{r_2}{r_1} - \frac{k}{hr_2}} \frac{r_2}{r_1}$$

Substituting C_1 and C_2 into the general solution, the variation of temperature is determined to be

$$\begin{aligned} T(r) &= -\frac{C_1}{r} + T_1 + \frac{C_1}{r_1} = C_1 \left(\frac{1}{r_1} - \frac{1}{r} \right) + T_1 = \frac{T_1 - T_\infty}{1 - \frac{r_2}{r_1} - \frac{k}{hr_2}} \left(\frac{r_2}{r_1} - \frac{r_2}{r} \right) + T_1 \\ &= \frac{(-183 - 20)^\circ\text{C}}{1 - \frac{2.1}{2} - \frac{18 \text{ W/m}\cdot^\circ\text{C}}{(25 \text{ W/m}^2\cdot^\circ\text{C})(2.1 \text{ m})}} \left(\frac{2.1}{2} - \frac{2.1}{r} \right) + (-183)^\circ\text{C} = 516.7(1.05 - 2.1/r) - 183 \end{aligned}$$

(c) The rate of heat transfer through the wall and the rate of evaporation of nitrogen are determined from

$$\begin{aligned} \dot{Q} &= -kA \frac{dT}{dr} = -k(4\pi r^2) \frac{C_1}{r^2} = -4\pi k C_1 = -4\pi k \frac{r_2(T_1 - T_\infty)}{1 - \frac{r_2}{r_1} - \frac{k}{hr_2}} \\ &= -4\pi(18 \text{ W/m}\cdot^\circ\text{C}) \frac{(2.1 \text{ m})(-183 - 20)^\circ\text{C}}{1 - \frac{2.1}{2} - \frac{18 \text{ W/m}\cdot^\circ\text{C}}{(25 \text{ W/m}^2\cdot^\circ\text{C})(2.1 \text{ m})}} = -245,450 \text{ W (to the tank since negative)} \\ \dot{m} &= \frac{\dot{Q}}{h_{fg}} = \frac{245,450 \text{ J/s}}{213,000 \text{ J/kg}} = 1.15 \text{ kg/s} \end{aligned}$$

2-128 A large plane wall is subjected to convection, radiation, and specified temperature on the right surface and no conditions on the left surface. The mathematical formulation, the variation of temperature in the wall, and the left surface temperature are to be determined for steady one-dimensional heat transfer.

Assumptions 1 Heat conduction is steady and one-dimensional since the wall is large relative to its thickness, and the thermal conditions on both sides of the wall are uniform. 2 Thermal conductivity is constant. 3 There is no heat generation in the wall.

Properties The thermal conductivity and emissivity are given to be $k = 8.4 \text{ W/m}\cdot^\circ\text{C}$ and $\varepsilon = 0.7$.

Analysis (a) Taking the direction normal to the surface of the wall to be the x direction with $x = 0$ at the left surface, and the mathematical formulation of this problem can be expressed as

$$\frac{d^2T}{dx^2} = 0$$

$$\text{and} \quad -k \frac{dT(L)}{dx} = h[T(L) - T_\infty] + \varepsilon\sigma[T(L)^4 - T_{\text{surr}}^4] = h[T_2 - T_\infty] + \varepsilon\sigma[(T_2 + 273)^4 - T_{\text{surr}}^4]$$

$$T(L) = T_2 = 45^\circ\text{C}$$

(b) Integrating the differential equation twice with respect to x yields

$$\frac{dT}{dx} = C_1$$

$$T(x) = C_1x + C_2$$

where C_1 and C_2 are arbitrary constants. Applying the boundary conditions give

$$\begin{aligned} \text{Convection at } x = L: \quad & -kC_1 = h[T_2 - T_\infty] + \varepsilon\sigma[(T_2 + 460)^4 - T_{\text{surr}}^4] \\ \rightarrow C_1 = & -\{h[T_2 - T_\infty] + \varepsilon\sigma[(T_2 + 460)^4 - T_{\text{surr}}^4]\} / k \end{aligned}$$

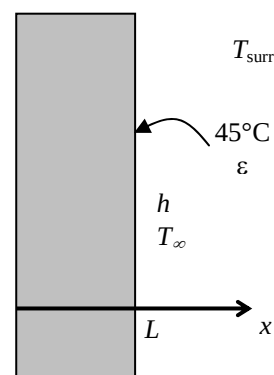
$$\text{Temperature at } x = L: \quad T(L) = C_1 \times L + C_2 = T_2 \rightarrow C_2 = T_2 - C_1L$$

Substituting C_1 and C_2 into the general solution, the variation of temperature is determined to be

$$\begin{aligned} T(x) &= C_1x + (T_2 - C_1L) = T_2 - (L - x)C_1 = T_2 + \frac{h[T_2 - T_\infty] + \varepsilon\sigma[(T_2 + 273)^4 - T_{\text{surr}}^4]}{k}(L - x) \\ &= 45^\circ\text{C} + \frac{(14 \text{ W/m}^2 \cdot ^\circ\text{C})(45 - 25)^\circ\text{C} + 0.7(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)[(318 \text{ K})^4 - (290 \text{ K})^4]}{8.4 \text{ W/m} \cdot ^\circ\text{C}}(0.4 - x) \text{ m} \\ &= 45 + 48.23(0.4 - x) \end{aligned}$$

(c) The temperature at $x = 0$ (the left surface of the wall) is

$$T(0) = 45 + 48.23(0.4 - 0) = \mathbf{64.3^\circ\text{C}}$$



2-129 The base plate of an iron is subjected to specified heat flux on the left surface and convection and radiation on the right surface. The mathematical formulation, and an expression for the outer surface temperature and its value are to be determined for steady one-dimensional heat transfer.

Assumptions 1 Heat conduction is steady and one-dimensional. 2 Thermal conductivity is constant. 3 There is no heat generation. 4 Heat loss through the upper part of the iron is negligible.

Properties The thermal conductivity and emissivity are given to be $k = 2.3 \text{ W/m}\cdot^\circ\text{C}$ and $\varepsilon = 0.7$.

Analysis (a) Noting that the upper part of the iron is well insulated and thus the entire heat generated in the resistance wires is transferred to the base plate, the heat flux through the inner surface is determined to be

$$\dot{q}_0 = \frac{\dot{Q}_0}{A_{\text{base}}} = \frac{1000 \text{ W}}{150 \times 10^{-4} \text{ m}^2} = 66,667 \text{ W/m}^2$$

Taking the direction normal to the surface of the wall to be the x direction with $x = 0$ at the left surface, the mathematical formulation of this problem can be expressed as

$$\frac{d^2T}{dx^2} = 0$$

$$\text{and} \quad -k \frac{dT(0)}{dx} = \dot{q}_0 = 66,667 \text{ W/m}^2$$

$$-k \frac{dT(L)}{dx} = h[T(L) - T_\infty] + \varepsilon\sigma[T(L)^4 - T_{\text{surr}}^4] = h[T_2 - T_\infty] + \varepsilon\sigma[(T_2 + 273)^4 - T_{\text{surr}}^4]$$

(b) Integrating the differential equation twice with respect to x yields

$$\frac{dT}{dx} = C_1$$

$$T(x) = C_1x + C_2$$

where C_1 and C_2 are arbitrary constants. Applying the boundary conditions give

$$x = 0: \quad -kC_1 = \dot{q}_0 \rightarrow C_1 = -\frac{\dot{q}_0}{k}$$

$$x = L: \quad -kC_1 = h[T_2 - T_\infty] + \varepsilon\sigma[(T_2 + 273)^4 - T_{\text{surr}}^4]$$

Eliminating the constant C_1 from the two relations above gives the following expression for the outer surface temperature T_2 ,

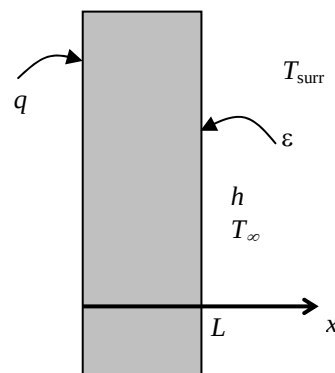
$$h(T_2 - T_\infty) + \varepsilon\sigma[(T_2 + 273)^4 - T_{\text{surr}}^4] = \dot{q}_0$$

(c) Substituting the known quantities into the implicit relation above gives

$$(30 \text{ W/m}^2 \cdot ^\circ\text{C})(T_2 - 22) + 0.7(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)[(T_2 + 273)^4 - 290^4] = 66,667 \text{ W/m}^2$$

Using an equation solver (or a trial and error approach), the outer surface temperature is determined from the relation above to be

$$T_2 = 758^\circ\text{C}$$



2-130 The base plate of an iron is subjected to specified heat flux on the left surface and convection and radiation on the right surface. The mathematical formulation, and an expression for the outer surface temperature and its value are to be determined for steady one-dimensional heat transfer.

Assumptions 1 Heat conduction is steady and one-dimensional. 2 Thermal conductivity is constant. 3 There is no heat generation. 4 Heat loss through the upper part of the iron is negligible.

Properties The thermal conductivity and emissivity are given to be $k = 2.3 \text{ W/m}\cdot^\circ\text{C}$ and $\varepsilon = 0.7$.

Analysis (a) Noting that the upper part of the iron is well insulated and thus the entire heat generated in the resistance wires is transferred to the base plate, the heat flux through the inner surface is determined to be

$$\dot{q}_0 = \frac{\dot{Q}_0}{A_{\text{base}}} = \frac{1500 \text{ W}}{150 \times 10^{-4} \text{ m}^2} = 100,000 \text{ W/m}^2$$

Taking the direction normal to the surface of the wall to be the x direction with $x = 0$ at the left surface, the mathematical formulation of this problem can be expressed as

$$\frac{d^2 T}{dx^2} = 0$$

$$\text{and} \quad -k \frac{dT(0)}{dx} = \dot{q}_0 = 80,000 \text{ W/m}^2$$

$$-k \frac{dT(L)}{dx} = h[T(L) - T_\infty] + \varepsilon \sigma [T(L)^4 - T_{\text{surr}}^4] = h[T_2 - T_\infty] + \varepsilon \sigma [(T_2 + 273)^4 - T_{\text{surr}}^4]$$

(b) Integrating the differential equation twice with respect to x yields

$$\frac{dT}{dx} = C_1$$

$$T(x) = C_1 x + C_2$$

where C_1 and C_2 are arbitrary constants. Applying the boundary conditions give

$$x = 0: \quad -kC_1 = \dot{q}_0 \rightarrow C_1 = -\frac{\dot{q}_0}{k}$$

$$x = L: \quad -kC_1 = h[T_2 - T_\infty] + \varepsilon \sigma [(T_2 + 273)^4 - T_{\text{surr}}^4]$$

Eliminating the constant C_1 from the two relations above gives the following expression for the outer surface temperature T_2 ,

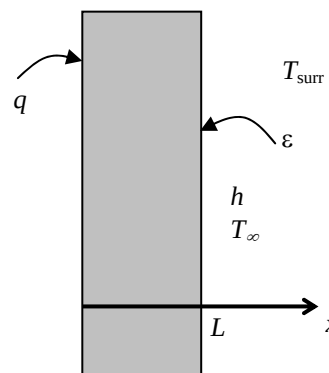
$$h(T_2 - T_\infty) + \varepsilon \sigma [(T_2 + 273)^4 - T_{\text{surr}}^4] = \dot{q}_0$$

(c) Substituting the known quantities into the implicit relation above gives

$$(30 \text{ W/m}^2 \cdot ^\circ\text{C})(T_2 - 22) + 0.7(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)[(T_2 + 273)^4 - 290^4] = 100,000 \text{ W/m}^2$$

Using an equation solver (or a trial and error approach), the outer surface temperature is determined from the relation above to be

$$T_2 = 895.8^\circ\text{C}$$



2-131E The concrete slab roof of a house is subjected to specified temperature at the bottom surface and convection and radiation at the top surface. The temperature of the top surface of the roof and the rate of heat transfer are to be determined when steady operating conditions are reached.

Assumptions 1 Steady operating conditions are reached. 2 Heat transfer is one-dimensional since the roof area is large relative to its thickness, and the thermal conditions on both sides of the roof are uniform. 3 Thermal properties are constant. 4 There is no heat generation in the wall.

Properties The thermal conductivity and emissivity are given to be $k = 1.1 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$ and $\varepsilon = 0.9$.

Analysis In steady operation, heat conduction through the roof must be equal to net heat transfer from the outer surface. Therefore, taking the outer surface temperature of the roof to be T_2 (in $^\circ\text{F}$),

$$kA \frac{T_1 - T_2}{L} = hA(T_2 - T_\infty) + \varepsilon A \sigma [(T_2 + 460)^4 - T_{\text{sky}}^4]$$

Canceling the area A and substituting the known quantities,

$$(1.1 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}) \frac{(62 - T_2)^\circ\text{F}}{0.8 \text{ ft}} = (3.2 \text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F})(T_2 - 50)^\circ\text{F} + 0.8(0.1714 \times 10^{-8} \text{ Btu/h}\cdot\text{ft}^2\cdot\text{R}^4)[(T_2 + 460)^4 - 310^4] \text{ R}^4$$

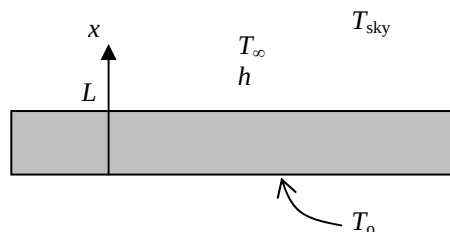
Using an equation solver (or the trial and error method), the outer surface temperature is determined to be

$$T_2 = 38^\circ\text{F}$$

Then the rate of heat transfer through the roof becomes

$$\dot{Q} = kA \frac{T_1 - T_2}{L} = (1.1 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F})(25 \times 35 \text{ ft}^2) \frac{(62 - 38)^\circ\text{F}}{0.8 \text{ ft}} = 28,875 \text{ Btu/h}$$

Discussion The positive sign indicates that the direction of heat transfer is from the inside to the outside. Therefore, the house is losing heat as expected.



2-132 The surface and interface temperatures of a resistance wire covered with a plastic layer are to be determined.

Assumptions 1 Heat transfer is steady since there is no change with time. 2 Heat transfer is one-dimensional since this two-layer heat transfer problem possesses symmetry about the center line and involves no change in the axial direction, and thus $T = T(r)$. 3 Thermal conductivities are constant. 4 Heat generation in the wire is uniform.

Properties It is given that $k_{\text{wire}} = 15 \text{ W/m}\cdot^\circ\text{C}$ and $k_{\text{plastic}} = 1.2 \text{ W/m}\cdot^\circ\text{C}$.

Analysis Letting T_i denote the unknown interface temperature, the mathematical formulation of the heat transfer problem in the wire can be expressed as

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dT}{dr} \right) + \frac{\dot{g}}{k} = 0$$

$$\text{with } T(r_1) = T_i \text{ and } \frac{dT(0)}{dr} = 0$$

Multiplying both sides of the differential equation by r , rearranging, and integrating give

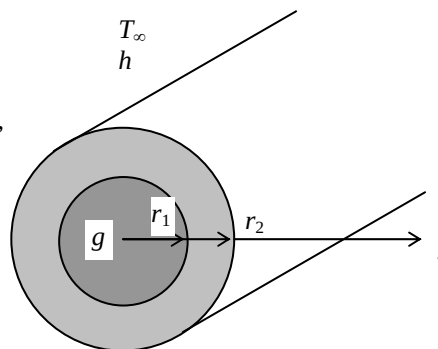
$$\frac{d}{dr} \left(r \frac{dT}{dr} \right) = -\frac{\dot{g}}{k} r \rightarrow r \frac{dT}{dr} = -\frac{\dot{g}}{k} \frac{r^2}{2} + C_1 \quad (a)$$

Applying the boundary condition at the center ($r = 0$) gives

$$\text{B.C. at } r = 0: 0 \times \frac{dT(0)}{dr} = -\frac{\dot{g}}{2k} \times 0 + C_1 \rightarrow C_1 = 0$$

Dividing both sides of Eq. (a) by r to bring it to a readily integrable form and integrating,

$$\frac{dT}{dr} = -\frac{\dot{g}}{2k} r \rightarrow T(r) = -\frac{\dot{g}}{4k} r^2 + C_2 \quad (b)$$



Applying the other boundary condition at $r = r_1$,

$$\text{B. C. at } r = r_1: \quad T_I = -\frac{\dot{q}}{4k}r_1^2 + C_2 \quad \rightarrow \quad C_2 = T_I + \frac{\dot{q}}{4k}r_1^2$$

Substituting this C_2 relation into Eq. (b) and rearranging give

$$T_{\text{wire}}(r) = T_I + \frac{\dot{q}}{4k_{\text{wire}}}(r_1^2 - r^2) \quad (c)$$

Plastic layer The mathematical formulation of heat transfer problem in the plastic can be expressed as

$$\frac{d}{dr} \left(r \frac{dT}{dr} \right) = 0$$

$$\text{with } T(r_1) = T_I \quad \text{and} \quad -k \frac{dT(r_2)}{dr} = h[T(r_2) - T_\infty]$$

The solution of the differential equation is determined by integration to be

$$r \frac{dT}{dr} = C_1 \quad \rightarrow \quad \frac{dT}{dr} = \frac{C_1}{r} \quad \rightarrow \quad T(r) = C_1 \ln r + C_2$$

where C_1 and C_2 are arbitrary constants. Applying the boundary conditions give

$$r = r_1: \quad C_1 \ln r_1 + C_2 = T_I \quad \rightarrow \quad C_2 = T_I - C_1 \ln r_1$$

$$r = r_2: \quad -k \frac{C_1}{r_2} = h[(C_1 \ln r_2 + C_2) - T_\infty] \quad \rightarrow \quad C_1 = \frac{T_\infty - T_I}{\ln \frac{r_2}{r_1} + \frac{k}{hr_2}}$$

Substituting C_1 and C_2 into the general solution, the variation of temperature in plastic is determined to be

$$T_{\text{plastic}}(r) = C_1 \ln r + T_I - C_1 \ln r_1 = T_I + \frac{T_\infty - T_I}{\ln \frac{r_2}{r_1} + \frac{k_{\text{plastic}}}{hr_2}} \ln \frac{r}{r_1}$$

We have already utilized the first interface condition by setting the wire and ceramic layer temperatures equal to T_I at the interface $r = r_1$. The interface temperature T_I is determined from the second interface condition that the heat flux in the wire and the plastic layer at $r = r_1$ must be the same:

$$-k_{\text{wire}} \frac{dT_{\text{wire}}(r_1)}{dr} = -k_{\text{plastic}} \frac{dT_{\text{plastic}}(r_1)}{dr} \quad \rightarrow \quad \frac{\dot{q}r_1}{2} = -k_{\text{plastic}} \frac{T_\infty - T_I}{\ln \frac{r_2}{r_1} + \frac{k}{hr_2}} \frac{1}{r_1}$$

Solving for T_I and substituting the given values, the interface temperature is determined to be

$$\begin{aligned} T_I &= \frac{\dot{q}r_1^2}{2k_{\text{plastic}}} \left(\ln \frac{r_2}{r_1} + \frac{k_{\text{plastic}}}{hr_2} \right) + T_\infty \\ &= \frac{(1.5 \times 10^6 \text{ W/m}^3)(0.003 \text{ m})^2}{2(1.8 \text{ W/m} \cdot ^\circ\text{C})} \left(\ln \frac{0.007 \text{ m}}{0.003 \text{ m}} + \frac{1.8 \text{ W/m} \cdot ^\circ\text{C}}{(14 \text{ W/m}^2 \cdot ^\circ\text{C})(0.007 \text{ m})} \right) + 25^\circ\text{C} = \mathbf{97.1^\circ\text{C}} \end{aligned}$$

Knowing the interface temperature, the temperature at the center line ($r = 0$) is obtained by substituting the known quantities into Eq. (c),

$$T_{\text{wire}}(0) = T_I + \frac{\dot{q}r_1^2}{4k_{\text{wire}}} = 97.1^\circ\text{C} + \frac{(1.5 \times 10^6 \text{ W/m}^3)(0.003 \text{ m})^2}{4 \times (18 \text{ W/m} \cdot ^\circ\text{C})} = \mathbf{97.3^\circ\text{C}}$$

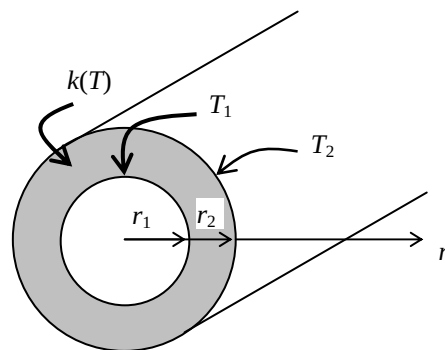
Thus the temperature of the centerline will be slightly above the interface temperature.

2-133 A cylindrical shell with variable conductivity is subjected to specified temperatures on both sides. The rate of heat transfer through the shell is to be determined.

Assumptions 1 Heat transfer is given to be steady and one-dimensional. 2 Thermal conductivity varies quadratically. 3 There is no heat generation.

Properties The thermal conductivity is given to be $k(T) = k_0(1 + \beta T^2)$.

Analysis When the variation of thermal conductivity with temperature $k(T)$ is known, the average value of the thermal conductivity in the temperature range between T_1 and T_2 is determined from



$$k_{\text{ave}} = \frac{\int_{T_1}^{T_2} k(T) dT}{T_2 - T_1} = \frac{\int_{T_1}^{T_2} k_0(1 + \beta T^2) dT}{T_2 - T_1} = \frac{k_0 \left(T + \frac{\beta}{3} T^3 \right) \Big|_{T_1}^{T_2}}{T_2 - T_1} = \frac{k_0 \left[(T_2 - T_1) + \frac{\beta}{3} (T_2^3 - T_1^3) \right]}{T_2 - T_1}$$

$$= k_0 \left[1 + \frac{\beta}{3} (T_2^2 + T_1 T_2 + T_1^2) \right]$$

This relation is based on the requirement that the rate of heat transfer through a medium with constant average thermal conductivity k_{ave} equals the rate of heat transfer through the same medium with variable conductivity $k(T)$.

Then the rate of heat conduction through the cylindrical shell can be determined from Eq. 2-77 to be

$$\dot{Q}_{\text{cylinder}} = 2\pi k_{\text{ave}} L \frac{T_1 - T_2}{\ln(r_2 / r_1)} = 2\pi k_0 \left[1 + \frac{\beta}{3} (T_2^2 + T_1 T_2 + T_1^2) \right] L \frac{T_1 - T_2}{\ln(r_2 / r_1)}$$

Discussion We would obtain the same result if we substituted the given $k(T)$ relation into the second part of Eq. 2-77, and performed the indicated integration.

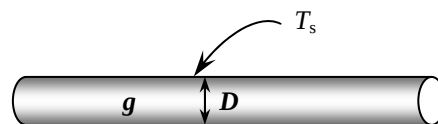
2-134 Heat is generated uniformly in a cylindrical uranium fuel rod. The temperature difference between the center and the surface of the fuel rod is to be determined.

Assumptions 1 Heat transfer is steady since there is no indication of any change with time. 2 Heat transfer is one-dimensional since there is thermal symmetry about the center line and no change in the axial direction. 3 Thermal conductivity is constant. 4 Heat generation is uniform.

Properties The thermal conductivity of uranium at room temperature is $k = 27.6 \text{ W/m}\cdot^\circ\text{C}$ (Table A-3).

Analysis The temperature difference between the center and the surface of the fuel rods is determined from

$$T_o - T_s = \frac{\dot{g} r_o^2}{4k} = \frac{(4 \times 10^7 \text{ W/m}^3)(0.016 \text{ m})^2}{4(27.6 \text{ W/m}\cdot^\circ\text{C})} = 92.8^\circ\text{C}$$



2-135 A large plane wall is subjected to convection on the inner and outer surfaces. The mathematical formulation, the variation of temperature, and the temperatures at the inner and outer surfaces to be determined for steady one-dimensional heat transfer.

Assumptions 1 Heat conduction is steady and one-dimensional. 2 Thermal conductivity is constant. 3 There is no heat generation.

Properties The thermal conductivity is given to be $k = 0.77 \text{ W/m}\cdot^\circ\text{C}$.

Analysis (a) Taking the direction normal to the surface of the wall to be the x direction with $x = 0$ at the inner surface, the mathematical formulation of this problem can be expressed as

$$\frac{d^2T}{dx^2} = 0$$

and

$$h_1[T_{\infty 1} - T(0)] = -k \frac{dT(0)}{dx}$$

$$-k \frac{dT(L)}{dx} = h_2[T(L) - T_{\infty 2}]$$

(b) Integrating the differential equation twice with respect to x yields

$$\frac{dT}{dx} = C_1$$

$$T(x) = C_1x + C_2$$

where C_1 and C_2 are arbitrary constants. Applying the boundary conditions give

$$x = 0: \quad h_1[T_{\infty 1} - (C_1 \times 0 + C_2)] = -kC_1$$

$$x = L: \quad -kC_1 = h_2[(C_1L + C_2) - T_{\infty 2}]$$

Substituting the given values, these equations can be written as

$$5(27 - C_2) = -0.77C_1$$

$$-0.77C_1 = (12)(0.2C_1 + C_2 - 8)$$

Solving these equations simultaneously give

$$C_1 = -45.44 \quad C_2 = 20$$

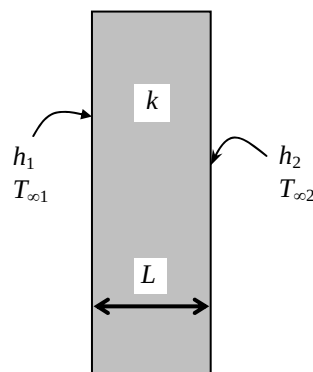
Substituting C_1 and C_2 into the general solution, the variation of temperature is determined to be

$$T(x) = 20 - 45.44x$$

(c) The temperatures at the inner and outer surfaces are

$$T(0) = 20 - 45.44 \times 0 = \mathbf{20^\circ\text{C}}$$

$$T(L) = 20 - 45.44 \times 0.2 = \mathbf{10.9^\circ\text{C}}$$



2-136 A hollow pipe is subjected to specified temperatures at the inner and outer surfaces. There is also heat generation in the pipe. The variation of temperature in the pipe and the center surface temperature of the pipe are to be determined for steady one-dimensional heat transfer.

Assumptions 1 Heat conduction is steady and one-dimensional since the pipe is long relative to its thickness, and there is thermal symmetry about the centerline. **2** Thermal conductivity is constant.

Properties The thermal conductivity is given to be $k = 20 \text{ W/m}\cdot^\circ\text{C}$.

Analysis The rate of heat generation is determined from

$$\dot{g} = \frac{\dot{W}}{V} = \frac{\dot{W}}{\pi(D_2^2 - D_1^2)L/4} = \frac{25,000 \text{ W}}{\pi[(0.4 \text{ m})^2 - (0.3 \text{ m})^2](12 \text{ m})/4} = 37,894 \text{ W/m}^3$$

Noting that heat transfer is one-dimensional in the radial r direction, the mathematical formulation of this problem can be expressed as

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dT}{dr} \right) + \frac{\dot{g}}{k} = 0$$

and $T(r_1) = T_1 = 60^\circ\text{C}$

$$T(r_2) = T_2 = 80^\circ\text{C}$$

Rearranging the differential equation

$$\frac{d}{dr} \left(r \frac{dT}{dr} \right) = \frac{-\dot{g}r}{k} = 0$$

and then integrating once with respect to r ,

$$r \frac{dT}{dr} = \frac{-\dot{g}r^2}{2k} + C_1$$

Rearranging the differential equation again

$$\frac{dT}{dr} = \frac{-\dot{g}r}{2k} + \frac{C_1}{r}$$

and finally integrating again with respect to r , we obtain

$$T(r) = \frac{-\dot{g}r^2}{4k} + C_1 \ln r + C_2$$

where C_1 and C_2 are arbitrary constants. Applying the boundary conditions give

$$r = r_1: \quad T(r_1) = \frac{-\dot{g}r_1^2}{4k} + C_1 \ln r_1 + C_2$$

$$r = r_2: \quad T(r_2) = \frac{-\dot{g}r_2^2}{4k} + C_1 \ln r_2 + C_2$$

Substituting the given values, these equations can be written as

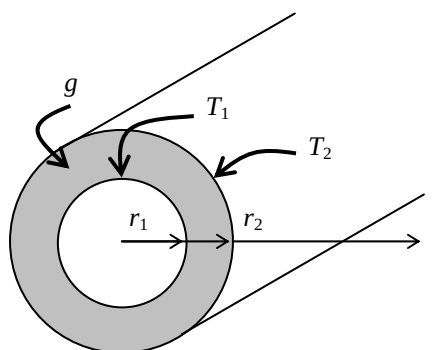
$$60 = \frac{-(37,894)(0.15)^2}{4(20)} + C_1 \ln(0.15) + C_2$$

$$80 = \frac{-(37,894)(0.20)^2}{4(20)} + C_1 \ln(0.20) + C_2$$

Solving for C_1 and C_2 simultaneously gives

$$C_1 = 98.34 \quad C_2 = 257.2$$

Substituting C_1 and C_2 into the general solution, the variation of temperature is determined to be



$$T(r) = \frac{-37,894r^2}{4(20)} + 98.34 \ln r + 257.2 = 257.2 - 473.68r^2 + 98.34 \ln r$$

The temperature at the center surface of the pipe is determined by setting radius r to be 17.5 cm, which is the average of the inner radius and outer radius.

$$T(r) = 257.2 - 473.68(0.175)^2 + 98.34 \ln(0.175) = \mathbf{71.2^\circ\text{C}}$$

2-137 A spherical ball in which heat is generated uniformly is exposed to iced-water. The temperatures at the center and at the surface of the ball are to be determined.

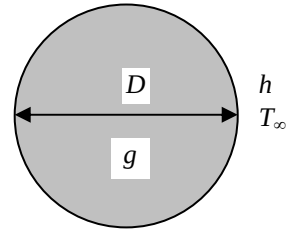
Assumptions **1** Heat transfer is steady since there is no indication of any change with time. **2** Heat transfer is one-dimensional, and there is thermal symmetry about the center point. **3** Thermal conductivity is constant. **4** Heat generation is uniform.

Properties The thermal conductivity is given to be $k = 45 \text{ W/m}\cdot^\circ\text{C}$.

Analysis The temperatures at the center and at the surface of the ball are determined directly from

$$T_s = T_\infty + \frac{\dot{g}r_0}{3h} = 0^\circ\text{C} + \frac{(2.6 \times 10^6 \text{ W/m}^3)(0.15 \text{ m})}{3(1200 \text{ W/m}^2 \cdot ^\circ\text{C})} = \mathbf{108.3^\circ\text{C}}$$

$$T_0 = T_s + \frac{\dot{g}r_0^2}{6k} = 108.3^\circ\text{C} + \frac{(2.6 \times 10^6 \text{ W/m}^3)(0.15 \text{ m})^2}{6(45 \text{ W/m}\cdot^\circ\text{C})} = \mathbf{325^\circ\text{C}}$$



2-138 2-141 Design and Essay Problems
